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Detecting *Babesia* species nucleic acid in a sample

Abstract

There is described herein a method for specifically detecting *Babesia* species nucleic acid in a sample, which in one aspect comprises: (1) contacting a sample, said sample suspected of containing *Babesia* species nucleic acid, with at least two oligomers for amplifying a target region of a *Babesia* species target nucleic acid, wherein the at least two amplification oligomers comprise: (a) a first amplification oligomer comprising a first target-hybridizing sequence (i) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:66 and comprises SEQ ID NO:56 or 57; or (ii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:96 and comprises SEQ ID NO:101; or (iii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:97 and comprises SEQ ID NO:101; (iv) comprises or consists of SEQ ID NO:8; (v) comprises or consists of SEQ ID NO:83 and (b) a second amplification oligomer comprising a second target-hybridizing sequence that is from about 15 to about 33 contiguous nucleotides in length, and (i) is contained in SEQ ID NO:68 and comprises SEQ ID NO:52, SEQ ID NO:53, SEQ ID NO:54, SEQ ID NO:55, or SEQ ID NO:85; or (ii) is contained in SEQ ID NO:67 and comprises SEQ ID NO:45 or SEQ ID NO:52; or (iii) is contained in SEQ ID NO:70 and comprises SEQ ID NO:46, SEQ ID NO:47, SEQ ID NO:48, SEQ ID NO:49, SEQ ID NO:50, or SEQ ID NO:51; (2) performing an in vitro nucleic acid amplification reaction, wherein any *Babesia* target nucleic acid present in said sample is used as a template for generating an amplification product; and (3) detecting the presence or absence of the amplification product, thereby indicating the presence or absence of *Babesia* species target nucleic acid in said sample.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4581333	12/1985	Kourilsky et al.	N/A	N/A
4683195	12/1986	Mullis et al.	N/A	N/A
4683202	12/1986	Mullis et al.	N/A	N/A
4786600	12/1987	Kramer et al.	N/A	N/A
4800159	12/1988	Mullis et al.	N/A	N/A
4868105	12/1988	Urdea et al.	N/A	N/A
5118801	12/1991	Lizardi et al.	N/A	N/A
5124246	12/1991	Urdea et al.	N/A	N/A
5130238	12/1991	Malek et al.	N/A	N/A
5283174	12/1993	Arnold, Jr. et al.	N/A	N/A
5312728	12/1993	Lizardi et al.	N/A	N/A
5378825	12/1994	Cook et al.	N/A	N/A
5422252	12/1994	Walker et al.	N/A	N/A
5424413	12/1994	Hogan et al.	N/A	N/A
5427930	12/1994	Birkenmeyer et al.	N/A	N/A
5437990	12/1994	Burg et al.	N/A	N/A
5451506	12/1994	Shoyab et al.	N/A	N/A

5516663	12/1995	Backman et al.	N/A	N/A
5547842	12/1995	Hogan et al.	N/A	N/A
5547861	12/1995	Nadeau et al.	N/A	N/A
5554516	12/1995	Kacian et al.	N/A	N/A
5585481	12/1995	Arnold, Jr. et al.	N/A	N/A
5614402	12/1996	Dahlberg et al.	N/A	N/A
5648211	12/1996	Fraiser et al.	N/A	N/A
5658737	12/1996	Nelson et al.	N/A	N/A
5823174	12/1997	Andress	N/A	N/A
5846717	12/1997	Brow et al.	N/A	N/A
5849481	12/1997	Urdea et al.	N/A	N/A
5925517	12/1998	Tyagi et al.	N/A	N/A
6150097	12/1999	Tyagi et al.	N/A	N/A
6180340	12/2000	Nelson	N/A	N/A
6350579	12/2001	Nelson	N/A	N/A
6361945	12/2001	Becker et al.	N/A	N/A
6534274	12/2002	Becker et al.	N/A	N/A
6706471	12/2003	Brow et al.	N/A	N/A
6835542	12/2003	Becker et al.	N/A	N/A
6849412	12/2004	Becker et al.	N/A	N/A
6949367	12/2004	Dempey et al.	N/A	N/A
7374885	12/2007	Becker et al.	N/A	N/A
11667978	12/2022	Bres	435/6.11	C12Q 1/6806
2006/0068417	12/2005	Becker et al.	N/A	N/A
2006/0194240	12/2005	Arnold et al.	N/A	N/A
2013/0157874	12/2012	Dowd	N/A	N/A
2016/0201144	12/2015	Chelliserry	435/6.11	C12Q 1/6806

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2922689	12/2014	CA	N/A
102181556	12/2010	CN	N/A
102230008	12/2010	CN	N/A
103103286	12/2013	CN	N/A
WO 88/01302	12/1987	WO	N/A
WO 88/10315	12/1987	WO	N/A
WO 89/002476	12/1988	WO	N/A
WO 93/13121	12/1992	WO	N/A
WO 95/03430	12/1994	WO	N/A
WO 95/32305	12/1994	WO	N/A
WO 99/60009	12/1998	WO	N/A
WO 2014/071946	12/2013	WO	N/A
WO 2014/197607	12/2013	WO	N/A
WO 2016/064887	12/2015	WO	N/A
WO 2016/183282	12/2015	WO	N/A

OTHER PUBLICATIONS

Allawi, et al., Invader Plus Method Detects Herpes Simplex Virus in Cerebrospinal Fluid and Simultaneously Differentiates Types 1 and 2. *J. Clin. Microbiol.* 44:3443-3447, 2006. cited by applicant

Allawi, et al., Modeling of Flap Endonuclease Interactions with DNA Substrate, *J. Mol. Biol.* (2003) 328, 537-554. doi:10.1016/S0022-2836(03)00351-6. cited by applicant

Berger, et al., *Methods in Enzymology*, vol. 152, Guide to Molecular Cloning Techniques; pp. 20-24, 33-48, 94-112, 173-180, 215-219, 248-254, 307-329, 343-349, 432-442, 522-538 and 556-562 (Academic Press, Inc., San Diego, Calif., 1987). cited by applicant

Cagnin, et al., Overview of Electrochemical DNA Biosensors: New Approaches to Detect the Expression of Life, *Sensors* 2009, 9, 3122-3148; doi:10.3390/s90403122. cited by applicant

Daniels, et al., Label-Free Impedance Biosensors: Opportunities and Challenges, *Electroanalysis*. May 16, 2007; 19(12): 1239-1257. cited by applicant

Drummond, et al., Electrochemical DNA sensors, *Nat. Biotechnol.* (2003) 21: 1192. cited by applicant

Eis, et al., An invasive cleavage assay for direct quantitation of specific RNAs, Nature Publishing Group, <http://biotech.nature.com> (Jul. 2001) vol. 19. cited by applicant

Feng, et al, Roles of divalent metal ions in flap endonuclease-substrate interactions, *Nat. Struct. Mol. Biol.* 11 :450-456, 2004. cited by applicant

Gooding, J. Justin, Electrochemical DNA Hybridization Biosensors, *Electroanalysis* 14: 1149, 2002. cited by applicant

Katz, et al., Probing Biomolecular Interactions at Conductive and Semiconductive Surfaces by Impedance Spectroscopy: Routes to Impedimetric Immunosensors, DNA-Sensors, and Enzyme Biosensors, *Electroanalysis* 2003, 15, No. 11. cited by applicant

Kaiser, et al., A Comparison of Eubacterial and Archaeal Structure-specific 5'- Exonucleases*, *The Journal of Biological Chemistry*, vol. 274, No. 30, Issue of Jul. 23, pp. 21387-21394, 1999. cited by applicant

Levin, et al., Transfusion-transmitted babesiosis: Is it time to screen the blood supply? *Curr Opin Hematol* (2016) 23(6):573-580. cited by applicant

Lyamichev, et al., Polymorphism identification and quantitative detection of genomic DNA by invasive cleavage of oligonucleotide probes, *Nat. Biotechnol.*(1999) 17:292-296. cited by applicant

Lyamichev, et al., Structure-Specific Endonucleolytic Cleavage of Nucleic Acids by Eubacterial DNA Polymerases, *Science* (May 1993) vol. 20, 7. cited by applicant

Ma, et al., RNA Template-dependent 5' Nuclease Activity of *Thermus aquaticus* and *Thermus thermophilus* DNA Polymerases*, *The Journal of Biological Chemistry* (2000) vol. 275, No. 32, Issue of Aug. 11, pp. 24693-24700. cited by applicant

Moritz, et al., Screening for *Babesia microti* in the U.S.Blood Supply *N Engl J Med* . (2016) 8, 375(23):2236-2245. cited by applicant

Ryan, et al., Non-PCR-Dependent Detection of the Factor V Leiden Mutation from Genomic DNA Using a Homogeneous Invader Microtiter Plate Assay, *Mol. Diagn.* (1999) 4: 135-144. cited by applicant

Sharma, et al., The Exonucleolytic and Endonucleolytic Cleavage Activities of Human Exonuclease 1 Are Stimulated by an Interaction with the Carboxyl-terminal Region of the Werner Syndrome Protein* *J. Biol. Chem.* (2003) vol. 278, No. 26, Issue of Jun. 27, pp. 23487-23496. cited by applicant

Shen, et al., Flap endonuclease homologs in archaeobacteria exist as independent proteins, *TIBS* 23, May 1998. cited by applicant

Vester, et al., LNA (Locked Nucleic Acid): High-Affinity Targeting of Complementary RNA and DNA†, *Biochemistry* (2004) 43: 13233-41. cited by applicant

Wang, Joseph, Electrochemical nucleic acid biosensors, *Anal. Chim. Acta* (2002) 469:63. cited by applicant

applicant

Examination Report dated Oct. 3, 2024, in corresponding Canadian Application No. 3,054,603.

cited by applicant

Blaschitz, et al., *Babesia* Species Occurring in Austrian Ixodes Ricinus Ticks, *Applied and environmental microbiology*; 2008: pp. 4841-4846. doi:10.1128/AEM.00035-08. cited by applicant

Bloch, et al. Development of real-time polymerase chain reaction assay for detection and quantitation of *Babesia microti* infection. *Transfusion*, 2013, vol. 53, Issue 10. cited by applicant

Cornillot, et al. Sequencing of the smallest *Apicomplexan* genome from the human pathogen

Babesia microti, *Nucleic Acids Research*, 2012, vol. 40, No. 18. cited by applicant

Criado-Fornelio, A., "A review of nucleic-acid-based diagnostic tests for *Babesia* and *Theileria*, with emphasis on bovine piroplasms", Italy May 1, 2007 (May 1, 2007), p. 39. cited by applicant

Eshoo, et al., Broad-Range Survey of Tick-Borne Pathogens in Southern Germany Reveals a High Prevalence of *Babesia microti* and a Diversity of Other Tick-Borne Pathogens Vector-Borne and zoonotic diseases, *Vector-Borne and Zoonotic Diseases* (2014) vol. 14; No. 8, pp. 584-591. cited by

applicant

Teal, et al., A new real-time PCR assay for improved detection of the parasite *Babesia microti*,

Journal of Clinical Microbiology (2012) vol. 50, No. 3, pp. 903-908. cited by applicant

Tonnetti, et al., Transfusion-transmitted *Babesia microti* identified through hemovigilance,

Transfusion (2009) vol. 49, pp. 2557-2563. cited by applicant

International Search Report and Written Opinion mailed Aug. 9, 218, in corresponding application No. PCT/US2018/036214. cited by applicant

Examination Report dated Dec. 21, 2020, in corresponding European Application No. 18734394.2. cited by applicant

Examination Report dated Apr. 20, 2021, in corresponding Australian Application No. 2018281196. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 16/611,806, filed Nov. 7, 2019, which is a national stage entry of International Application No. PCT/US2018/036214, filed Jun. 6, 2018, which claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application Nos. 62/516,530, filed Jun. 7, 2017, and 62/520,793, filed Jun. 16, 2017. The entire contents of each of the foregoing applications are incorporated herein by reference in their entirety.

REFERENCE TO SEQUENCE LISTING

(1) The instant application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML Copy, created on Mar. 30, 2023, is named "GPR_7520 US_Seq_Listing" and is 158,798 bytes in size.

BACKGROUND

(2) Babesiosis is caused by infection of red blood cells by species of protozoan parasites of the genus *Babesia*. The species *Babesia microti* is responsible for most human babesiosis infections reported in the United States. Infections are typically asymptomatic in human individuals but can lead to severe illness or death, especially in elderly or immunosuppressed individuals. The parasite is transmitted to humans by exposure to deer ticks in endemic areas or by blood transfusion

(transfusion-transmitted babesiosis (TTB)). Over 100 cases of transfusion-transmitted babesiosis have been reported to the FDA since 1979. Despite being reported as the most unaddressed infectious risk to the United States blood supply, there is still no licensed test for screening for *B. microti* in donated blood (*N Engl J Med.* (2016) 8, 375(23):2236-2245). The threat of TTB has led to a consensus by the Food and Drug Administration (FDA) and the American Association of Blood Banks (AABB) that screening of blood donations for *Babesia* is urgently required for blood safety. Initiation of blood donor screening to prevent TTB should be given high priority (*Curr Opin Hematol.* (2016) 23(6):573-580).

(3) Therefore a specific and sensitive assay for detecting *Babesia* species in a sample is needed.

SUMMARY

(4) In one aspect, there is provided a method for specifically detecting *Babesia* species nucleic acid in a sample, said method comprising: (1) contacting a sample, said sample suspected of containing *Babesia* species nucleic acid, with at least two oligomers for amplifying a target region of a *Babesia* species target nucleic acid, wherein the at least two amplification oligomers comprise: (a) a first amplification oligomer comprising a first target-hybridizing sequence (i) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:66 and comprises SEQ ID NO:56 or 57; or (ii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:96 and comprises SEQ ID NO:101; or (iii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:97 and comprises SEQ ID NO:101; (iv) comprises or consists of SEQ ID NO:8; (v) comprises or consists of SEQ ID NO:83 and (b) a second amplification oligomer comprising a second target-hybridizing sequence that is from about 15 to about 33 contiguous nucleotides in length, and (i) is contained in SEQ ID NO:68 and comprises SEQ ID NO:52, SEQ ID NO:53, SEQ ID NO:54, SEQ ID NO:55, or SEQ ID NO:85; or (ii) is contained in SEQ ID NO:67 and comprises SEQ ID NO:45 or SEQ ID NO:52; (iii) is contained in SEQ ID NO:70 and comprises SEQ ID NO:46, SEQ ID NO:47, SEQ ID NO:48, SEQ ID NO:49, SEQ ID NO:50, or SEQ ID NO:51, or (iv) comprises or consists of SEQ ID NO:84; (2) performing an in vitro nucleic acid amplification reaction, wherein any *Babesia* target nucleic acid present in said sample is used as a template for generating an amplification product; and (3) detecting the presence or absence of the amplification product, thereby indicating the presence or absence of *Babesia* species target nucleic acid in said sample.

(5) Suitably, the first target-hybridizing sequence comprises or consists of a sequence selected from the group consisting of: SEQ ID NOs:2, and 4 and 6 and 8, suitably, wherein the first target-hybridizing sequence comprises or consists of a sequence selected from the group consisting of: SEQ ID NOs:2, and 4 and 8.

(6) Suitably, the second amplification oligomer comprises or consists of the sequence selected from the group consisting of: SEQ ID NOs:13, 16, 17, 18, 19, 20, 21, 27, 28, 29, 31, 32, 33, 34, 35, 36, 84 and 86.

(7) Suitably, the second amplification oligomer sequence comprises or consists of SEQ ID NO:21 or SEQ ID NO:27 or SEQ ID NO:34 or SEQ ID NO:84 or SEQ ID NO:86.

(8) Suitably, the first amplification oligomer is a promoter primer or promoter provider further comprising a promoter sequence located 5' to the first target-hybridizing sequence. Suitably, the promoter sequence is a T7 promoter sequence. Suitably, the T7 promoter sequence comprises or consists of SEQ ID NO:58.

(9) Suitably, the first amplification oligomer comprises or consists of a sequence selected from the group consisting of SEQ ID NOs:1 and 3 and 5 and 7 and 82, suitably, wherein the first amplification oligomer comprises or consists of a sequence selected from the group consisting of SEQ ID NOs:1 and 3 and 7 and 82.

(10) Suitably, the first and second target-hybridizing sequences respectively comprise or consist of the nucleotide sequences of: (a) SEQ ID NO:2 or 6 and SEQ ID NO:11; (b) SEQ ID

NO:4 or 6 and SEQ ID NO:13; (c) SEQ ID NO:4 and SEQ ID NO:16 or SEQ ID NO:17; (d) SEQ ID NO:4 and SEQ ID NO:18 or SEQ ID NO:19; (e) SEQ ID NO:4 and SEQ ID NO:20; (f) SEQ ID NO:4 or 6 or 8 and SEQ ID NO:21; (g) SEQ ID NO:2 or 4 or 8 and SEQ ID NO:27; (h) SEQ ID NO:4 and SEQ ID NO:28; (i) SEQ ID NO:4 and SEQ ID NO:29; (j) SEQ ID NO:4 and SEQ ID NO:31; (k) SEQ ID NO:8 and SEQ ID NO:32; (l) SEQ ID NO:8 and SEQ ID NO:33; (m) SEQ ID NO:8 and SEQ ID NO:34; (n) SEQ ID NO:8 and SEQ ID NO:35; (o) SEQ ID NO:8 and SEQ ID NO:36; (p) SEQ ID NO:8 and SEQ ID NO:84; (q) SEQ ID NO:8 and SEQ ID NO:86; (r) SEQ ID NO:83 and SEQ ID NO:34; (s) SEQ ID NO:83 and SEQ ID NO:35; (t) SEQ ID NO:83 and SEQ ID NO:36; (u) SEQ ID NO:83 and SEQ ID NO:84; (v) SEQ ID NO:83 and SEQ ID NO:86.

(11) Suitably, the first and second target-hybridizing sequences respectively comprise or consist of the nucleotide sequences of: (a) SEQ ID NO:2 and SEQ ID NO:27; (b) SEQ ID NO:4 and SEQ ID NO:21; (c) SEQ ID NO:8 and SEQ ID NO:21; (d) SEQ ID NO:8 and SEQ ID NO:34; (e) SEQ ID NO:8 and SEQ ID NO:84; (f) SEQ ID NO:8 and SEQ ID NO:86; (g) SEQ ID NO:83 and SEQ ID NO:34; (h) SEQ ID NO:83 and SEQ ID NO:84; (i) SEQ ID NO:83 and SEQ ID NO:86.

(12) Suitably, the method further comprises purifying the target nucleic acid from other components in the sample before step (1).

(13) Suitably, the purifying step comprises contacting the sample with at least one capture probe oligomer comprising a target-hybridizing sequence covalently attached to a sequence or moiety that binds to an immobilized probe, wherein said target-hybridizing sequence (i) is from about 15 to 21 contiguous nucleotides contained in the sequence of SEQ ID NO:78; or (ii) is from about 21 to 30 contiguous nucleotides comprising the sequence of SEQ ID NO:78; or (iii) is the sequence is SEQ ID NO:44. Suitably, the capture probe oligomer sequence comprises or consists of SEQ ID NO:43.

(14) Suitably, the detecting step (3) comprises contacting said in vitro nucleic acid amplification reaction with a detection probe oligomer configured to specifically hybridize to the amplification product under conditions whereby the presence or absence of the amplification product is determined, thereby indicating the presence or absence of *Babesia* species in said sample.

(15) Suitably, the detection probe oligomer comprises a target-hybridizing sequence that is from about 14 to about 40 nucleotides in length and is configured to specifically hybridize to a target sequence comprising or consisting of SEQ ID NO:59, the RNA equivalent of SEQ ID NO:59, the complement of SEQ ID NO:59, the RNA equivalent of the complement of SEQ ID NO:59, or SEQ ID NO:65, the DNA equivalent of SEQ ID NO:65, the complement of SEQ ID NO:65, or the DNA equivalent of the complement of SEQ ID NO:65.

(16) Suitably, the detection probe target-hybridizing sequence is contained in the sequence of SEQ ID NO:65 and includes at least the sequence of SEQ ID NO:37 or SEQ ID NO:42.

(17) Suitably, the detection probe target-hybridizing sequence is contained in the sequence of SEQ ID NO:41 and includes at least the sequence of SEQ ID NO:38 or SEQ ID NO:39.

(18) Suitably, the detection probe consists of the sequence selected from the group consisting of: SEQ ID NO:37, 38, 39, 41, 42, 91, 92, 94, or 99.

(19) Suitably, the detection probe comprises or consists of a nucleotide sequence selected from the group consisting of SEQ ID NOs:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 and 99.

(20) Suitably, the detection probe oligomer further comprises a 2' methoxy modification on at least one nucleotide residue member of the nucleotide sequence.

(21) Suitably, the first and second amplification oligomer target-hybridizing sequences and the detection probe oligomer target-hybridizing sequences respectively comprise or consist of the nucleotide sequences of: (a) SEQ ID NO:2 and SEQ ID NO:11 and SEQ ID NO:39 or SEQ ID NO:37; (b) SEQ ID NO:2 and SEQ ID NO:27 and SEQ ID NO:38 or SEQ ID NO:39; (c) SEQ ID NO:4 and SEQ ID NO:13 and SEQ ID NO:39 or SEQ ID NO:37; (d) SEQ ID NO:4 and SEQ ID NO:16 or SEQ ID NO:17 and SEQ ID NO:39; (e) SEQ ID NO:4 and SEQ ID NO:18 or SEQ ID NO:19 and SEQ ID NO:39 or SEQ ID NO:37; (f) SEQ ID NO:4 and SEQ ID NO:20 and SEQ ID NO:39 or SEQ ID NO:37; (g) SEQ ID NO:4 and SEQ ID NO:21 and SEQ ID NO:39 or SEQ ID

NO:37; (h) SEQ ID NO:4 and SEQ ID NO:27 and SEQ ID NO:39 or SEQ ID NO:38; (i) SEQ ID NO:4 and SEQ ID NO:28 and SEQ ID NO:39; (j) SEQ ID NO:4 and SEQ ID NO:29 and SEQ ID NO:39 or SEQ ID NO:37; (k) SEQ ID NO:4 and SEQ ID NO:31 and SEQ ID NO:39; (l) SEQ ID NO:6 and SEQ ID NO:11 and SEQ ID NO:37; (m) SEQ ID NO:6 and SEQ ID NO:13 and SEQ ID NO:37; (n) SEQ ID NO:6 and SEQ ID NO:21 and SEQ ID NO:37; (o) SEQ ID NO:8 and SEQ ID NO:21 and SEQ ID NO:39 or SEQ ID NO:37 or SEQ ID NO:42; (p) SEQ ID NO:8 and SEQ ID NO:27 and SEQ ID NO:39; (q) SEQ ID NO:8 and SEQ ID NO:32 and SEQ ID NO:37 or SEQ ID NO:42; (r) SEQ ID NO:8 and SEQ ID NO:33 and SEQ ID NO:37 or SEQ ID NO:42; (s) SEQ ID NO:8 and SEQ ID NO:34 and SEQ ID NO:37 or SEQ ID NO:42; (t) SEQ ID NO:8 and SEQ ID NO:35 and SEQ ID NO:37 or SEQ ID NO:42; (u) SEQ ID NO:8 and SEQ ID NO:36 and SEQ ID NO:37 or SEQ ID NO:42; (v) SEQ ID NO:8, and SEQ ID NO:34, 84, or 86, and SEQ ID NO:91, 92, or 93; (v) SEQ ID NO:8, and SEQ ID NO:34, 84, or 86, and SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99; or (x) SEQ ID NO:83, and SEQ ID NO:34, 84, or 86, and SEQ ID NO:91, 92, or 93.

(22) Suitably, the first and second amplification oligomer target-hybridizing sequences and the detection probe oligomer target-hybridizing sequences respectively comprise or consist of the nucleotide sequences of: (a) SEQ ID NO:2 and SEQ ID NO:27 and SEQ ID NO:38; (b) SEQ ID NO:4 and SEQ ID NO:21 and SEQ ID NO:39; (c) SEQ ID NO:8 and SEQ ID NO:21 and SEQ ID NO:37 or SEQ ID NO:39; (d) SEQ ID NO:8 and SEQ ID NO:34 and SEQ ID NO:37, 42, 91, 92 or 93; (e) SEQ ID NO:8 and SEQ ID NO:34, or 84 or 86 and SEQ ID NO:37, 42, 91, 92 or 93; (f) SEQ ID NO:83 and SEQ ID NO:34, 84 or 86 and SEQ ID NO:37, 42, 91, 92 or 93; (g) SEQ ID NO:8 and SEQ ID NO:34, 84 or 86 and SEQ ID NO:37, 42, 91, 92 or 93; (h) SEQ ID NO:83 and SEQ ID NO:34, 84 or 86 and SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99; or (i) SEQ ID NO:83 and SEQ ID NO:34, 84 or 86 and SEQ ID NO:37, 42, 91, 92 or 93.

(23) Suitably, the detection probe comprises a label. Suitably, the label is a chemiluminescent label or a fluorescent label. Suitably, the detecting step (3) occurs during the amplifying step (2). Suitably, the detection probe comprises a fluorescent label and a quencher. Suitably, the detection probe is selected from the group consisting of a molecular torch, a molecular beacon, and a TaqMan detection probe. Suitably, the detection probe further comprises a non-target-hybridizing sequence. Suitably, the detection probe is a molecular torch or a molecular beacon.

(24) Suitably, the amplification reaction at step (2) is an isothermal amplification reaction. Suitably, the amplification reaction is a transcription-mediated amplification (TMA) reaction. Suitably, the amplification reaction is a real-time amplification reaction.

(25) Suitably, the sample is a clinical sample. Suitably, the sample is a blood sample. Suitably, the sample is a lysed blood cell sample. Suitably, the lysed blood cell sample is a lysed red blood cell sample.

(26) Suitably, the amplification product has a length of from 180 to 220 contiguous nucleotides and contains SEQ ID NO:65 or the complement thereof.

(27) In a further aspect, there is described a method for specifically detecting *Babesia* species nucleic acid in a sample, said method comprising: (1) contacting a sample, said sample suspected of containing *Babesia* species nucleic acid, with at least two oligomers for amplifying a target region of a *Babesia* species target nucleic acid, wherein two of said at least two amplification oligomers are selected from the group consisting of: (a) a first amplification oligomer and a second amplification oligomer, wherein the first amplification oligomer comprises a first target-hybridizing sequence (i) that is from 15 to 33 contiguous nucleobases in length, is contained in SEQ ID NO:66 and contains SEQ ID NO:56 or SEQ ID NO:57, or (ii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:96 and comprises SEQ ID NO:101; or (iii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:97 and comprises SEQ ID NO:101; (iv) comprises/consists of SEQ ID NO:8; (v) comprises or consists of SEQ ID NO:83; or (b) a first

amplification oligomer and a second amplification oligomer, wherein the second amplification oligomer comprises a second target-hybridizing sequence that is from about 15 to about 33 contiguous nucleotides in length, and (i) is contained in SEQ ID NO:68 and contains SEQ ID NO:52, SEQ ID NO:53 SEQ ID NO:54, SEQ ID NO:55, or SEQ ID NO:85, or (ii) is contained in SEQ ID NO:67 and contains SEQ ID NO:45 or SEQ ID NO:69, or (iii) is contained in SEQ ID NO:70 and contains SEQ ID NO:46, SEQ ID NO:47, SEQ ID NO:48, SEQ ID NO:49, SEQ ID NO:50, or SEQ ID NO:51, or (iv) comprises or consists of SEQ ID NO:84; (2) performing an in vitro nucleic acid amplification reaction, wherein any *Babesia* target nucleic acid present in said sample is used as a template for generating an amplification product, wherein said amplification product has a length of from 180 to 220 contiguous nucleotides and contains SEQ ID NO:65 or the complement thereof; and (3) detecting the presence or absence of the amplification product, thereby indicating the presence or absence of *Babesia* species target nucleic acid in said sample.

(28) Suitably, the first target-hybridizing sequence comprises or consists of a sequence selected from the group consisting of: SEQ ID NOs:2, and 4 and 6 and 8, suitably, wherein the first target-hybridizing sequence comprises or consists of a sequence selected from the group consisting of: SEQ ID NOs:2, and 4 and 8.

(29) Suitably, the second amplification oligomer comprises or consists of the sequence selected from the group consisting of: SEQ ID NOs:13, 16, 17, 18, 19, 20, 21, 27, 28, 29, 31, 32, 33, 34, 35, 36, 84, and 86.

(30) Suitably, the second amplification oligomer sequence comprises or consists of SEQ ID NO:21 or SEQ ID NO:27 or SEQ ID NO:34 or SEQ ID NO:84 or SEQ ID NO:86.

(31) Suitably, the first amplification oligomer is a promoter primer or promoter provided further comprising a promoter sequence located 5' to the first target-hybridizing sequence. Suitably, the promoter sequence is a T7 promoter sequence. Suitably, the T7 promoter sequence comprises or consists of SEQ ID NO:58. Suitably, the first amplification oligomer comprises or consists of a sequence selected from the group consisting of SEQ ID NOs:1 and 3 and 5 and 7 and 82, suitably, wherein the first amplification oligomer comprises or consists of a sequence selected from the group consisting of SEQ ID NOs:1 and 3 and 7 and 82.

(32) Suitably, the first and second target-hybridizing sequences respectively comprise or consist of the nucleotide sequences of: (a) SEQ ID NO:2 or 6 and SEQ ID NO:11; (b) SEQ ID NO:4 or 6 and SEQ ID NO:13; (c) SEQ ID NO:4 and SEQ ID NO:16 or SEQ ID NO:17; (d) SEQ ID NO:4 and SEQ ID NO:18 or SEQ ID NO:19; (e) SEQ ID NO:4 and SEQ ID NO:20; (f) SEQ ID NO:4 or 6 or 8 and SEQ ID NO:21; (g) SEQ ID NO:2 or 4 or 8 and SEQ ID NO:27; (h) SEQ ID NO:4 and SEQ ID NO:28; (i) SEQ ID NO:4 and SEQ ID NO:29; (j) SEQ ID NO:4 and SEQ ID NO:31; (k) SEQ ID NO:8 and SEQ ID NO:32; (l) SEQ ID NO:8 and SEQ ID NO:33; (m) SEQ ID NO:8 and SEQ ID NO:34; (n) SEQ ID NO:8 and SEQ ID NO:35; (o) SEQ ID NO:8 and SEQ ID NO:36; (p) SEQ ID NO:8 and SEQ ID NO:84; (q) SEQ ID NO:8 and SEQ ID NO:86; (r) SEQ ID NO:83 and SEQ ID NO:34; (s) SEQ ID NO:83 and SEQ ID NO:84; (t) SEQ ID NOs:83 and SEQ ID NO:86.

(33) Suitably, the first and second target-hybridizing sequences respectively comprise or consist of the nucleotide sequences of: (a) SEQ ID NO:2 and SEQ ID NO:27; (b) SEQ ID NO:4 and SEQ ID NO:21; (c) SEQ ID NO:8 and SEQ ID NO:21; (d) SEQ ID NO:8 and SEQ ID NO:34; (e) SEQ ID NO:8 and SEQ ID NO:84; (f) SEQ ID NO:8 and SEQ ID NO:86; (g) SEQ ID NO:83 and SEQ ID NO:34; (h) SEQ ID NO:83 and SEQ ID NO:84; (i) SEQ ID NO:83 and SEQ ID NO:86.

(34) Suitably, the method further comprises purifying the target nucleic acid from other components in the sample before step (1).

(35) Suitably, the purifying step comprises contacting the sample with at least one capture probe oligomer comprising a target-hybridizing sequence covalently attached to a sequence or moiety that binds to an immobilized probe, wherein said target-hybridizing sequence (i) is from about 15 to 21 contiguous nucleotides contained in the sequence of SEQ ID NO:78; or (ii) is from about 21 to about 30 contiguous nucleotides comprising the sequence of SEQ ID NO:78; or (iii) the sequence

consists of SEQ ID NO:44.

(36) Suitably, the capture probe oligomer sequence comprises or consists of SEQ ID NO:43.

(37) Suitably, the detecting step (3) comprises contacting said in vitro nucleic acid amplification reaction with a detection probe oligomer configured to specifically hybridize to the amplification product under conditions whereby the presence or absence of the amplification product is determined, thereby indicating the presence or absence of *Babesia* species in said sample.

(38) Suitably, the detection probe oligomer comprises a target-hybridizing sequence that is from about 14 to about 40 nucleotides in length and is configured to specifically hybridize to a target sequence comprising or consisting of SEQ ID NO:59, the RNA equivalent of SEQ ID NO:59, the complement of SEQ ID NO:59, the RNA equivalent of the complement of SEQ ID NO:59, SEQ ID NO:65, the DNA equivalent of SEQ ID NO:65, the complement of SEQ ID NO:65, or the DNA equivalent of the complement of SEQ ID NO:65.

(39) Suitably, the detection probe target-hybridizing sequence is contained in the sequence of SEQ ID NO:59 and includes at least the sequence of SEQ ID NO:42, 92, 94, or 99.

(40) Suitably, the detection probe target-hybridizing sequence is contained in the sequence of SEQ ID NO:41 and includes at least the sequence of SEQ ID NO:38 or SEQ ID NO:60.

(41) Suitably, the detection probe oligomer comprises a nucleotide sequence that is from 16 to 25 contiguous nucleotides in length and specifically hybridizes to SEQ ID NO:65, or the DNA equivalent thereof; or specifically hybridizes to the complement of SEQ ID NO:65, or the DNA equivalent thereof.

(42) Suitably, the detection probe oligomer sequence further comprises a nucleotide sequence containing SEQ ID NO:59 or SEQ ID NO:60.

(43) Suitably, the detection probe oligomer further comprises a nucleotide sequence consisting of SEQ ID NO:37, 38, 39, 42, or 99.

(44) Suitably, the detection probe target-hybridizing sequence consists of the sequence selected from the group consisting of: SEQ ID NOs:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 and 99.

(45) Suitably, the detection probe oligomer further comprises a 2' methoxy modification on at least one nucleotide residue member of the nucleotide sequence.

(46) Suitably, the first and second amplification oligomer target-hybridizing sequences and the detection probe oligomer target-hybridizing sequences respectively comprise or consist of the nucleotide sequences of: (a) SEQ ID NO:2 and SEQ ID NO:11 and SEQ ID NO:39 or SEQ ID NO:37; (b) SEQ ID NO:2 and SEQ ID NO:27 and SEQ ID NO:38 or SEQ ID NO:39; (c) SEQ ID NO:4 and SEQ ID NO:13 and SEQ ID NO:39 or SEQ ID NO:37; (d) SEQ ID NO:4 and SEQ ID NO:16 or SEQ ID NO:17 and SEQ ID NO:39; (e) SEQ ID NO:4 and SEQ ID NO:18 or SEQ ID NO:19 and SEQ ID NO:39 or SEQ ID NO:37; (f) SEQ ID NO:4 and SEQ ID NO:20 and SEQ ID NO:39 or SEQ ID NO:37; (g) SEQ ID NO:4 and SEQ ID NO:21 and SEQ ID NO:39 or SEQ ID NO:37; (h) SEQ ID NO:4 and SEQ ID NO:27 and SEQ ID NO:39 or SEQ ID NO:38; (i) SEQ ID NO:4 and SEQ ID NO:28 and SEQ ID NO:39; (j) SEQ ID NO:4 and SEQ ID NO:29 and SEQ ID NO:39 or SEQ ID NO:37; (k) SEQ ID NO:4 and SEQ ID NO:31 and SEQ ID NO:39; (l) SEQ ID NO:6 and SEQ ID NO:11 and SEQ ID NO:37; (m) SEQ ID NO:6 and SEQ ID NO:13 and SEQ ID NO:37; (n) SEQ ID NO:6 and SEQ ID NO:21 and SEQ ID NO:37; (o) SEQ ID NO:8 and SEQ ID NO:21 and SEQ ID NO:39 or SEQ ID NO:37 or SEQ ID NO:42; (p) SEQ ID NO:8 and SEQ ID NO:27 and SEQ ID NO:39; (q) SEQ ID NO:8 and SEQ ID NO:32 and SEQ ID NO:37 or SEQ ID NO:42; (r) SEQ ID NO:8 and SEQ ID NO:33 and SEQ ID NO:37 or SEQ ID NO:42; (s) SEQ ID NO:8 and SEQ ID NO:34 and SEQ ID NO:37 or SEQ ID NO:42; (t) SEQ ID NO:8 and SEQ ID NO:35 and SEQ ID NO:37 or SEQ ID NO:42; or (u) SEQ ID NO:8 and SEQ ID NO:36 and SEQ ID NO:37 or SEQ ID NO:42; (v) SEQ ID NO:8, and SEQ ID NO:84, and SEQ ID NOs:91, 92 and/or 93; (w) SEQ ID NO:8, and SEQ ID NO:86, and SEQ ID NOs:91, 92 and/or 93; (x) SEQ ID NO:83, and SEQ ID NO:34, and SEQ ID NOs:91, 92 and/or 93; (y) SEQ ID NOs:83, and SEQ ID NO:84, and SEQ ID NOs:91, 92 and/or 93; (z) SEQ ID NOs:83, and SEQ ID NO:86, and SEQ ID

SEQ ID NOs:91, 92 and/or 93; or (aa) SEQ ID NO:8 or 83, SEQ ID NO:34, 84 or 86, and SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(47) Suitably, the first and second amplification oligomer target-hybridizing sequences and the detection probe oligomer target-hybridizing sequences respectively comprise or consist of the nucleotide sequences of: (a) SEQ ID NO:2 and SEQ ID NO:27 and SEQ ID NO:38; (b) SEQ ID NO:4 and SEQ ID NO:21 and SEQ ID NO:39; (c) SEQ ID NO:8 and SEQ ID NO:21 and SEQ ID NO:37 or SEQ ID NO:39; or (d) SEQ ID NO:8 and SEQ ID NO:34 and SEQ ID NO:37 or SEQ ID NO:42; (e) SEQ ID NO:8, and SEQ ID NO:84, and SEQ ID NOs:91, 92 and/or 93; (f) SEQ ID NO:8, and SEQ ID NO:86, and SEQ ID NOs:91, 92 and/or 93; (g) SEQ ID NO:83, and SEQ ID NO:34, and SEQ ID NOs:91, 92 and/or 93; (h) SEQ ID NOs:83, and SEQ ID NO:84, and SEQ ID NOs:91, 92 and/or 93; (i) SEQ ID NOs:83, and SEQ ID NO:86, and SEQ ID NOs:91, 92 and/or 93; or (j) SEQ ID NO:8 or 83, SEQ ID NO:34, 84 or 86, and SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(48) Suitably, the detection probe comprises a label. Suitably, the label is a chemiluminescent label or a fluorescent label. Suitably, the detecting step (3) occurs during the amplifying step (2). Suitably, the detection probe comprises a fluorescent label and a quencher. Suitably, the detection probe is selected from the group consisting of a molecular torch, a molecular beacon, and a TaqMan detection probe. Suitably, the detection probe further comprises a non-target-hybridizing sequence. Suitably, the detection probe is a molecular torch or a molecular beacon.

(49) Suitably, the amplification reaction at step (2) is an isothermal amplification reaction. Suitably, the amplification reaction is a transcription-mediated amplification (TMA) reaction. Suitably, the amplification reaction is a real-time amplification reaction.

(50) Suitably, the sample is a clinical sample. Suitably, the sample is a blood sample. Suitably, the sample is a lysed blood cell sample.

(51) Suitably, the lysed blood cell sample is a lysed red blood cell sample.

(52) In a further aspect, there is described a combination of at least two oligomers for determining the presence or absence of *Babesia* in a sample, said oligomer combination comprising first and second amplification oligomers for amplifying a target region of *Babesia* target nucleic acid, wherein (a) the first amplification oligomer comprises a first target-hybridizing sequence (i) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:66 and comprises SEQ ID NO:56 or 57; or (ii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:96 and comprises SEQ ID NO:101; or (iii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:97 and comprises SEQ ID NO:101; (iv) comprises or consists of SEQ ID NO:8; (v) comprises or consists of SEQ ID NO:83 and (b) the second amplification oligomer comprises a second target-hybridizing sequence that is from about 15 to about 33 contiguous nucleotides in length, and (i) is contained in SEQ ID NO:68 and comprises SEQ ID NO:52, SEQ ID NO:53, SEQ ID NO:54, SEQ ID NO:55, or SEQ ID NO:85; or (ii) is contained in SEQ ID NO:67 and comprises SEQ ID NO:45 or SEQ ID NO:52; or (iii) is contained in SEQ ID NO:70 and comprises SEQ ID NO:46, SEQ ID NO:47, SEQ ID NO:48, SEQ ID NO:49, SEQ ID NO:50, or SEQ ID NO:51; or (iv) comprises or consists of SEQ ID NO:84.

(53) Suitably, the first amplification comprises or consists of the sequence selected from the group consisting of: SEQ ID NOs:2 and 4 and 6 and 8 and 83, suitably, wherein the first amplification comprises or consists of the sequence selected from the group consisting of: SEQ ID NOs:2 and 4 and 8 and 83.

(54) Suitably, the second amplification oligomer comprises or consists of the sequence selected from the group consisting of: SEQ ID NOs:13, 16, 17, 18, 19, 20, 21, 27, 28, 29, 31, 32, 33, 34, 35, 36, 84, and 86.

(55) Suitably, the second amplification oligomer sequence comprises or consists of SEQ ID NO:21 or SEQ ID NO:27 or SEQ ID NO:34 or SEQ ID NO:84 or SEQ ID NO:86.

(56) Suitably, the first amplification oligomer is a promoter primer or promoter provider further comprising a promoter sequence located 5' to the first target-hybridizing sequence. Suitably, the promoter sequence is a T7 promoter sequence. Suitably, the T7 promoter sequence comprises or consists of SEQ ID NO:58.

(57) Suitably, the first amplification oligomer comprises or consists of a sequence selected from the group consisting of SEQ ID NOs:1 and 3 and 5 and 7 and 82, suitably, wherein the first amplification oligomer comprises or consists of a sequence selected from the group consisting of SEQ ID NOs:1 and 3 and 7 and 82.

(58) Suitably, the first and second target-hybridizing sequences respectively comprise or consist of the nucleotide sequences of: (a) SEQ ID NO:2 or 6 and SEQ ID NO:11; (b) SEQ ID NO:4 or 6 and SEQ ID NO:13; (c) SEQ ID NO:4 and SEQ ID NO:16 or SEQ ID NO:17; (d) SEQ ID NO:4 and SEQ ID NO:18 or SEQ ID NO:19; (e) SEQ ID NO:4 and SEQ ID NO:20; (f) SEQ ID NO:4 or 6 or 8 and SEQ ID NO:21; (g) SEQ ID NO:2 or 4 or 8 and SEQ ID NO:27; (h) SEQ ID NO:4 and SEQ ID NO:28; (i) SEQ ID NO:4 and SEQ ID NO:29; (j) SEQ ID NO:4 and SEQ ID NO:31; (k) SEQ ID NO:8 and SEQ ID NO:32; (l) SEQ ID NO:8 and SEQ ID NO:33; (m) SEQ ID NO:8 and SEQ ID NO:34; (n) SEQ ID NO:8 and SEQ ID NO:35; (o) SEQ ID NO:8 and SEQ ID NO:36; (p) SEQ ID NO:8 and SEQ ID NO:84; (q) SEQ ID NO:8 and SEQ ID NO:86; (r) SEQ ID NO:83 and SEQ ID NO:34; (s) SEQ ID NO:83 and SEQ ID NO:84; or (t) SEQ ID NO:83 and SEQ ID NO:86.

(59) Suitably, the first and second target-hybridizing sequences respectively comprise or consist of the nucleotide sequences of: (a) SEQ ID NO:2 and SEQ ID NO:27; (b) SEQ ID NO:4 and SEQ ID NO:21; (c) SEQ ID NO:8 and SEQ ID NO:21; (d) SEQ ID NO:8 and SEQ ID NO:34; (e) SEQ ID NO:8 and SEQ ID NO:84; (f) SEQ ID NO:8 and SEQ ID NO:86; (g) SEQ ID NO:83 and SEQ ID NO:34; (h) SEQ ID NO:83 and SEQ ID NO:84; or (i) SEQ ID NO:83 and SEQ ID NO:86.

(60) Suitably, the combination further comprises at least one capture probe oligomer.

(61) Suitably, the at least one capture probe oligomer comprises a target-hybridizing sequence covalently attached to a sequence or moiety that binds to an immobilized probe, wherein said target-hybridizing sequence (i) is from about 15 to about 21 contiguous nucleotides contained in the sequence of SEQ ID NO:78, or (ii) is about 21 to 30 contiguous nucleotides comprising the sequence of SEQ ID NO:78; or (iii) the sequence consists of SEQ ID NO:44.

(62) Suitably, the capture probe oligomer sequence comprises or consists of SEQ ID NO:43.

(63) Suitably, the combination further comprises a detection probe oligomer.

(64) Suitably, the detection probe oligomer comprises a target-hybridizing sequence that is from about 14 to about 40 nucleotides in length and is configured to specifically hybridize to a target sequence contained in SEQ ID NO:59, the RNA equivalent of SEQ ID NO:59, the complement of SEQ ID NO:59, the RNA equivalent of the complement of SEQ ID NO:59, SEQ ID NO:65, the DNA equivalent of SEQ ID NO:65, the complement of SEQ ID NO:65, or the DNA equivalent of the complement of SEQ ID NO:65.

(65) Suitably, the detection probe target-hybridizing sequence contains the sequence of SEQ ID NO:59 and includes at least the sequence of SEQ ID NO:42, 92, 94, or 99.

(66) Suitably, the detection probe target-hybridizing sequence contains the sequence of SEQ ID NO:60 and includes at least the sequence of SEQ ID NO:38 or SEQ ID NO:39.

(67) Suitably, the detection probe target hybridising sequence consists of the sequence selected from the group consisting of: SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(68) Suitably, the detection probe oligomer comprises a nucleotide sequence that is from 16 to 25 contiguous nucleotides in length and specifically hybridizes to SEQ ID NO:65, or the DNA equivalent thereof; or specifically hybridizes to the complement of SEQ ID NO:65, or the DNA equivalent thereof.

(69) Suitably, the detection probe oligomer sequence further comprises a nucleotide sequence containing SEQ ID NO:59 or SEQ ID NO:60.

(70) Suitably, the detection probe oligomer further comprises a nucleotide sequence consisting of

SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(71) Suitably, the detection probe oligomer further comprises a 2' methoxy modification on at least one nucleotide residue member of the nucleotide sequence.

(72) Suitably, the first and second amplification oligomer target-hybridizing sequences and the detection probe oligomer target-hybridizing sequences respectively comprise or consist of the nucleotide sequences of: (a) SEQ ID NO:2 and SEQ ID NO:11 and SEQ ID NO:39 or SEQ ID NO:37; (b) SEQ ID NO:2 and SEQ ID NO:27 and SEQ ID NO:38 or SEQ ID NO:39; (c) SEQ ID NO:4 and SEQ ID NO:13 and SEQ ID NO:39 or SEQ ID NO:37; (d) SEQ ID NO:4 and SEQ ID NO:16 or SEQ ID NO:17 and SEQ ID NO:39; (e) SEQ ID NO:4 and SEQ ID NO:18 or SEQ ID NO:19 and SEQ ID NO:39 or SEQ ID NO:37; (f) SEQ ID NO:4 and SEQ ID NO:20 and SEQ ID NO:39 or SEQ ID NO:37; (g) SEQ ID NO:4 and SEQ ID NO:21 and SEQ ID NO:39 or SEQ ID NO:37; (h) SEQ ID NO:4 and SEQ ID NO:27 and SEQ ID NO:39 or SEQ ID NO:38; (i) SEQ ID NO:4 and SEQ ID NO:28 and SEQ ID NO:39; (j) SEQ ID NO:4 and SEQ ID NO:29 and SEQ ID NO:39 or SEQ ID NO:37; (k) SEQ ID NO:4 and SEQ ID NO:31 and SEQ ID NO:39; (l) SEQ ID NO:6 and SEQ ID NO:11 and SEQ ID NO:37; (m) SEQ ID NO:6 and SEQ ID NO:13 and SEQ ID NO:37; (n) SEQ ID NO:6 and SEQ ID NO:21 and SEQ ID NO:37; (o) SEQ ID NO:8 and SEQ ID NO:21 and SEQ ID NO:39 or SEQ ID NO:37 or SEQ ID NO:42; (p) SEQ ID NO:8 and SEQ ID NO:27 and SEQ ID NO:39; (q) SEQ ID NO:8 and SEQ ID NO:32 and SEQ ID NO:37 or SEQ ID NO:42; (r) SEQ ID NO:8 and SEQ ID NO:33 and SEQ ID NO:37 or SEQ ID NO:42; (s) SEQ ID NO:8 and SEQ ID NO:34 and SEQ ID NO:37 or SEQ ID NO:42; (t) SEQ ID NO:8 and SEQ ID NO:35 and SEQ ID NO:37 or SEQ ID NO:42; (u) SEQ ID NO:8 and SEQ ID NO:36 and SEQ ID NO:37 or SEQ ID NO:42; (v) SEQ ID NO:8, and SEQ ID NO:84, and SEQ ID NOs:91, 92 and/or 93; (w) SEQ ID NO:8, and SEQ ID NO:86, and SEQ ID NOs:91, 92 and/or 93; (x) SEQ ID NO:83, and SEQ ID NO:34, and SEQ ID NOs:91, 92 and/or 93; (y) SEQ ID NOs:83, and SEQ ID NO:84, and SEQ ID NOs:91, 92 and/or 93; (z) SEQ ID NOs:83, and SEQ ID NO:86, and SEQ ID NOs:91, 92 and/or 93; or (aa) SEQ ID NO:8 or 83, SEQ ID NO:34, 84 or 86, and SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(73) Suitably, the first and second amplification oligomer target-hybridizing sequences and the detection probe oligomer target-hybridizing sequences respectively comprise or consist of the nucleotide sequences of: (a) SEQ ID NO:2 and SEQ ID NO:27 and SEQ ID NO:38; (b) SEQ ID NO:4 and SEQ ID NO:21 and SEQ ID NO:39; (c) SEQ ID NO:8 and SEQ ID NO:21 and SEQ ID NO:37 or SEQ ID NO:39; (d) SEQ ID NO:8 and SEQ ID NO:34 and SEQ ID NO:37 or SEQ ID NO:42; (e) SEQ ID NO:8, and SEQ ID NO:84, and SEQ ID NOs:91, 92 and/or 93; (f) SEQ ID NO:8, and SEQ ID NO:86, and SEQ ID NOs:91, 92 and/or 93; (g) SEQ ID NO:83, and SEQ ID NO:34, and SEQ ID NOs:91, 92 and/or 93; (h) SEQ ID NOs:83, and SEQ ID NO:84, and SEQ ID NOs:91, 92 and/or 93; (i) SEQ ID NOs:83, and SEQ ID NO:86, and SEQ ID NOs:91, 92 and/or 93; or (j) SEQ ID NO:8 or 83, SEQ ID NO:34, 84 or 86, and SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(74) Suitably, the detection probe comprises a label. Suitably, the label is a chemiluminescent label or a fluorescent label. Suitably, the detection probe comprises a fluorescent label and a quencher. Suitably, the detection probe is selected from the group consisting of a molecular torch, a molecular beacon, and a TaqMan detection probe. Suitably, the detection probe further comprises a non-target-hybridizing sequence. Suitably, the detection probe is a molecular torch or a molecular beacon.

(75) In a further aspect, there is described a detection probe oligomer comprising a target-hybridizing sequence that is from about 14 to about 40 nucleotides in length and is configured to specifically hybridize to a target sequence contained in SEQ ID NO:59, the RNA equivalent of SEQ ID NO:59, the complement of SEQ ID NO:59, the RNA equivalent of the complement of SEQ ID NO:59, SEQ ID NO:65, the RNA equivalent of SEQ ID NO:65, the complement of SEQ ID NO:65, or the RNA equivalent of the complement of SEQ ID NO:65.

- (76) Suitably, the detection probe target-hybridizing sequence is contained in the sequence of SEQ ID NO:59 and includes at least the sequence of SEQ ID NO:42, 92, 94 or 99.
- (77) Suitably, the detection probe target-hybridizing sequence is contained in the sequence of SEQ ID NO:65 and includes at least the sequence of SEQ ID NO:59 or SEQ ID NO:94.
- (78) Suitably, the detection probe target hybridising sequence consists of the sequence selected from the group consisting of: SEQ ID NOs:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 and 99.
- (79) Suitably, the detection probe oligomer further comprises a 2' methoxy modification on at least one nucleotide residue member of the nucleotide sequence.
- (80) Suitably, the detection probe comprises a label. Suitably, the label is a chemiluminescent label or a fluorescent label. Suitably, the detection probe comprises a fluorescent label and a quencher. Suitably, the detection probe is selected from the group consisting of a molecular torch, a molecular beacon, and a TaqMan detection probe.
- (81) Suitably, the detection probe further comprises a non-target-hybridizing sequence.
- (82) Suitably, the detection probe is a molecular torch or a molecular beacon.
- (83) In a further aspect, there is described a capture probe oligomer for specifically isolating *Babesia* species nucleic acid from a sample, said capture probe oligomer comprising a target-hybridizing sequence covalently attached to a sequence or moiety that binds to an immobilized probe, wherein said target-hybridizing sequence is from about 15 to about 30 contiguous nucleotides contained in the sequence of SEQ ID NO:78.
- (84) Suitably, the capture probe oligomer sequence comprises or consists of SEQ ID NO:43.
- (85) In a further aspect, there is described a kit comprising the combination of at least two oligomers.
- (86) In a further aspect, there is described a reaction mixture comprising the combination of at least two oligomers.
- (87) In a further aspect, there is described the use of the combination of at least two oligomers for specifically amplifying *Babesia* species nucleic acid in a sample.
- (88) In a further aspect, there is described the use of the detection probe oligomer for specifically detecting *Babesia* species nucleic acid in a sample.
- (89) In a further aspect, there is described the use of the capture probe oligomer for specifically capturing *Babesia* species nucleic acid from a sample.
- (90) Suitably, *Babesia microti* and/or *Babesia divergens* and/or *Babesia duncani* and/or *Babesia venatorum* are detected.

Definitions

- (91) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art pertinent to the methods and compositions described. As used herein, the following terms and phrases have the meanings ascribed to them unless specified otherwise.
- (92) The terms “a,” “an,” and “the” include plural referents, unless the context clearly indicates otherwise.
- (93) “Sample” includes any specimen that may contain, or is suspected of containing, *Babesia* nucleic acid or components thereof, such as nucleic acids or fragments of *Babesia* nucleic acids. The sample may be an isolated sample. Samples include “biological samples” which include any tissue or material derived from a living or dead human that may contain the *Babesia* parasite or components thereof (e.g., a target nucleic acid derived therefrom), including, e.g., blood, peripheral blood and red blood cells. The use of other sample types that may contain the *Babesia* parasite or components thereof (e.g., a target nucleic acid derived therefrom)—such as plasma, serum, lymph node, gastrointestinal tissue, faeces, urine, semen or other body fluids or materials—is also contemplated. The biological sample may be treated to physically or mechanically disrupt tissue or cell structure, thus releasing intracellular components into a solution which may further contain enzymes, buffers, salts, detergents and the like, which are used to prepare, using standard methods,

a biological sample for analysis. Also, samples may include processed samples, such as those obtained from passing samples over or through a filtering device, or following centrifugation, or by adherence to a medium, matrix, or support.

(94) “Nucleic acid” refers to a multimeric compound comprising two or more covalently bonded nucleosides or nucleoside analogs having nitrogenous heterocyclic bases, or base analogs, where the nucleosides are linked together by phosphodiester bonds or other linkages to form a polynucleotide. Nucleic acids include RNA, DNA, or chimeric DNA-RNA polymers or oligonucleotides, and analogs thereof. A nucleic acid “backbone” may be made up of a variety of linkages, including one or more of sugar-phosphodiester linkages, peptide-nucleic acid bonds (in “peptide nucleic acids” or PNAs, see WO95/32305), phosphorothioate linkages, methylphosphonate linkages, or combinations thereof. Sugar moieties of the nucleic acid may be either ribose or deoxyribose, or similar compounds having known substitutions, e.g., 2' methoxy substitutions and 2' halide substitutions (e.g., 2'-F). Nitrogenous bases may be conventional bases (A, G, C, T, U), analogs thereof (e.g., inosine, 5-methylisocytosine, isoguanine; *The Biochemistry of the Nucleic Acids* 5-36, Adams et al, ed., 11th ed., 1992, *BioTechniques* (2007) 43: 617-24), which include derivatives of purine or pyrimidine bases (e.g., N4-methyl deoxygaunosine, deaza- or aza-purines, deaza- or aza-pyrimidines, pyrimidine bases having substituent groups at the 5 or 6 position, purine bases having an altered or replacement substituent at the 2, 6 and/or 8 position, such as 2-amino-6-methylaminopurine, 06-methylguanine, 4-thio-pyrimidines, 4-amino-pyrimidines, 4-dimethylhydrazine-pyrimidines, and 04-alkyl-pyrimidines, and pyrazolo-compounds, such as unsubstituted or 3-substituted pyrazolo[3,4-d]pyrimidine; U.S. Pat. Nos. 5,378,825, 6,949,367 and PCT No. WO 93/13121). Nucleic acids may include “abasic” residues in which the backbone does not include a nitrogenous base for one or more residues (U.S. Pat. No. 5,585,481). A nucleic acid may comprise only conventional sugars, bases, and linkages as found in RNA and DNA, or may include conventional components and substitutions (e.g., conventional bases linked by a 2' methoxy backbone, or a nucleic acid including a mixture of conventional bases and one or more base analogs). Nucleic acids may include “locked nucleic acids” (LNA), in which one or more nucleotide monomers have a bicyclic furanose unit locked in an RNA mimicking sugar conformation, which enhances hybridization affinity toward complementary sequences in single-stranded RNA (ssRNA), single-stranded DNA (ssDNA), or double-stranded DNA (dsDNA) (*Biochemistry* (2004) 43: 13233-41). Nucleic acids may include modified bases to alter the function or behavior of the nucleic acid, e.g., addition of a 3'-terminal dideoxynucleotide to block additional nucleotides from being added to the nucleic acid. Synthetic methods for making nucleic acids in vitro are well-known in the art although nucleic acids may be purified from natural sources using routine techniques.

(95) The term “polynucleotide,” as used herein, denotes a nucleic acid chain. Throughout this application, nucleic acids are designated by the 5'-terminus to the 3'-terminus. Standard nucleic acids, e.g., DNA and RNA, are typically synthesized “5'-to-3',” i.e., by the addition of nucleotides to the 3'-terminus of a growing nucleic acid.

(96) A “nucleotide,” as used herein, is a subunit of a nucleic acid consisting of a phosphate group, a 5-carbon sugar and a nitrogenous base. The 5-carbon sugar found in RNA is ribose. In DNA, the 5-carbon sugar is 2'-deoxyribose. The term also includes analogs of such subunits, such as a methoxy group at the 2' position of the ribose (2'-O-Me).

(97) A “nucleic-acid-based detection assay,” as used herein, is an assay for the detection of a target sequence within a target nucleic acid and utilizing one more oligonucleotides that specifically hybridize to the target sequence.

(98) In certain embodiments, a nucleic-acid-based detection assay is an “amplification-based assay,” i.e., an assay that utilizes one or more steps for amplifying a nucleic acid target sequence. Various amplification methods for use in detection assays are known in the art, several of which are summarized further herein. For the sake of clarity, an amplification-based assay may include one or

more steps that do not amplify a target sequence, such as, for example, steps used in non-amplification-based assay methods (e.g., a hybridization assay or a cleavage-based assay).

(99) In other embodiments, a nucleic-acid-based detection assay is a “non-amplification-based assay,” i.e., an assay that does not rely on any step for amplifying a nucleic acid target sequence. For the sake of clarity, a nucleic-acid-based detection assay that includes a reaction for extension of a primer in the absence of any corresponding downstream amplification oligomer (e.g., extension of a primer by a reverse transcriptase to generate an RNA:DNA duplex followed by an RNase digestion of the RNA, resulting in a single-stranded cDNA complementary to an RNA target but without generating copies of the cDNA) is understood to be a non-amplification-based assay.

(100) An exemplary non-amplification-based assay is a “cleavage-based assay,” which is an assay that relies on the specific cleavage, by a flap endonuclease, of a linear duplex cleavage structure formed by the specific hybridization of overlapping oligonucleotides to a target nucleic acid. In these assays, a probe oligonucleotide containing a non-target-hybridizing flap region is cleaved in an overlap-dependent manner by the flap endonuclease to release a cleavage product that is then detected. The principles of cleavage-based assays are well-known in the art, and exemplary assays are described in, for example, *Nat. Biotechnol.* (1999) 17:292-296, *Mol. Diagn.* (1999) 4: 135-144, *J. Clin. Microbiol.* (2006) 44:3443-3447, and U.S. Pat. Nos. 5,846,717, 6,706,471 and 5,614,402. Cleavage-based assays include, e.g., the commercially available Invader® assays (Hologic, Inc., Madison, WI).

(101) A “target nucleic acid,” as used herein, is a nucleic acid comprising a target sequence to be detected. Target nucleic acids may be DNA or RNA as described herein, and may be either single-stranded or double-stranded. The target nucleic acid may include other sequences besides the target sequence.

(102) By “isolated” it is meant that a sample containing a target nucleic acid is taken from its natural milieu, but the term does not connote any degree of purification.

(103) The term “target sequence,” as used herein, refers to the particular nucleotide sequence of a target nucleic acid that is to be detected. The “target sequence” includes the complexing sequences to which oligonucleotides (e.g., probe oligonucleotide, priming oligonucleotides and/or promoter oligonucleotides) complex during a detection process (e.g., an amplification-based detection assay such as, for example, TMA or PCR, or a non-amplification-based detection assay such as, for example, a cleavage-based assay). Where the target nucleic acid is originally single-stranded, the term “target sequence” will also refer to the sequence complementary to the “target sequence” as present in the target nucleic acid. Where the target nucleic acid is originally double-stranded, the term “target sequence” refers to both the sense (+) and antisense (−) strands. In choosing a target sequence, the skilled artisan will understand that a “unique” sequence should be chosen so as to distinguish between unrelated or closely related target nucleic acids.

(104) “Target-hybridizing sequence” is used herein to refer to the portion of an oligomer that is configured to hybridize with a target nucleic acid sequence. Preferably, the target-hybridizing sequences are configured to specifically hybridize with a target nucleic acid sequence. Target-hybridizing sequences may be 100% complementary to the portion of the target sequence to which they are configured to hybridize, but not necessarily. Target-hybridizing sequences may also include inserted, deleted and/or substituted nucleotide residues relative to a target sequence. Less than 100% complementarity of a target-hybridizing sequence to a target sequence may arise, for example, when the target nucleic acid is a plurality strains within a species, such as would be the case for an oligomer configured to hybridize to the various strains of *Babesia*. It is understood that other reasons exist for configuring a target-hybridizing sequence to have less than 100% complementarity to a target nucleic acid.

(105) The term “targets a sequence,” as used herein in reference to a region of *Babesia* sp. nucleic acid, refers to a process whereby an oligonucleotide hybridizes to the target sequence in a manner that allows for detection as described herein. In one embodiment, the oligonucleotide is

complementary with the targeted *Babesia* sp. nucleic acid sequence and contains no mismatches. In another embodiment, the oligonucleotide is complementary but contains 1, 2, 3, 4, or 5 mismatches with the targeted *Babesia* sp. nucleic acid sequence. Preferably, the oligonucleotide that hybridizes to the target nucleic acid sequence includes at least 10 to as many as 50 nucleotides complementary to the target sequence. It is understood that at least 10 and as many as 50 is an inclusive range such that 10, 50 and each whole number there between are included. Preferably, the oligomer specifically hybridizes to the target sequence.

(106) The term “configured to” denotes an actual arrangement of the polynucleotide sequence configuration of a referenced oligonucleotide target-hybridizing sequence. For example, oligonucleotides that are configured to specifically hybridize to a target sequence have a polynucleotide sequence that specifically hybridizes to the referenced sequence under stringent hybridization conditions.

(107) The term “configured to specifically hybridize to” as used herein means that the target-hybridizing region of an oligonucleotide is designed to have a polynucleotide sequence that could target a sequence of the referenced *Babesia* sp. target region. Such an oligonucleotide is not limited to targeting that sequence only, but is rather useful as a composition, in a kit or in a method for targeting a *Babesia* sp. target nucleic acid. The oligonucleotide is designed to function as a component of an assay for detection of *Babesia* sp. from a sample, and therefore is designed to target *Babesia* sp. in the presence of other nucleic acids commonly found in testing samples.

“Specifically hybridize to” does not mean exclusively hybridize to, as some small level of hybridization to non-target nucleic acids may occur, as is understood in the art. Rather, “specifically hybridize to” means that the oligonucleotide is configured to function in an assay to primarily hybridize the target so that an accurate detection of target nucleic acid in a sample can be determined. The term “configured to” denotes an actual arrangement of the polynucleotide sequence configuration of the oligonucleotide target-hybridizing sequence.

(108) The term “fragment,” as used herein in reference to a *Babesia* sp. targeted nucleic acid, refers to a piece of contiguous nucleic acid.

(109) The term “region,” as used herein, refers to a portion of a nucleic acid wherein said portion is smaller than the entire nucleic acid. For example, when the nucleic acid in reference is an oligonucleotide promoter primer, the term “region” may be used refer to the smaller promoter portion of the entire oligonucleotide. As a non-limiting example, when the nucleic acid in reference is an amplicon, the term region may be used to refer to the smaller nucleotide sequence identified for hybridization by the target-hybridizing sequence of a probe.

(110) The interchangeable terms “oligomer,” “oligo,” and “oligonucleotide” refer to a nucleic acid having generally less than 1,000 nucleotide (nt) residues, including polymers in a range having a lower limit of about 5 nt residues and an upper limit of about 500 to 900 nt residues. In some embodiments, oligonucleotides are in a size range having a lower limit of about 12 to 15 nt and an upper limit of about 50 to 600 nt, and other embodiments are in a range having a lower limit of about 15 to 20 nt and an upper limit of about 22 to 100 nt. Oligonucleotides may be purified from naturally occurring sources or may be synthesized using any of a variety of well-known enzymatic or chemical methods. The term oligonucleotide does not denote any particular function to the reagent; rather, it is used generically to cover all such reagents described herein. An oligonucleotide may serve various different functions. For example, it may function as a primer if it is specific for and capable of hybridizing to a complementary strand and can further be extended in the presence of a nucleic acid polymerase; it may function as a primer and provide a promoter if it contains a sequence recognized by an RNA polymerase and allows for transcription (e.g., a T7 Primer); and it may function to detect a target nucleic acid if it is capable of hybridizing to the target nucleic acid, or an amplicon thereof, and further provides a detectible moiety (e.g., an acridinium-ester compound).

(111) As used herein, an oligonucleotide can “substantially correspond to” a specified reference

nucleic acid sequence, which means that the oligonucleotide is sufficiently similar to the reference nucleic acid sequence such that the oligonucleotide has similar hybridization properties to the reference nucleic acid sequence in that it would hybridize with the same target nucleic acid sequence under stringent hybridization conditions. One skilled in the art will understand that “substantially corresponding oligonucleotides” can vary from a reference sequence and still hybridize to the same target nucleic acid sequence. It is also understood that a first nucleic acid corresponding to a second nucleic acid includes the RNA and DNA thereof and includes the complements thereof, unless the context clearly dictates otherwise. This variation from the nucleic acid may be stated in terms of a percentage of identical bases within the sequence or the percentage of perfectly complementary bases between the probe or primer and its target sequence. Thus, in certain embodiments, an oligonucleotide “substantially corresponds” to a reference nucleic acid sequence if these percentages of base identity or complementarity are from 100% to about 80%. In preferred embodiments, the percentage is from 100% to about 85%. In more preferred embodiments, this percentage is from 100% to about 90%; in other preferred embodiments, this percentage is from 100% to about 95%. Similarly, a region of a nucleic acid or amplified nucleic acid can be referred to herein as corresponding to a reference nucleic acid sequence. One skilled in the art will understand the various modifications to the hybridization conditions that might be required at various percentages of complementarity to allow hybridization to a specific target sequence without causing an unacceptable level of non-specific hybridization.

(112) An “amplification oligomer” is an oligomer, at least the 3'-end of which is complementary to a target nucleic acid, and which hybridizes to a target nucleic acid, or its complement, and participates in a nucleic acid amplification reaction. An example of an amplification oligomer is a “primer” that hybridizes to a target nucleic acid and contains a 3' OH end that is extended by a polymerase in an amplification process. Another example of an amplification oligomer is an oligomer that is not extended by a polymerase (e.g., because it has a 3' blocked end) but participates in or facilitates amplification. For example, the 5' region of an amplification oligonucleotide—such as a first amplification oligomer as described herein—may include a promoter sequence that is non-complementary to the target nucleic acid (which may be referred to as a “promoter primer” or “promoter provider”). Those skilled in the art will understand that an amplification oligomer that functions as a primer may be modified to include a 5' promoter sequence, and thus function as a promoter primer. Incorporating a 3' blocked end further modifies the promoter primer, which is now capable of hybridizing to a target nucleic acid and providing an upstream promoter sequence that serves to initiate transcription, but does not provide a primer for oligo extension. Such a modified oligo is referred to herein as a “promoter provider” oligomer. Size ranges for amplification oligonucleotides include those that are about 10 to about 70 nt long (not including any promoter sequence or poly-A tails) and contain at least about 10 contiguous bases, or even at least 12 contiguous bases that are complementary to a region of the target nucleic acid sequence (or a complementary strand thereof). The contiguous bases are at least 80%, or at least 90%, or completely complementary to the target sequence to which the amplification oligomer binds. An amplification oligomer may optionally include modified nucleotides or analogs, or additional nucleotides that participate in an amplification reaction but are not complementary to or contained in the target nucleic acid, or template sequence. It is understood that when referring to ranges for the length of an oligonucleotide, amplicon, or other nucleic acid, that the range is inclusive of all whole numbers (e.g., 19-25 contiguous nucleotides in length includes 19, 20, 21, 22, 23, 24 & 25).

(113) As used herein, a “promoter” is a specific nucleic acid sequence that is recognized by a DNA-dependent RNA polymerase (“transcriptase”) as a signal to bind to the nucleic acid and begin the transcription of RNA at a specific site.

(114) As used herein, a “promoter provider” or “provider” refers to an oligonucleotide comprising first and second regions, and which is modified to prevent the initiation of DNA synthesis from its

3'-terminus. The "first region" of a promoter provider oligonucleotide comprises a base sequence which hybridizes to a DNA template, where the hybridizing sequence is situated 3', but not necessarily adjacent to, a promoter region. The hybridizing portion of a promoter oligonucleotide is typically at least 10 nucleotides in length, and may extend up to 50 or more nucleotides in length. The "second region" comprises a promoter sequence for an RNA polymerase. A promoter oligonucleotide is engineered so that it is incapable of being extended by an RNA- or DNA-dependent DNA polymerase, e.g., reverse transcriptase, preferably comprising a blocking moiety at its 3'-terminus as described above. As referred to herein, a "T7 Provider" is a blocked promoter provider oligonucleotide that provides an oligonucleotide sequence that is recognized by T7 RNA polymerase.

(115) "Amplification" refers to any known procedure for obtaining multiple copies of a target nucleic acid sequence or its complement or fragments thereof. The multiple copies may be referred to as amplicons or amplification products. Known amplification methods include both thermal cycling and isothermal amplification methods. In some embodiments, isothermal amplification methods are preferred. Replicase-mediated amplification, polymerase chain reaction (PCR), ligase chain reaction (LCR), strand-displacement amplification (SDA), and transcription-mediated or transcription-associated amplification are non-limiting examples of nucleic acid amplification methods. Replicase-mediated amplification uses self-replicating RNA molecules, and a replicase such as QB-replicase (e.g., U.S. Pat. No. 4,786,600). PCR amplification uses a DNA polymerase, pairs of primers, and thermal cycling to synthesize multiple copies of two complementary strands of dsDNA or from a cDNA (e.g., U.S. Pat. Nos. 4,683,195, 4,683,202, and 4,800,159). LCR amplification uses four or more different oligonucleotides to amplify a target and its complementary strand by using multiple cycles of hybridization, ligation, and denaturation (e.g., U.S. Pat. Nos. 5,427,930 and 5,516,663). SDA uses a primer that contains a recognition site for a restriction endonuclease and an endonuclease that nicks one strand of a hemimodified DNA duplex that includes the target sequence, whereby amplification occurs in a series of primer extension and strand displacement steps (e.g., U.S. Pat. Nos. 5,422,252; 5,547,861; and 5,648,211). Preferred embodiments use an amplification method suitable for the amplification of RNA target nucleic acids, such as transcription-mediated amplification (TMA) or NASBA, but it will be apparent to persons of ordinary skill in the art that oligomers disclosed herein may be readily used as primers in other amplification methods.

(116) "Transcription-associated amplification," also referred to herein as "transcription-mediated amplification" (TMA), refers to nucleic acid amplification that uses an RNA polymerase to produce multiple RNA transcripts from a nucleic acid template. These methods generally employ an RNA polymerase, a DNA polymerase, deoxyribonucleoside triphosphates, ribonucleoside triphosphates, and a template complementary oligonucleotide that includes a promoter sequence, and optionally may include one or more other oligonucleotides. TMA methods are embodiments of amplification methods used for amplifying and detecting HSV target sequences as described herein. Variations of transcription-associated amplification are well-known in the art as previously disclosed in detail (e.g., U.S. Pat. Nos. 4,868,105; 5,124,246; 5,130,238; 5,437,990; 5,554,516; and 7,374,885; and PCT Pub. Nos. WO 88/01302, WO 88/10315, and WO 95/03430). The person of ordinary skill in the art will appreciate that the disclosed compositions may be used in amplification methods based on extension of oligomer sequences by a polymerase.

(117) As used herein, the term "real-time TMA" refers to single-primer transcription-mediated amplification ("TMA") of target nucleic acid that is monitored by real-time detection means.

(118) The term "amplicon," which is used interchangeably with "amplification product," refers to the nucleic acid molecule generated during an amplification procedure that is complementary or homologous to a sequence contained within the target sequence. These terms can be used to refer to a single strand amplification product, a double strand amplification product or one of the strands of a double strand amplification product.

(119) “Probe,” “detection probe,” “detection oligonucleotide,” and “detection probe oligomer” are used interchangeably herein to refer to a nucleic acid oligomer that hybridizes specifically to a target sequence in a nucleic acid, or in an amplified nucleic acid, under conditions that promote hybridization to allow detection of the target sequence or amplified nucleic acid. Detection may either be direct (e.g., a probe hybridized directly to its target sequence) or indirect (e.g., a probe linked to its target via an intermediate molecular structure). Probes may be DNA, RNA, analogs thereof or combinations thereof and they may be labeled or unlabeled. A probe’s “target sequence” generally refers to a smaller nucleic acid sequence within a larger nucleic acid sequence that hybridizes specifically to at least a portion of a probe oligomer by standard base pairing. A probe may comprise target-specific sequences and other sequences that contribute to the three-dimensional conformation of the probe (e.g., U.S. Pat. Nos. 5,118,801; 5,312,728; 6,849,412; 6,835,542; 6,534,274; and 6,361,945; and US Pub. No. 20060068417). In a preferred embodiment, the detection probe comprises a 2’ methoxy backbone which can result in a higher signal being obtained.

(120) The term “TaqMan® probe” refers to detection oligonucleotides that contain a fluorescent dye, typically on the 5’ base, and a non-fluorescent quenching dye (quencher), typically on the 3’ base. When irradiated, the excited fluorescent dye transfers energy to the nearby quenching dye molecule rather than fluorescing, resulting in a non-fluorescent substrate. During amplification, the exonuclease activity of the polymerase cleaves the TaqMan probe to separate the fluorophore from the quencher, thereby allowing an unquenched signal to be emitted from the fluorophore as an indicator of amplification.

(121) As used herein, a “label” refers to a moiety or compound joined directly or indirectly to a probe that is detected or leads to a detectable signal. Direct labelling can occur through bonds or interactions that link the label to the probe, including covalent bonds or non-covalent interactions, e.g., hydrogen bonds, hydrophobic and ionic interactions, or formation of chelates or coordination complexes. Indirect labelling can occur through use of a bridging moiety or “linker” such as a binding pair member, an antibody or additional oligomer, which is either directly or indirectly labeled, and which may amplify the detectable signal. Labels include any detectable moiety, such as a radionuclide, ligand (e.g., biotin, avidin), enzyme or enzyme substrate, reactive group, or chromophore (e.g., dye, particle, or bead that imparts detectable color), luminescent compound (e.g., bioluminescent, phosphorescent, or chemiluminescent labels), or fluorophore. Labels may be detectable in a homogeneous assay in which bound labeled probe in a mixture exhibits a detectable change different from that of an unbound labeled probe, e.g., instability or differential degradation properties. A “homogeneous detectable label” can be detected without physically removing bound from unbound forms of the label or labeled probe (e.g., U.S. Pat. Nos. 5,283,174, 5,656,207, and 5,658,737). Labels include chemiluminescent compounds, e.g., acridinium ester (“AE”) compounds that include standard AE and derivatives (e.g., U.S. Pat. Nos. 5,656,207, 5,658,737, and 5,639,604). Synthesis and methods of attaching labels to nucleic acids and detecting labels are well known (e.g., Sambrook et al., *Molecular Cloning, A Laboratory Manual*, 2nd ed. (Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY, 1989), Chapter 10; U.S. Pat. Nos. 5,658,737, 5,656,207, 5,547,842, 5,283, 174, and 4,581,333). More than one label, and more than one type of label, may be present on a particular probe, or detection may use a mixture of probes in which each probe is labeled with a compound that produces a detectable signal (e.g., U.S. Pat. Nos. 6,180,340 and 6,350,579).

(122) As used herein, structures referred to as “molecular torches” are designed to include distinct regions of self-complementarity (“the closing domain”) which are connected by a joining region (“the target binding domain”) and which hybridize to one another under predetermined hybridization assay conditions. All or part of the nucleotide sequences comprising target closing domains may also function as target binding domains. Thus, target closing sequences can include, target binding sequences, non-target binding sequences, and combinations thereof.

(123) “Capture probe,” “capture oligonucleotide,” “target capture oligonucleotide,” and “capture probe oligomer” are used interchangeably herein to refer to a nucleic acid oligomer that specifically hybridizes to a target sequence in a target nucleic acid by standard base pairing and joins to a binding partner on an immobilized probe to capture the target nucleic acid to a support. One example of a capture oligomer includes an oligonucleotide comprising two binding regions: a target hybridizing sequence and an immobilized probe-binding region. A variation of this example, the two regions may be present on two different oligomers joined together by one or more linkers. Another embodiment of a capture oligomer the target hybridizing sequence is a sequence that includes random or non-random poly-GU, poly-GT, or poly U sequences to bind non-specifically to a target nucleic acid and link it to an immobilized probe on a support (see, e.g., WO 2008/016988). The immobilized probe binding region can be a nucleic acid sequence, referred to as a tail. Tails include a substantially homopolymeric tail of about 10 to 40 nucleotides (e.g., A₁₀ to A₄₀), or of about 14 to 33 nt (e.g., T_{3A14} to T_{3A30}), that bind to a complementary immobilized sequence attached to the support particle or support matrix. Thus, a non-limiting example of preferred nucleic acid tails can in some embodiments include T₀-4A₁₀₄₀ sequences. Another example of a capture oligomer comprises two regions, a target hybridizing sequence and a binding pair member that is not a nucleic acid sequence.

(124) As used herein, an “immobilized oligonucleotide,” “immobilized probe” or “immobilized nucleic acid” refers to a nucleic acid binding partner that joins a capture oligomer to a support, directly or indirectly. An immobilized probe joined to a support facilitates separation of a capture probe bound target from unbound material in a sample. One embodiment of an immobilized probe is an oligomer joined to a support that facilitates separation of bound target sequence from unbound material in a sample. Supports may include known materials, such as matrices and particles free in solution, which may be made of nitrocellulose, nylon, glass, polyacrylate, mixed polymers, polystyrene, silane, polypropylene, metal, or other compositions, of which one embodiment is magnetically attractable particles. Supports may be monodisperse magnetic spheres (e.g., uniform size+5%), to which an immobilized probe is joined directly (via covalent linkage, chelation, or ionic interaction), or indirectly (via one or more linkers), where the linkage or interaction between the probe and support is stable during hybridization conditions.

Description

DESCRIPTION

(1) The present disclosure is generally directed to methods and compositions for determining the presence or absence of the protozoan parasite *Babesia* sp. in a sample—such as a blood sample. Suitably, the methods and compositions described herein are able to detect the presence of *Babesia microti* and/or *Babesia divergens* and/or *Babesia duncani* and/or *Babesia venatorum*. In some embodiments, the present disclosure provides methods and compositions for diagnosing Babesiosis in a subject. In other, non-mutually exclusive embodiments, the present disclosure provides methods for the detection of *Babesia* sp. in a sample, where the method includes performing amplification-based detection of a target nucleic from *Babesia* sp. The present disclosure further provides compositions (including reaction mixtures) and kits comprising a combination of oligomers for detecting *Babesia* sp.—including *Babesia microti* and/or *Babesia divergens* and/or *Babesia duncani* and/or *Babesia venatorum*—in a sample. The oligomer combination generally includes at least two amplification oligomers for detecting *Babesia* sp.—including *Babesia microti* and/or *Babesia divergens* and/or *Babesia duncani* and/or *Babesia venatorum*—in a sample, and may further include one or more additional oligomers as described herein for performing amplification-based detection of *Babesia* sp.—including *Babesia microti* and/or *Babesia divergens* and/or *Babesia duncani* and/or *Babesia venatorum*—such as, for example, a capture probe and/or a

detection probe.

(2) The methods for diagnosing *Babesiosis* sp. generally include detecting the presence or absence of *Babesia* sp. in a sample from a subject. The sample may be suspected of being infected with or containing *Babesia* sp. The subject may be suspected of being infected with *Babesia* sp. or having Babesiosis. In particular, an assay is performed for the specific detection in the sample of *Babesia* sp. nucleic acid. Based on the results from the detection assay, a status of either positive or negative is assigned for the *Babesia* sp. The presence or absence of Babesiosis in the subject can be determined based on the *Babesia* sp. status.

(3) While *Babesia* sp. nucleic acid may be detected using any suitable method, it is preferred that these protozoan parasites are detected using a nucleic-acid-based detection assay. Nucleic-acid-based detection assays generally utilize oligonucleotides that specifically hybridize to a target nucleic acid of *Babesia* sp. with minimal cross-reactivity to other nucleic acids suspected of being in a sample. Accordingly, oligonucleotides for nucleic-acid-based detection of *Babesia* sp. will have minimal cross-reactivity to other nucleic acids including, for example, *P. falciparum*.

(4) A positive signal from a nucleic-acid-based detection assay in accordance with the present disclosure is indicative of the presence of one or more of *Babesia microti*, *Babesia divergens*, *Babesia duncani* and/or *Babesia venatorum* in a sample.

(5) In some embodiments of a method comprising the use of a nucleic-acid-base detection assay—such as an amplification-based assay—is used to detect *Babesia* sp. Such methods generally include amplifying a target sequence within a target nucleic acid utilizing an in vitro nucleic acid amplification reaction and detecting the amplified product by, for example, specifically hybridizing the amplified product with a nucleic acid detection probe that provides a signal to indicate the presence of a target in the sample. The amplification step includes contacting the sample with two or more amplification oligomers specific for a target sequence in a target nucleic acid to produce an amplified product if the target nucleic acid is present in the sample. Amplification synthesizes additional copies of the target sequence or its complement such as, e.g., by using at least one nucleic acid polymerase to extend the sequence from an amplification oligomer (a primer) using a template strand. One embodiment for detecting the amplified product uses a hybridizing step that includes contacting the amplified product with at least one probe specific for a sequence amplified by the selected amplification oligomers, e.g., a sequence contained in the target sequence flanked by a pair of selected amplification oligomers. Suitable amplification methods include, for example, replicase-mediated amplification, polymerase chain reaction (PCR), ligase chain reaction (LCR), strand-displacement amplification (SDA), and transcription-mediated or transcription-associated amplification (TMA). Such amplification methods are well-known in the art (see, e.g., discussion of amplification methods in Definitions section, supra) and are readily used in accordance with the methods of the present disclosure.

(6) For example, some amplification methods that use TMA amplification include the following steps. Briefly, the target nucleic acid that contains the sequence to be amplified is provided as single stranded nucleic acid (e.g., ssRNA or ssDNA). Those skilled in the art will appreciate that conventional melting of double stranded nucleic acid (e.g., dsDNA) may be used to provide single-stranded target nucleic acids. A promoter primer binds specifically to the target nucleic acid at its target sequence and a reverse transcriptase (RT) extends the 3' end of the promoter primer using the target strand as a template to create a cDNA copy of the target sequence strand, resulting in an RNA:DNA duplex. An RNase digests the RNA strand of the RNA:DNA duplex and a second primer binds specifically to its target sequence, which is located on the cDNA strand downstream from the promoter primer end. RT synthesizes a new DNA strand by extending the 3' end of the second primer using the first cDNA template to create a dsDNA that contains a functional promoter sequence. An RNA polymerase specific for the promoter sequence then initiates transcription to produce RNA transcripts that are about 100 to 1000 amplified copies (“amplicons”) of the initial target strand in the reaction. Amplification continues when the second primer binds specifically to

its target sequence in each of the amplicons and RT creates a DNA copy from the amplicon RNA template to produce an RNA:DNA duplex. RNase in the reaction mixture digests the amplicon RNA from the RNA:DNA duplex and the promoter primer binds specifically to its complementary sequence in the newly synthesized DNA. RT extends the 3' end of the promoter primer to create a dsDNA that contains a functional promoter to which the RNA polymerase binds to transcribe additional amplicons that are complementary to the target strand. The autocatalytic cycles of making more amplicon copies repeat during the course of the reaction resulting in about a billion-fold amplification of the target nucleic acid present in the sample. The amplified products may be detected in real-time during amplification, or at the end of the amplification reaction by using a probe that binds specifically to a target sequence contained in the amplified products. Detection of a signal resulting from the bound probes indicates the presence of the target nucleic acid in the sample.

(7) In some embodiments, the method utilizes a “reverse” TMA reaction. In such variations, the initial or “forward” amplification oligomer is a priming oligonucleotide that hybridizes to the target nucleic acid in the vicinity of the 3'-end of the target region. A reverse transcriptase (RT) synthesizes a cDNA strand by extending the 3'-end of the primer using the target nucleic acid as a template. The second or “reverse” amplification oligomer is a promoter primer or promoter provider having a target-hybridizing sequence configured to hybridize to a target-sequence contained within the synthesized cDNA strand. Where the second amplification oligomer is a promoter primer, RT extends the 3' end of the promoter primer using the cDNA strand as a template to create a second, cDNA copy of the target sequence strand, thereby creating a dsDNA that contains a functional promoter sequence. Amplification then continues essentially as described above for initiation of transcription from the promoter sequence utilizing an RNA polymerase. Alternatively, where the second amplification oligomer is a promoter provider, a terminating oligonucleotide, which hybridizes to a target sequence that is in the vicinity to the 5'-end of the target region, is typically utilized to terminate extension of the priming oligomer at the 3'-end of the terminating oligonucleotide, thereby providing a defined 3'-end for the initial cDNA strand synthesized by extension from the priming oligomer. The target-hybridizing sequence of the promoter provider then hybridizes to the defined 3'-end of the initial cDNA strand, and the 3'-end of the cDNA strand is extended to add sequence complementary to the promoter sequence of the promoter provider, resulting in the formation of a double-stranded promoter sequence. The initial cDNA strand is then used as a template to transcribe multiple RNA transcripts complementary to the initial cDNA strand, not including the promoter portion, using an RNA polymerase that recognizes the double-stranded promoter and initiates transcription therefrom. Each of these RNA transcripts is then available to serve as a template for further amplification from the first priming amplification oligomer.

(8) In one aspect, there is provided a method for specifically detecting *Babesia* species nucleic acid in a sample, which comprises the use of at least two amplification oligomers comprising (a) a first amplification oligomer comprising a first target-hybridizing sequence (i) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:66 and comprises SEQ ID NO:56 or 57; or (ii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:96 and comprises SEQ ID NO:101; or (iii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:97 and comprises SEQ ID NO:101; (iv) comprises or consists of SEQ ID NO:8; (v) comprises or consists of SEQ ID NO:83 and (b) a second amplification oligomer comprising a second target-hybridizing sequence that is from about 15 to about 33 contiguous nucleotides in length, and (i) is contained in SEQ ID NO:68 and comprises SEQ ID NO:52, SEQ ID NO:53, SEQ ID NO:54, SEQ ID NO:55, or SEQ ID NO:85; or (ii) is contained in SEQ ID NO:67 and comprises SEQ ID NO:45 or SEQ ID NO:52; or (iii) is contained in SEQ ID NO:70 and comprises SEQ ID NO:46, SEQ ID NO:47, SEQ ID NO:48, SEQ ID NO:49, SEQ ID NO:50, or SEQ ID NO:51; or (iv) comprises or consists of SEQ ID NO:84. In certain embodiments comprising an amplification-

based detection assay, a combination of at least two amplification oligomers is therefore utilized for the detection of a *Babesia* sp. nucleic acid.

(9) Suitably, the first amplification oligomer comprises or consists of a sequence selected from the group consisting of: SEQ ID NOs:2 and 4 and 6 and 8 and 83, more suitably, wherein the first amplification oligomer comprises or consists of the sequence selected from the group consisting of: SEQ ID NOs:2 and 4 and 8 and 83.

(10) The first amplification oligomer of the combination may be a promoter primer or promoter provider further comprising a promoter sequence located 5' to the first target-hybridizing sequence. Suitably, the promoter sequence is a T7 promoter sequence which optionally comprises or consists of SEQ ID NO:58. According to this embodiment, the first amplification oligomer may comprise or consist of a sequence selected from the group consisting of SEQ ID NOs:1 and 3 and 5 and 7 and 82, suitably, wherein the first amplification oligomer comprises or consists of a sequence selected from the group consisting of SEQ ID NOs:1 and 3 and 7 and 82.

(11) Suitably, the second amplification oligomer comprises or consists of a sequence selected from the group consisting of: SEQ ID NOs:13, 16, 17, 18, 19, 20, 21, 27, 28, 29, 31, 32, 33, 34, 35, 36, 84, and 86. More suitably, the second amplification oligomer sequence comprises or consists of SEQ ID NO:21 or SEQ ID NO:27 or SEQ ID NO:34 or SEQ ID NO:84 or SEQ ID NO:86.

(12) In one embodiment, the first and second target-hybridizing sequences respectively comprise or consist of the nucleotide sequences of: (a) SEQ ID NO:2 or 6 and SEQ ID NO:11; (b) SEQ ID NO:4 or 6 and SEQ ID NO:13; (c) SEQ ID NO:4 and SEQ ID NO:16 or SEQ ID NO:17; (d) SEQ ID NO:4 and SEQ ID NO:18 or SEQ ID NO:19; (e) SEQ ID NO:4 and SEQ ID NO:20; (f) SEQ ID NO:4 or 6 or 8 and SEQ ID NO:21; (g) SEQ ID NO:2 or 4 or 8 and SEQ ID NO:27; (h) SEQ ID NO:4 and SEQ ID NO:28; (i) SEQ ID NO:4 and SEQ ID NO:29; (j) SEQ ID NO:4 and SEQ ID NO:31; (k) SEQ ID NO:8 and SEQ ID NO:32; (l) SEQ ID NO:8 and SEQ ID NO:33; (m) SEQ ID NO:8 and SEQ ID NO:34; (n) SEQ ID NO:8 and SEQ ID NO:35; (o) SEQ ID NO:8 and SEQ ID NO:36; (p) SEQ ID NO:8 and SEQ ID NO:84; (q) SEQ ID NO:8 and SEQ ID NO:86; (r) SEQ ID NO:83 and SEQ ID NO:34; (s) SEQ ID NO:83 and SEQ ID NO:84; or (t) SEQ ID NO:83 and SEQ ID NO:86.

(13) In another embodiment, the first and second target-hybridizing sequences respectively comprise or consist of the nucleotide sequences of: (a) SEQ ID NO:2 and SEQ ID NO:27; (b) SEQ ID NO:4 and SEQ ID NO:21; (c) SEQ ID NO:8 and SEQ ID NO:21; (d) SEQ ID NO:8 and SEQ ID NO:34; (e) SEQ ID NO:8 and SEQ ID NO:84; (f) SEQ ID NO:8 and SEQ ID NO:86; (g) SEQ ID NO:83 and SEQ ID NO:34; (h) SEQ ID NO:83 and SEQ ID NO:84; or (i) SEQ ID NO:83 and SEQ ID NO:86.

(14) In embodiments where the combination further includes one or more detection probe oligomers, the first and second amplification oligomer target-hybridizing sequences and the detection probe oligomer target-hybridizing sequences respectively may comprise or consist of the nucleotide sequences of: (a) SEQ ID NO:2 and SEQ ID NO:11 and SEQ ID NO:39 or SEQ ID NO:37; (b) SEQ ID NO:2 and SEQ ID NO:27 and SEQ ID NO:38 or SEQ ID NO:39; (c) SEQ ID NO:4 and SEQ ID NO:13 and SEQ ID NO:39 or SEQ ID NO:37; (d) SEQ ID NO:4 and SEQ ID NO:16 or SEQ ID NO:17 and SEQ ID NO:39; (e) SEQ ID NO:4 and SEQ ID NO:18 or SEQ ID NO:19 and SEQ ID NO:39 or SEQ ID NO:37; (f) SEQ ID NO:4 and SEQ ID NO:20 and SEQ ID NO:39 or SEQ ID NO:37; (g) SEQ ID NO:4 and SEQ ID NO:21 and SEQ ID NO:39 or SEQ ID NO:37; (h) SEQ ID NO:4 and SEQ ID NO:27 and SEQ ID NO:39 or SEQ ID NO:38; (i) SEQ ID NO:4 and SEQ ID NO:28 and SEQ ID NO:39; (j) SEQ ID NO:4 and SEQ ID NO:29 and SEQ ID NO:39 or SEQ ID NO:37; (k) SEQ ID NO:4 and SEQ ID NO:31 and SEQ ID NO:39; (l) SEQ ID NO:6 and SEQ ID NO:11 and SEQ ID NO:37; (m) SEQ ID NO:6 and SEQ ID NO:13 and SEQ ID NO:37; (n) SEQ ID NO:6 and SEQ ID NO:21 and SEQ ID NO:37; (o) SEQ ID NO:8 and SEQ ID NO:21 and SEQ ID NO:39 or SEQ ID NO:37 or SEQ ID NO:42; (p) SEQ ID NO:8 and SEQ ID NO:27 and SEQ ID NO:39; (q) SEQ ID NO:8 and SEQ ID NO:32 and SEQ ID NO:37 or SEQ ID

NO:42; (r) SEQ ID NO:8 and SEQ ID NO:33 and SEQ ID NO:37 or SEQ ID NO:42; (s) SEQ ID NO:8 and SEQ ID NO:34 and SEQ ID NO:37 or SEQ ID NO:42; (t) SEQ ID NO:8 and SEQ ID NO:35 and SEQ ID NO:37 or SEQ ID NO:42; (u) SEQ ID NO:8 and SEQ ID NO:36 and SEQ ID NO:37 or SEQ ID NO:42; (v) SEQ ID NO:8, and SEQ ID NO:84, and SEQ ID NOs:91, 92 and/or 93; (w) SEQ ID NO:8, and SEQ ID NO:86, and SEQ ID NOs:91, 92 and/or 93; (x) SEQ ID NO:83, and SEQ ID NO:34, and SEQ ID NOs:91, 92 and/or 93; (y) SEQ ID NOs:83, and SEQ ID NO:84, and SEQ ID NOs:91, 92 and/or 93; (z) SEQ ID NOs:83, and SEQ ID NO:86, and SEQ ID NOs:91, 92 and/or 93; or (aa) SEQ ID NO:8 or 83, SEQ ID NO:34, 84 or 86, and SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(15) In other embodiments where the combination further include one or more detection probe oligomers, the first and second amplification oligomer target-hybridizing sequences and the detection probe oligomer target-hybridizing sequences respectively may comprise or consist of the nucleotide sequences of: (a) SEQ ID NO:2 and SEQ ID NO:27 and SEQ ID NO:38; (b) SEQ ID NO:4 and SEQ ID NO:21 and SEQ ID NO:39; (c) SEQ ID NO:8 and SEQ ID NO:21 and SEQ ID NO:37 or SEQ ID NO:39; (d) SEQ ID NO:8 and SEQ ID NO:34 and SEQ ID NO:37 or SEQ ID NO:42; (e) SEQ ID NO:8, and SEQ ID NO:84, and SEQ ID NOs:91, 92 and/or 93; (f) SEQ ID NO:8, and SEQ ID NO:86, and SEQ ID NOs:91, 92 and/or 93; (g) SEQ ID NO:83, and SEQ ID NO:34, and SEQ ID NOs:91, 92 and/or 93; (h) SEQ ID NOs:83, and SEQ ID NO:84, and SEQ ID NOs:91, 92 and/or 93; (i) SEQ ID NOs:83, and SEQ ID NO:86, and SEQ ID NOs:91, 92 and/or 93; or (j) SEQ ID NO:8 or 83, SEQ ID NO:34, 84 or 86, and SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(16) As will be appreciated, the present disclosure contemplates the use of various combinations of first and second amplification oligomers, including: a first amplification oligomer comprising a first target-hybridizing sequence (i) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:66 and comprises SEQ ID NO:56 or (ii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:96 and comprises SEQ ID NO:101; or (iii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:97 and comprises SEQ ID NO:101, and a second amplification oligomer comprising a second target-hybridizing sequence that is from about 15 to about 33 contiguous nucleotides in length, and is contained in SEQ ID NO:68 and comprises SEQ ID NO:52; a first amplification oligomer comprising a first target-hybridizing sequence (i) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:66 and comprises SEQ ID NO:56 or (ii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:96 and comprises SEQ ID NO:101; or (iii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:97 and comprises SEQ ID NO:101, and a second amplification oligomer comprising a second target-hybridizing sequence that is from about 15 to about 33 contiguous nucleotides in length, and is contained in SEQ ID NO:68 and comprises SEQ ID NO:53; a first amplification oligomer comprising a first target-hybridizing sequence (i) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:66 and comprises SEQ ID NO:56 or (ii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:96 and comprises SEQ ID NO:101; or (iii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:97 and comprises SEQ ID NO:101, and a second amplification oligomer comprising a second target-hybridizing sequence that is from about 15 to about 33 contiguous nucleotides in length, and is contained in SEQ ID NO:68 and comprises SEQ ID NO:54; a first amplification oligomer comprising a first target-hybridizing sequence (i) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:66 and comprises SEQ ID NO:56 or (ii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:96 and comprises SEQ ID

[illegible]

[illegible]

[illegible]

second target-hybridizing sequence comprises or consists of SEQ ID NO:86.

(23) The present disclosure also contemplates the use of other combinations of first and second amplification oligomers, including amplification oligomers in which: the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:13; the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:16; the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:17; the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:18; the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:19; the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:20; the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:21; the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:27; the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:28; the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:29; the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:31; the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:32; the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:33; the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:34; the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:35; the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:36; the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:84; and the first target-hybridizing sequence comprises or consists of SEQ ID NO:83 and the second target-hybridizing sequence comprises or consists of SEQ ID NO:86.

(24) The present disclosure also contemplates the use of other combinations of first and second amplification oligomers, wherein the first and second target-hybridizing sequences respectively comprise or consist of the nucleotide sequences of: SEQ ID NO:2 and SEQ ID NO:11; SEQ ID NO:6 and SEQ ID NO:11; SEQ ID NO:4 and SEQ ID NO:13; SEQ ID NO:6 and SEQ ID NO:13; SEQ ID NO:4 and SEQ ID NO:16; SEQ ID NO:4 and SEQ ID NO:17; SEQ ID NO:4 and SEQ ID NO:18; SEQ ID NO:4 and SEQ ID NO:19; SEQ ID NO:4 and SEQ ID NO:20; SEQ ID NO:4 and SEQ ID NO:21; SEQ ID NO:6 and SEQ ID NO:21; SEQ ID NO:8 and SEQ ID NO:21; SEQ ID NO:2 and SEQ ID NO:27; SEQ ID NO:4 and SEQ ID NO:27; SEQ ID NO:8 and SEQ ID NO:27; SEQ ID NO:4 and SEQ ID NO:28; SEQ ID NO:4 and SEQ ID NO:29; SEQ ID NO:4 and SEQ ID NO:31; SEQ ID NO:8 and SEQ ID NO:32; SEQ ID NO:8 and SEQ ID NO:33; SEQ ID NO:8 and SEQ ID NO:34; SEQ ID NO:8 and SEQ ID NO:84; SEQ ID NO:8 and SEQ ID NO:86; SEQ ID NO:8 and SEQ ID NO:35; SEQ ID NO:8 and SEQ ID NO:36; SEQ ID NO:83 and SEQ ID NO:32; SEQ ID NO:83 and SEQ ID NO:33; SEQ ID NO:83 and SEQ ID NO:34; SEQ ID NO:83 and SEQ ID NO:84; SEQ ID NO:83 and SEQ ID NO:86; SEQ ID NO:83 and SEQ ID NO:35; or SEQ ID NO:83 and SEQ ID NO:36.

(25) The present disclosure also contemplates the use of combinations of a first amplification oligomer, a second amplification oligomer and a detection probe in which the first and second

amplification oligomer target-hybridizing sequences and the detection probe oligomer target-hybridizing sequences respectively comprise or consist of the nucleotide sequences of: SEQ ID NO:2 and SEQ ID NO:11 and SEQ ID NO:39; SEQ ID NO:2 and SEQ ID NO:11 and SEQ ID NO:37; SEQ ID NO:2 and SEQ ID NO:27 and SEQ ID NO:38; SEQ ID NO:2 and SEQ ID NO:27 and SEQ ID NO:39; SEQ ID NO:4 and SEQ ID NO:13 and SEQ ID NO:39; SEQ ID NO:4 and SEQ ID NO:13 and SEQ ID NO:37; SEQ ID NO:4 and SEQ ID NO:16 and SEQ ID NO:39; SEQ ID NO:4 and SEQ ID NO:17 and SEQ ID NO:39; SEQ ID NO:4 and SEQ ID NO:18 and SEQ ID NO:39; SEQ ID NO:4 and SEQ ID NO:19 and SEQ ID NO:39; SEQ ID NO:4 and SEQ ID NO:18 and SEQ ID NO:37; SEQ ID NO:4 and SEQ ID NO:19 and SEQ ID NO:37; SEQ ID NO:4 and SEQ ID NO:20 and SEQ ID NO:39; SEQ ID NO:4 and SEQ ID NO:20 and SEQ ID NO:37; SEQ ID NO:4 and SEQ ID NO:21 and SEQ ID NO:39; SEQ ID NO:4 and SEQ ID NO:21 and SEQ ID NO:37; SEQ ID NO:4 and SEQ ID NO:27 and SEQ ID NO:39; SEQ ID NO:4 and SEQ ID NO:27 and SEQ ID NO:38; SEQ ID NO:4 and SEQ ID NO:28 and SEQ ID NO:39; SEQ ID NO:4 and SEQ ID NO:29 and SEQ ID NO:39; SEQ ID NO:4 and SEQ ID NO:29 and SEQ ID NO:37; SEQ ID NO:4 and SEQ ID NO:31 and SEQ ID NO:39; SEQ ID NO:6 and SEQ ID NO:11 and SEQ ID NO:37; SEQ ID NO:6 and SEQ ID NO:13 and SEQ ID NO:37; SEQ ID NO:6 and SEQ ID NO:21 and SEQ ID NO:37; SEQ ID NO:8 and SEQ ID NO:21 and SEQ ID NO:39; SEQ ID NO:8 and SEQ ID NO:21 and SEQ ID NO:37; SEQ ID NO:8 and SEQ ID NO:21 and SEQ ID NO:42; SEQ ID NO:8 and SEQ ID NO:27 and SEQ ID NO:39; SEQ ID NO:8 and SEQ ID NO:32 and SEQ ID NO:37; SEQ ID NO:8 and SEQ ID NO:32 and SEQ ID NO:42; SEQ ID NO:8 and SEQ ID NO:33 and SEQ ID NO:37; SEQ ID NO:8 and SEQ ID NO:33 and SEQ ID NO:42; SEQ ID NO:8 and SEQ ID NO:34 and SEQ ID NO:37; SEQ ID NO:8 and SEQ ID NO:34 and SEQ ID NO:42; SEQ ID NO:8 and SEQ ID NO:35 and SEQ ID NO:37; SEQ ID NO:8 and SEQ ID NO:35 and SEQ ID NO:42; SEQ ID NO:8 and SEQ ID NO:36 and SEQ ID NO:37; or SEQ ID NO:8 and SEQ ID NO:36 and SEQ ID NO:42; SEQ ID NO:8, and SEQ ID NO:84, and SEQ ID NOs:91, 92 and/or 93; SEQ ID NO:8, and SEQ ID NO:86, and SEQ ID NOs:91, 92 and/or 93; SEQ ID NO:83, and SEQ ID NO:34, and SEQ ID NOs:91, 92 and/or 93; SEQ ID NOs:83, and SEQ ID NO:84, and SEQ ID NOs:91, 92 and/or 93; SEQ ID NOs:83, and SEQ ID NO:86, and SEQ ID NOs:91, 92 and/or 93; or SEQ ID NO:8 or 83, SEQ ID NO:34, 84 or 86, and SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(26) In a further aspect, there is described a method for specifically detecting *Babesia* species nucleic acid in a sample, said method comprising: (1) contacting a sample, said sample suspected of containing *Babesia* species nucleic acid, with at least two oligomers for amplifying a target region of a *Babesia* species target nucleic acid, wherein two of said at least two amplification oligomers are selected from the group consisting of: (a) a first amplification oligomer and a second amplification oligomer, wherein the first amplification oligomer comprises a first target-hybridizing sequence (i) that is from 15 to 33 contiguous nucleobases in length, is contained in SEQ ID NO:66 and contains SEQ ID NO:56 or SEQ ID NO:57, or (ii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:96 and comprises SEQ ID NO:101; or (iii) that is from about 15 to about 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:97 and comprises SEQ ID NO:101; (iv) comprises/consists of SEQ ID NO:8 or 83; or (b) a first amplification oligomer and a second amplification oligomer, wherein the second amplification oligomer comprises a second target-hybridizing sequence that is from about 15 to about 33 contiguous nucleotides in length, and is (i) contained in SEQ ID NO:68 and contains SEQ ID NO:52, SEQ ID NO:53, SEQ ID NO:54, SEQ ID NO:55, or SEQ ID NO:85, or (ii) is contained in SEQ ID NO:67 and contains SEQ ID NO:45 or SEQ ID NO:69, or (iii) is contained in SEQ ID NO:70 and contains SEQ ID NO:46, SEQ ID NO:47, SEQ ID NO:48, SEQ ID NO:49, SEQ ID NO:50, or SEQ ID NO:51; or (iv) comprises or consists of SEQ ID NO:84; (2) performing an in vitro nucleic acid amplification reaction, wherein any *Babesia* target nucleic acid present in said sample is used as a template for generating an

amplification product, wherein said amplification product has a length of from 180 to 220 contiguous nucleotides and contains SEQ ID NO:65 or the complement thereof; and (3) detecting the presence or absence of the amplification product, thereby indicating the presence or absence of *Babesia* species target nucleic acid in said sample.

(27) In those embodiments in which the amplification product has a length of from 180 to 220 contiguous nucleotides and contains SEQ ID NO:65 or the complement thereof it is also contemplated that the amplification product has a length of from 180 to 210 contiguous nucleotides and contains SEQ ID NO:65 or the complement thereof. It is also contemplated that the amplification product has a length of from 180 to 200 contiguous nucleotides and contains SEQ ID NO:65 or the complement thereof. It is also contemplated that the amplification product has a length of from 180 to 190 contiguous nucleotides and contains SEQ ID NO:65 or the complement thereof. It is also contemplated that the amplification product has a length of from 190 to 220 contiguous nucleotides and contains SEQ ID NO:65 or the complement thereof. It is also contemplated that the amplification product has a length of from 200 to 220 contiguous nucleotides and contains SEQ ID NO:65 or the complement thereof. It is also contemplated that the amplification product has a length of from 210 to 220 contiguous nucleotides and contains SEQ ID NO:65 or the complement thereof.

(28) Further combinations of first and second amplification oligomers therefore also include those in which the first amplification oligomer comprises a first target-hybridizing sequence that is from 15 to 33 contiguous nucleobases in length, is contained in SEQ ID NO:66 and contains SEQ ID NO:56; or comprises a first target-hybridizing sequence that is from 15 to 33 contiguous nucleobases in length, is contained in SEQ ID NO:66 and contains SEQ ID NO:57; or comprises a first target hybridizing sequence that is from 15 to 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:96 and contains SEQ ID NO:101; or comprises a first target-hybridizing sequence from 15 to 33 contiguous nucleotides in length, is contained in the sequence of SEQ ID NO:97 and contains SEQ ID NO:101; or comprises or consists of SEQ ID NO:8.

(29) Further combinations of first and second amplification oligomers therefore also include those in which the second amplification oligomer comprises a second target-hybridizing sequence that is from about 15 to about 33 contiguous nucleotides in length and is contained in SEQ ID NO:68 and contains SEQ ID NO:52, or contains SEQ ID NO:53 or contains SEQ ID NO:54 or contains SEQ ID NO:55 or contains SEQ ID NO:85.

(30) Further combinations of first and second amplification oligomers therefore also include those in which the second amplification oligomer comprises a second target-hybridizing sequence that is from about 15 to about 33 contiguous nucleotides in length and is contained in SEQ ID NO:67 and contains SEQ ID NO:45 or contains SEQ ID NO:69.

(31) Further combinations of first and second amplification oligomers therefore also include those in which the second amplification oligomer comprises a second target-hybridizing sequence that is from about 15 to about 33 contiguous nucleotides in length and is contained in SEQ ID NO:70 and contains SEQ ID NO:46 or contains SEQ ID NO:47 or contains SEQ ID NO:48 or contains SEQ ID NO:49 or contains SEQ ID NO:50 or contains SEQ ID NO:51.

(32) Further combinations of first and second amplification oligomers therefore also include those in which the second amplification oligomer comprises a second target-hybridizing sequence that is from about 15 to about 33 contiguous nucleotides in length and contains SEQ ID NO:84.

(33) In some embodiments, combinations of certain first and second amplification oligomers are preferred.

(34) One preferred combination is a first amplification oligomer comprising a first target-hybridizing sequence that is from 15 to 33 contiguous nucleobases in length, is contained in SEQ ID NO:66 and comprises SEQ ID NO:56 or SEQ ID NO:57 and a second amplification oligomer comprising a second target-hybridizing sequence that is contained in SEQ ID NO:67 and comprises SEQ ID NO:45. A more preferred combination is a first amplification oligomer comprising or

consisting of the sequence set forth in SEQ ID NO:1 or comprising or consisting of the target-hybridizing sequence set forth in SEQ ID NO:2 and a second amplification oligomer comprising or consisting of the target-hybridizing sequence set forth in SEQ ID NO:27. According to this embodiment, these combinations of first and second amplification oligomers can be used in combination with a detection probe, suitably, a detection probe in which the target hybridising sequence comprises or consists of the sequence set forth in SEQ ID NO:38.

(35) Another preferred combination is a first amplification oligomer comprising a first target-hybridizing sequence that is from 15 to 33 contiguous nucleobases in length, is contained in the sequence of SEQ ID NO:66 and comprises SEQ ID NO:56 and a second amplification oligomer comprising a second target-hybridizing sequence that is contained in SEQ ID NO:70 and comprises SEQ ID NO:49 or SEQ ID NO:50, or SEQ ID NO:51. A more preferred combination is a first amplification oligomer comprising or consisting of the sequence set forth in SEQ ID NO:3 or the target-hybridising sequence set forth in SEQ ID NO:4 and a second amplification oligomer comprising or consisting of the target-hybridising sequence set forth in SEQ ID NO:21. According to this embodiment, these combinations of first and second amplification oligomers can be used in combination with a detection probe, suitably, a detection probe in which the target hybridising sequence comprises or consists of the sequence set forth in SEQ ID NO:39.

(36) Another preferred combination is a first amplification oligomer comprising a first target-hybridizing sequence that is from 15 to 33 contiguous nucleobases in length, is contained in the sequence of SEQ ID NO:96 and comprises SEQ ID NO:101 and a second amplification oligomer comprising a second target-hybridizing sequence that is contained in SEQ ID NO:68 and comprises SEQ ID NO:52, SEQ ID NO:53, SEQ ID NO:54, SEQ ID NO:55, or SEQ ID NO:85. A more preferred combination is a first amplification oligomer comprising or consisting of the sequence set forth in SEQ ID NO:7 or 82 or the target-hybridising sequence set forth in SEQ ID NO:8 or 83 and a second amplification oligomer comprising or consisting of the target-hybridising sequence set forth in SEQ ID NO:13, 16, 17, 18, 19, 20, 21, 27, 28, 29, 31, 32, 33, 34, 35, 36, 84, or 86.

According to this embodiment, these combinations of first and second amplification oligomers can be used in combination with a detection probe, suitably, a detection probe in which the target hybridising sequence comprises or consists of the sequence set forth in SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(37) Another preferred combination is a first amplification oligomer comprising a first target-hybridizing sequence that is from 15 to 33 contiguous nucleobases in length, is contained in the sequence of SEQ ID NO:96 and comprises SEQ ID NO:101 and a second amplification oligomer comprising a second target-hybridizing sequence that is contained in SEQ ID NO:70 and comprises SEQ ID NO:49 or SEQ ID NO:50, or SEQ ID NO:51. A more preferred combination is a first amplification oligomer comprising or consisting of the sequence set forth in SEQ ID NO:7 or 82 or the target-hybridising sequence set forth in SEQ ID NO:8 or 83 and a second amplification oligomer comprising or consisting of the target-hybridising sequence set forth in SEQ ID NO:13, 16, 17, 18, 19, 20, 21, 27, 28, 29, 31, 32, 33, 34, 35, 36, 84, or 86. According to this embodiment, these combinations of first and second amplification oligomers can be used in combination with a detection probe, suitably, a detection probe in which the target hybridising sequence comprises or consists of the sequence set forth in SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(38) Another preferred combination is a first amplification oligomer comprising a first target-hybridizing sequence that is from 15 to 33 contiguous nucleobases in length, is contained in the sequence of SEQ ID NO:96 and comprises SEQ ID NO:101 and a second amplification oligomer comprising a second target-hybridizing sequence comprising or consisting of SEQ ID NO:84. A more preferred combination is a first amplification oligomer comprising or consisting of the sequence set forth in SEQ ID NO:7 or 82 or the target-hybridising sequence set forth in SEQ ID NO:8 or 83 and a second amplification oligomer comprising or consisting of the target-hybridising

sequence set forth in SEQ ID NO:13, 16, 17, 18, 19, 20, 21, 27, 28, 29, 31, 32, 33, 34, 35, 36, 84, or 86. According to this embodiment, these combinations of first and second amplification oligomers can be used in combination with a detection probe, suitably, a detection probe in which the target hybridising sequence comprises or consists of the sequence set forth in SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(39) Another preferred combination is a first amplification oligomer comprising a first target-hybridizing sequence that is from 15 to 33 contiguous nucleobases in length, is contained in the sequence of SEQ ID NO:97 and comprises SEQ ID NO:101 and a second amplification oligomer comprising a second target-hybridizing sequence that is contained in SEQ ID NO:68 and comprises SEQ ID NO:52, SEQ ID NO:53, SEQ ID NO:54, SEQ ID NO:55, or SEQ ID NO:85. A more preferred combination is a first amplification oligomer comprising or consisting of the sequence set forth in SEQ ID NO:7 or 82 or the target-hybridising sequence set forth in SEQ ID NO:8 or 83 and a second amplification oligomer comprising or consisting of the target-hybridising sequence set forth in SEQ ID NO:13, 16, 17, 18, 19, 20, 21, 27, 28, 29, 31, 32, 33, 34, 35, 36, 84, or 86.

According to this embodiment, these combinations of first and second amplification oligomers can be used in combination with a detection probe, suitably, a detection probe in which the target hybridising sequence comprises or consists of the sequence set forth in SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(40) Another preferred combination is a first amplification oligomer comprising a first target-hybridizing sequence that is from 15 to 33 contiguous nucleobases in length, is contained in the sequence of SEQ ID NO:97 and comprises SEQ ID NO:101 and a second amplification oligomer comprising a second target-hybridizing sequence that is contained in SEQ ID NO:70 and comprises SEQ ID NO:49 or SEQ ID NO:50, or SEQ ID NO:51. A more preferred combination is a first amplification oligomer comprising or consisting of the sequence set forth in SEQ ID NO:7 or 82 or the target-hybridising sequence set forth in SEQ ID NO:8 or 83 and a second amplification oligomer comprising or consisting of the target-hybridising sequence set forth in SEQ ID NO:13, 16, 17, 18, 19, 20, 21, 27, 28, 29, 31, 32, 33, 34, 35, 36, 84, or 86. According to this embodiment, these combinations of first and second amplification oligomers can be used in combination with a detection probe, suitably, a detection probe in which the target hybridising sequence comprises or consists of the sequence set forth in SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(41) Another preferred combination is a first amplification oligomer comprising a first target-hybridizing sequence that is from 15 to 33 contiguous nucleobases in length, is contained in the sequence of SEQ ID NO:97 and comprises SEQ ID NO:101 and a second amplification oligomer comprising a second target-hybridizing sequence comprising or consisting of SEQ ID NO:84. A more preferred combination is a first amplification oligomer comprising or consisting of the sequence set forth in SEQ ID NO:7 or 82 or the target-hybridising sequence set forth in SEQ ID NO:8 or 83 and a second amplification oligomer comprising or consisting of the target-hybridising sequence set forth in SEQ ID NO:13, 16, 17, 18, 19, 20, 21, 27, 28, 29, 31, 32, 33, 34, 35, 36, 84, or 86. According to this embodiment, these combinations of first and second amplification oligomers can be used in combination with a detection probe, suitably, a detection probe in which the target hybridising sequence comprises or consists of the sequence set forth in SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(42) Another preferred combination is a first amplification oligomer in which the target-hybridizing sequence comprises or consists of SEQ ID NO:8 or SEQ ID NO:83 and a second amplification oligomer comprising a second target-hybridizing sequence that is contained in SEQ ID NO:70 and comprises SEQ ID NO:49 or SEQ ID NO:50, or SEQ ID NO:51. A more preferred combination is a first amplification oligomer comprising or consisting of the sequence set forth in SEQ ID NO:7 or SEQ ID NO:8 or SEQ ID NO:82 or SEQ ID NO:83 and a second amplification oligomer comprising or consisting of the sequence set forth in SEQ ID NO:21. According to this

embodiment, these combinations of first and second amplification oligomers can be used in combination with a detection probe, suitably, a detection probe in which the target hybridising sequence comprises or consists of the sequence set forth in SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(43) Another preferred combination is a first amplification oligomer in which the target-hybridizing sequence comprises or consists of SEQ ID NO:8 or SEQ ID NO:83 and a second amplification oligomer comprising a second target-hybridizing sequence that is contained in SEQ ID NO:70 and comprises SEQ ID NO:46 or SEQ ID NO:47 or SEQ ID NO:48 or SEQ ID NO:49 or SEQ ID NO:50 or SEQ ID NO:51. A more preferred combination is a first amplification oligomer comprising or consisting of the sequence set forth in SEQ ID NO:7 or SEQ ID NO:8 or SEQ ID NO:82 or SEQ ID NO:83 and a second amplification oligomer comprising or consisting of the sequence set forth in SEQ ID NO:34 or SEQ ID NO:84 or SEQ ID NO:86. According to this embodiment, these combinations of first and second amplification oligomers can be used in combination with a detection probe, suitably, a detection probe in which the target hybridising sequence comprises or consists of the sequence set forth in SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(44) Another preferred combination is a first amplification oligomer comprising a first target-hybridizing sequence that is from 15 to 33 contiguous nucleobases in length, is contained in the sequence of SEQ ID NO:96 and comprises SEQ ID NO:101 and a second amplification oligomer comprising a second target-hybridizing sequence that is from about 15 to about 33 contiguous nucleotides in length, and is contained in SEQ ID NO:68 and comprises SEQ ID NO:85. A more preferred combination is a first amplification oligomer comprising or consisting of the sequence set forth in SEQ ID NO:7 or SEQ ID NO:8 or SEQ ID NO:82 or SEQ ID NO:83 and a second amplification oligomer comprising or consisting of the sequence set forth in SEQ ID NO:34 or SEQ ID NO:84 or SEQ ID NO:86. According to this embodiment, these combinations of first and second amplification oligomers can be used in combination with a detection probe, suitably, a detection probe in which the target hybridising sequence comprises or consists of the sequence set forth in SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(45) Another preferred combination is a first amplification oligomer comprising a first target-hybridizing sequence that is from 15 to 33 contiguous nucleobases in length, is contained in the sequence of SEQ ID NO:97 and comprises SEQ ID NO:101 and a second amplification oligomer comprising a second target-hybridizing sequence that is from about 15 to about 33 contiguous nucleotides in length, and is contained in SEQ ID NO:68 and comprises SEQ ID NO:85. A more preferred combination is a first amplification oligomer comprising or consisting of the sequence set forth in SEQ ID NO:7 or SEQ ID NO:8 or SEQ ID NO:82 or SEQ ID NO:83 and a second amplification oligomer comprising or consisting of the sequence set forth in SEQ ID NO:34 or SEQ ID NO:84 or SEQ ID NO:86. According to this embodiment, these combinations of first and second amplification oligomers can be used in combination with a detection probe, suitably, a detection probe in which the target hybridising sequence comprises or consists of the sequence set forth in SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(46) Detection of the amplified products may be accomplished by a variety of methods to detect a signal specifically associated with the amplified target sequence. The nucleic acids may be associated with a surface that results in a physical change, such as a detectable electrical change. Amplified nucleic acids may be detected by concentrating them in or on a matrix and detecting the nucleic acids or dyes associated with them (e.g., an intercalating agent such as ethidium bromide or cyber green), or detecting an increase in dye associated with nucleic acid in solution phase. Other methods of detection may use nucleic acid detection probes that are configured to specifically hybridize to a sequence in the amplified product and detecting the presence of the probe:product complex, or by using a complex of probes that may amplify the detectable signal associated with the amplified products (e.g., U.S. Pat. Nos. 5,424,413; 5,451,503; and 5,849,481). Directly or

indirectly labeled probes that specifically associate with the amplified product provide a detectable signal that indicates the presence of the target nucleic acid in the sample.

(47) Detection probes (where labelled) that hybridize to the complementary amplified sequences may be DNA or RNA oligomers, or oligomers that contain a combination of DNA and RNA nucleotides, or oligomers synthesized with a modified backbone, e.g., an oligomer that includes one or more 2'-methoxy substituted ribonucleotides. Probes used for detection of the amplified sequences may be unlabeled and detected indirectly (e.g., by binding of another binding partner to a moiety on the probe) or may be labeled with a variety of detectable labels. In some embodiments of the method for diagnosing BV, such as in certain embodiments using transcription-mediated amplification (TMA), the detection probe is a linear chemiluminescently labeled probe such as, e.g., a linear acridinium ester (AE) labeled probe. The detection step may also provide additional information on the amplified sequence, such as, e.g., all or a portion of its nucleic acid base sequence. Detection may be performed after the amplification reaction is completed, or may be performed simultaneously with amplifying the target region, e.g., in real time. In one embodiment, the detection step allows homogeneous detection, e.g., detection of the hybridized probe without removal of unhybridized probe from the mixture (see, e.g., U.S. Pat. Nos. 5,639,604 and 5,283,174).

(48) In embodiments that detect the amplified product near or at the end of the amplification step, a linear detection probe may be used to provide a signal to indicate hybridization of the probe to the amplified product. One example of such detection uses a luminescently labeled probe that hybridizes to target nucleic acid. Luminescent label is then hydrolyzed from non-hybridized probe. Detection is performed by chemiluminescence using a luminometer. (see, e.g., International Patent Application Pub. No. WO 89/002476). In other embodiments that use real-time detection, the detection probe may be a hairpin probe such as, for example, a molecular beacon, molecular torch, or hybridization switch probe that is labeled with a reporter moiety that is detected when the probe binds to amplified product. Such probes may comprise target-hybridizing sequences and non-target-hybridizing sequences. Various forms of such probes have been described previously (see, e.g., U.S. Pat. Nos. 5,118,801; 5,312,728; 5,925,517; 6,150,097; 6,849,412; 6,835,542; 6,534,274; and 6,361,945; and US Patent Application Pub. Nos. 20060068417A1 and 20060194240A1).

(49) In certain embodiments comprising an amplification-based detection assay targeting *Babesia* sp., the method utilizes one or more detection probes that specifically hybridizes to a *Babesia* sp. amplification product. In particular variations, a *Babesia* sp.-specific detection probe oligomer comprises a target-hybridizing sequence that is from about 14 to about 40 nucleotides in length and is configured to specifically hybridize to a target sequence contained in SEQ ID NO:59, the RNA equivalent of SEQ ID NO:59, the complement of SEQ ID NO:59, the RNA equivalent of the complement of SEQ ID NO:59, SEQ ID NO:65, the DNA equivalent of SEQ ID NO:65, the complement of SEQ ID NO:65, or the DNA equivalent of the complement of SEQ ID NO:65.

(50) Suitably, the detection probe target-hybridizing sequence contains the sequence of SEQ ID NO:59 and includes at least the sequence of SEQ ID NO:37, 42, or 99.

(51) Suitably, the detection probe target-hybridizing sequence is contained in the sequence of SEQ ID NO:65 and includes at least the sequence of SEQ ID NO:59, 94, or 99.

(52) Suitably, the detection probe oligomer comprises a nucleotide sequence that is from 16 to 25 contiguous nucleotides in length and specifically hybridizes to SEQ ID NO:65, or the DNA equivalent thereof; or specifically hybridizes to the complement of SEQ ID NO:65, or the DNA equivalent thereof.

(53) Suitably, the detection probe oligomer sequence further comprises a nucleotide sequence comprising or consisting of SEQ ID NO:59, 94, or 99.

(54) Suitably, the detection probe target hybridising sequence consists of the sequence selected from the group consisting of: SEQ ID NOs:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(55) Suitably, the detection probe oligomer further comprises a 2' methoxy modification on at least

one nucleotide residue member of the nucleotide sequence.

(56) In some embodiments of a method comprising the use of a nucleic-acid-base detection assay, a non-amplification-based assay is used to detect *Babesia* sp. In some such embodiments, the non-amplification-based assay is a hybridization assay comprising the hybridization of a specific detection probe to a target nucleic acid. Methods for conducting polynucleotide hybridization assays have been well developed in the art. Hybridization assay procedures and conditions will vary depending on the application and are selected in accordance with the general binding methods known, including those referred to in, e.g., Maniatis et al, Molecular Cloning: A Laboratory Manual (3rd ed. Cold Spring Harbor, N.Y., 2002), and Berger and Kimmel, Methods in Enzymology, Vol. 152, Guide to Molecular Cloning Techniques (Academic Press, Inc., San Diego, Calif., 1987). Generally, the probe and sample are mixed under conditions that will permit specific nucleic acid hybridization, and specific hybridization of the probe to its respective target is then detected. Nucleic acid hybridization is adaptable to a variety of assay formats. One suitable format is the sandwich assay format, which is particularly adaptable to hybridization under non-denaturing conditions. A primary component of a sandwich-type assay is a solid support, which has adsorbed to it or covalently coupled to it immobilized nucleic acid probe that is unlabeled and complementary to one portion of the DNA sequence. Target nucleic acid is hybridized to the immobilized probe, and a second, labeled detection probe—which is complementary to a second and different region of the same DNA strand to which the immobilized, unlabeled nucleic acid probe is hybridized—is hybridized to the [target nucleic acid]:[immobilized probe] duplex to detect the target nucleic acid. Another exemplary format utilizes electrochemical detection of target nucleic acids hybridized to unlabeled detection probes immobilized on a suitable electrode surface as a signal transducer. See, e.g., Drummond et al., Nat. Biotechnol. 21: 1192, 2003; Gooding, Electroanalysis 14: 1149, 2002; Wang, Anal. Chim Acta 469:63, 2002; Cagnin et al., Sensors 9:3122, 2009; Katz and Willner, Electroanalysis 15:913, 2003; Daniels and Pourmand, Electroanalysis 19: 1239, 2007.

(57) In certain embodiments comprising a hybridization assay, a detection probe is utilized for the detection of a *Babesia* sp. In such embodiments, a detection probe oligomer for detecting *Babesia* sp. comprises a target-hybridizing sequence that is from about 14 to about 40 nucleotides in length and is configured to specifically hybridize to a target sequence contained in SEQ ID NO:59, the RNA equivalent of SEQ ID NO:59, the complement of SEQ ID NO:59, the RNA equivalent of the complement of SEQ ID NO:59, SEQ ID NO:65, the DNA equivalent of SEQ ID NO:65, the complement of SEQ ID NO:65, or the DNA equivalent of the complement of SEQ ID NO:65.

(58) Suitably, the detection probe target-hybridizing sequence is contained in the sequence of SEQ ID NO:59 and includes at least the sequence of SEQ ID NO:42, 92, 94, or 99.

(59) Suitably, the detection probe target-hybridizing sequence contains the sequence of SEQ ID NO:59 and includes at least the sequence of SEQ ID NO:37, 42, or 99.

(60) Suitably, the detection probe target-hybridizing sequence is contained in the sequence of SEQ ID NO:65 and includes at least the sequence of SEQ ID NO:59, 94 or 99.

(61) Suitably, the detection probe oligomer comprises a nucleotide sequence that is from 16 to 25 contiguous nucleotides in length and specifically hybridizes to SEQ ID NO:65, or the DNA equivalent thereof; or specifically hybridizes to the complement of SEQ ID NO:65, or the DNA equivalent thereof.

(62) Suitably, the detection probe oligomer sequence further comprises a nucleotide sequence containing SEQ ID NO:59, 94, or 99.

(63) Suitably, the detection probe target hybridising sequence consists of the sequence selected from the group consisting of: SEQ ID NO:37, 38, 39, 40, 41, 42, 59, 60, 91, 92, 93, 94, 98 or 99.

(64) Suitably, the detection probe oligomer further comprises a 2' methoxy modification on at least one nucleotide residue member of the nucleotide sequence.

(65) In some embodiments, a non-amplification-based assay for detection of *Babesia* sp. is a

cleavage-based assay, in which a probe oligonucleotide containing a non-target-hybridizing flap region is cleaved in an overlap-dependent manner by a flap endonuclease to release a cleavage product that is then detected. Exemplary cleavage-based assay reagents are described in, e.g., Lyamichev et al. (Nat. Biotechnol. 17:292-296, 1999), Ryan et al. (Mol. Diagn. 4: 135-144, 1999), and Allawi et al. (J. Clin. Microbiol. 44:3443-3447, 2006).

(66) Appropriate conditions for flap endonuclease reactions are either known or can be readily determined using methods known in the art (see, e.g., Kaiser et al, J. Biol. Chem. 274:2138-721394, 1999). Exemplary flap endonucleases that may be used in the method include *Thermus aquaticus* DNA polymerase I, *Thermus thermophilus* DNA polymerase I, mammalian FEN-1, *Archaeoglobus fulgidus* FEN-1, *Methanococcus jannaschii* FEN-1, *Pyrococcus fiiriosus* FEN-1, *Methanobacterium thermoautotrophicum* FEN-1, *Thermus thermophilus* FEN-1, CLEAVASE® (Hologic, Inc., Madison, WI), *S. cerevisiae* RTH1, *S. cerevisiae* RAD27, *Schizosaccharomyces pombe* rad2, bacteriophage T5 5'-3' exonuclease, *Pyrococcus horikoshii* FEN-1, human endonuclease 1, calf thymus 5'-3' exonuclease, including homologs thereof in eubacteria, eukaryotes, and archaea, such as members of the class II family of structure-specific enzymes, as well as enzymatically active mutants or variants thereof. Descriptions of flap endonucleases can be found in, for example, Lyamichev et al., Science 260:778-783, 1993; Eis et al, Nat. Biotechnol. 19:673-676, 2001; Shen et al, Trends in Bio. Sci. 23: 171-173, 1998; Kaiser et al, J. Biol. Chem. 274:21387-21394, 1999; Ma et al, J. Biol. Chem. 275:24693-24700, 2000; Allawi et al, J. Mol. Biol. 328:537-554, 2003; Sharma et al, J. Biol. Chem. 278:23487-23496, 2003; and Feng et al, Nat. Struct. Mol. Biol. 11:450-456, 2004.

(67) In certain variations, a cleavage-based assay detects an RNA target nucleic acid of *Babesia* sp., and the cleavage-based assay utilizes a flap endonuclease that is capable of cleaving and RNA:DNA linear duplex structure. In some alternative embodiments, a cleavage-based assay detects a DNA target nucleic acid of *Babesia* sp., and the cleavage-based assay utilizes a flap endonuclease that is capable of cleaving and DNA:DNA linear duplex structure. Exemplary flap endonucleases capable of cleaving RNA:DNA duplexes include polymerase-deficient 5' nucleases of the genus *Thermus* as well as certain CLEAVASE® enzymes (Hologic, Inc., Madison, WI) such as, for example, CLEAVASE® BN (BstX-NotI deletion of Taq polymerase, see U.S. Pat. No. 5,614,402), CLEAVASE® II ("AG" mutant of full length Taq polymerase, see U.S. Pat. No. 5,614,402), CLEAVASE® VII (synthesis-deficient mutation of full length *Thermus thermophilus* polymerase), CLEAVASE® IX (polymerase deficient mutant of the Tth DNA polymerase), and CLEAVASE® XII (polymerase deficient chimeric polymerase constructed from fragments of taq DNA polymerase and Tth DNA polymerase). Exemplary flap endonucleases capable of cleaving DNA:DNA duplexes include the flap endonucleases indicated above, as well as CLEAVASE® 2.0 (*Archaeoglobus fulgidus* FEN-1), CLEAVASE® 2.1 (*Archaeoglobus fulgidus* FEN-1 with 6 histidines on the C-terminus), CLEAVASE® 3.0 (*Archaeoglobus veneficus* FEN-1), and CLEAVASE® 3.1 (*Archaeoglobus veneficus* FEN-1 with 6 histidines on the C-terminus).

(68) In some embodiments, a cleavage-based assay detects an RNA target nucleic acid of *Babesia* sp., and the assay includes a step for synthesizing a DNA complement of an RNA target region, which cDNA strand is then hybridized to overlapping first and second probe oligonucleotides to form a linear duplex cleavage structure for cleavage by the flap endonuclease. Reaction conditions for synthesizing cDNA from an RNA template, using an RNA-dependent DNA polymerase (reverse transcriptase), are well-known in the art.

(69) In certain embodiments utilizing a nucleic-acid-based detection assay, the method further includes purifying the *Babesia* sp. target nucleic acid from other components in the sample. Such purification may include methods of separating and/or concentrating organisms contained in a sample from other sample components. In particular embodiments, purifying the target nucleic acid includes capturing the target nucleic acid to specifically or non-specifically separate the target nucleic acid from other sample components. Non-specific target capture methods may involve

selective precipitation of nucleic acids from a substantially aqueous mixture, adherence of nucleic acids to a support that is washed to remove other sample components, or other means of physically separating nucleic acids from a mixture that contains *Babesia* sp. nucleic acid and other sample components.

(70) In some embodiments, a target nucleic acid of *Babesia* sp. is separated from other sample components by hybridizing the target nucleic acid to a capture probe oligomer. The capture probe oligomer comprises a target-hybridizing sequence configured to specifically or non-specifically hybridize to a target nucleic acid so as to form a [target nucleic acid]:[capture probe] complex that is separated from other sample components. Capture probes comprising target-hybridizing sequences suitable for non-specific capture of target nucleic acids are described in, e.g., WO 2008/016988. In some specific variations comprising target-hybridizing sequence(s) configured to specifically hybridize to a *Babesia* sp. target nucleic acid, a *Babesia*-specific capture probe comprises a target-hybridizing sequence that (i) is from about 15 to about 21 contiguous nucleotides contained in the sequence of SEQ ID NO:78, or (ii) is about 21 to 30 contiguous nucleotides comprising the sequence of SEQ ID NO:78; or (iii) the sequence consists of SEQ ID NO:44. In a preferred variation, the capture probe binds the [target nucleic acid]:[capture probe] complex to an immobilized probe to form a [target nucleic acid]:[capture probe]:[immobilized probe] complex that is separated from the sample and, optionally, washed to remove non-target sample components (see, e.g., U.S. Pat. Nos. 6,110,678; 6,280,952; and 6,534,273). In such variations, the capture probe oligomer further comprises a sequence or moiety that binds the capture probe, with its bound target sequence, to an immobilized probe attached to a solid support, thereby permitting the hybridized target nucleic acid to be separated from other sample components.

(71) In more specific embodiments, the capture probe oligomer includes a tail portion (e.g., a 3' tail) that is not complementary to target nucleic acid but that specifically hybridizes to a sequence on the immobilized probe, thereby serving as the moiety allowing the target nucleic acid to be separated from other sample components, such as previously described in, e.g., U.S. Pat. No. 6,110,678. Any sequence may be used in a tail region, which is generally about 5 to 50 nt long, and preferred embodiments include a substantially homopolymeric tail of about 10 to 40 nt (e.g., A10 to A40), more preferably about 14 to 33 nt (e.g., A14 to A30 or T3A14 to T3A30), that bind to a complementary immobilized sequence (e.g., poly-T) attached to a solid support, e.g., a matrix or particle. In some such embodiments comprising target-hybridizing sequence(s) configured to specifically hybridize to *Babesia* sp. target nucleic acid, a *Babesia*-specific capture probe comprises or consists of the nucleotide sequence of SEQ ID NO:43.

(72) Target capture typically occurs in a solution phase mixture that contains one or more capture probe oligomers that hybridize to the target nucleic acid under hybridizing conditions, usually at a temperature higher than the T_m of the [tail sequence] [immobilized probe sequence] duplex. For embodiments comprising a capture probe tail, the [target nucleic acid]:[capture probe] complex is captured by adjusting the hybridization conditions so that the capture probe tail hybridizes to the immobilized probe, and the entire complex on the solid support is then separated from other sample components. The support with the attached [immobilized probe]:[capture probe]:[target nucleic acid] may be washed one or more times to further remove other sample components. Preferred embodiments use a particulate solid support, such as paramagnetic beads, so that particles with the attached [target nucleic acid]:[capture probe]:[immobilized probe] complex may be suspended in a washing solution and retrieved from the washing solution, preferably by using magnetic attraction. In embodiments of the method comprising the use of an amplification-based detection assay, to limit the number of handling steps, a target nucleic acid may be amplified by simply mixing the target nucleic acid in the complex on the support with amplification oligomers and proceeding with amplification steps.

(73) In some embodiments of a method for diagnosing Babesiosis, where detection of *Babesia* sp.

indicates Babesiosis in a subject, the method further includes treating Babesiosis in the subject. Treatment regimens for Babesiosis are generally known in the art and include, for example, administration of anti-parasitic medications or red blood cell exchange transfusion as an adjunct therapy. In certain variations, the subject has not been previously diagnosed with Babesiosis. In other embodiments, the subject has been previously diagnosed with Babesiosis and is undergoing treatment for Babesiosis at the time a diagnostic method of the present disclosure is performed. Such variations are particularly useful for monitoring treatment of Babesiosis in a subject. For example, if the method indicates that Babesiosis is still present in the subject, then the subject may continue treatment. In some embodiments, the same treatment regime (i.e., the same treatment that the subject is undergoing at the time the present diagnostic method is performed) is re-administered to the subject. Alternatively, the continued presence of Babesiosis in the subject undergoing treatment may indicate that a change in the ongoing treatment is needed, and a different treatment regime (e.g., a different medication, or an increased dosage and/or frequency of a drug) is administered to the subject.

(74) In accordance with the present disclosure, detecting the presence or absence of *Babesia* sp. may be performed separately (e.g., in a separate reaction vessel), or performed together with another assay as a multiplex reaction system. Accordingly, in some embodiments, a method as described herein (e.g., a method for diagnosing Babesiosis) utilizes a multiplex reaction, where the reaction mix contains reagents for assaying multiple (e.g., at least two, three, four, or more) different target sequences in parallel. In these cases, a reaction mix may contain multiple different target-specific oligonucleotides for performing the detection assay. For example, in a method utilizing an amplification-based detection assay, a multiplex reaction may contain multiple sets (e.g., multiple pairs) of amplification oligomers (for example, multiple pairs of PCR primers or multiple pairs of TMA amplification oligomers (e.g., for TMA, multiple pairs of promoter primer and non-promoter primer, or multiple pairs of promoter primer and non-promoter primer)). In other embodiments utilizing a cleavage-based detection assay, a multiplex reaction may contain multiple probe oligonucleotides having different flaps, multiple different overlapping probe oligonucleotides, and multiple different FRET cassettes for detecting the different flaps, once they are cleaved.

(75) The oligomer combination described herein may be in the form of a reaction mixture or a kit comprising the oligomers. The reaction mixture or kit may further include a number of optional components such as, for example, capture probe nucleic acids or arrays of capture probe nucleic acids. For an amplification reaction mixture, the reaction mixture will typically include other reagents suitable for performing in vitro amplification such as, e.g., buffers, salt solutions, appropriate nucleotide triphosphates (e.g., dATP, dCTP, dGTP, dTTP, ATP, CTP, GTP and UTP), and/or enzymes (e.g., reverse transcriptase, and/or RNA polymerase), and will typically include test sample components, in which a *Babesia* sp. target nucleic acid may or may not be present. A kit comprising an oligomer combination for amplification of *Babesia* sp. may also include other reagents suitable for performing in vitro amplification such as, e.g., buffers, salt solutions, appropriate nucleotide triphosphates (e.g., dATP, dCTP, dGTP, dTTP, ATP, CTP, GTP and UTP), and/or enzymes (e.g., reverse transcriptase, and/or RNA polymerase). For an oligomer combination (e.g., reaction mixture or kit) that includes a detection probe together with an amplification oligomer combination targeting a common target nucleic acid, selection of amplification oligomers and detection probe oligomers are linked by a common target region (i.e., the combination will include a probe that binds to a sequence amplifiable by the amplification oligomer combination).

(76) The compositions, methods, reaction mixtures, systems, kits and the like for detection of *Babesia* nucleic acids are further illustrated by the following non-limiting examples.

EXAMPLES

(77) “Sample Transport Solution” generally refers to a solution formulated to preserve a sample, and in some instances formulated to at least partially lyse one or more cell types in a sample. One

exemplary sample transport solution comprises 15 mM sodium phosphate monobasic, 15 mM sodium phosphate dibasic, 1 mM EDTA, 1 mM EGTA, and 110 mM lithium lauryl sulfate (LLS), at pH 6.7. Another exemplary sample transport solution comprises an aqueous solution of 100 mM TRIS, 30 mM magnesium chloride, and 6% (v/v) LLS, at pH 7.5. A further exemplary sample transport solution comprises an aqueous solution of 14 mM sodium bicarbonate, 250 mM ammonium chloride, 5% (v/v) LLS, and 0.1 mM EDTA, at a pH of 7.4. Other formulations of sample transport solutions may function equally well.

(78) “Target Capture Reagent” generally refers to a solution containing a number of components that facilitate capture of a nucleic acid from a solution. One exemplary Target Capture Reagent comprises 250 mM HEPES, 310 mM lithium hydroxide, 1.88 M lithium chloride, 100 mM EDTA, at pH 6.4, and 250 µg/ml of magnetic particles (1 micron SERA-MAG[®] MG-CM particles, GE Healthcare Lifesciences) with dT.sub.14 oligomers covalently bound thereto. Another exemplary Target Capture Reagent comprises 790 mM HEPES, 453 mM lithium hydroxide, 10% w/v LLS, 230 mM Succinic Acid, 0.03% w/v Foam Ban MS-575, and 0.0125% w/v of magnetic particles (1 micron SERA-MAG[®] TM MG-CM particles, GE Healthcare Lifesciences) with dT.sub.14 oligomers covalently bound. Other formulations of Target Capture Reagent may function equally as well.

(79) “Wash Solution” generally refers to a solution containing 10 mM HEPES, 150 mM sodium chloride, 6.5 mM sodium hydroxide, 1 mM EDTA, 0.3% (v/v) ethanol, 0.02% (w/v) methyl paraben, 0.01% (w/v) propyl paraben, and 0.1% (w/v) sodium lauryl sulfate, at pH 7.5.

(80) “Probe Reagent” generally refers to a solution containing one or more labeled detection probes. One exemplary Probe Reagent is a solution made up of from about 75 to about 100 mM lithium succinate, 2% (w/v) LLS, 15 mM mercaptoethanesulfonate, 1.2 M lithium chloride, 20 mM EDTA, and 3% (v/v) ethanol, at pH 4.7. Another exemplary Probe Reagent is a solution made up of from about 75 to about 100 mM succinic acid, 3.5% (w/v) LLS, 75 mM lithium hydroxide, 15 mM aldrithiol-2, 1.0 M lithium chloride, 1 mM EDTA, and 3.0% (v/v) ethanol, at pH 4.1-4.3. Other formulations may perform equally as well.

(81) “Amplification Reagent” generally refers to a concentrated mixture of reaction components to facilitate amplification reactions. An Amplification Reagent will comprise a number of different reagents at various concentrations depending on factors such as for example amplification type (PCR, TMA, etc.), target nucleic acids (GC content), and the like. One exemplary Amplification Reagent comprises 47.6 mM Na-HEPES, 12.5 mM N-acetyl-L-cysteine, 2.5% TRITON[™] X-100, 54.8 mM KCl, 23 mM MgCl₂, 3 mM NaOH, 0.35 mM of each dNTP (dATP, dCTP, dGTP, dTTP), 7.06 mM rATP, 1.35 mM rCTP, 1.35 mM UTP, 8.85 mM rGTP, 0.26 mM Na₂EDTA, 5% v/v glycerol, 2.9% trehalose, 0.225% ethanol, 0.075% methylparaben, 0.015% propylparaben, and 0.002% Phenol Red, at pH 7.5-7.6. Another exemplary Amplification Reagent comprises 19.1 mM Trizma Base, 7.5 mM Trizma Hydrochloride, 23.3 mM KCl, 21.5 mM MgCl₂, 1 mM of each dNTP (dATP, dCTP, dGTP, dTTP), 6.5 mM rATP, 4.0 mM rCTP, 4.0 mM UTP, 6.5 mM rGTP, 3.33% v/v glycerol, 0.05 mM Zinc Acetate, 6 ppm Pro Clin 300 preservative, at pH 8.25-8.45. Other formulations of amplification reagent may function equally well. Primers may be added to the amplification reagent or added to amplification reactions separate from the amplification reagent. Enzymes in an amplification reagent can include one or more of Moloney Murine Leukemia Virus Reverse Transcriptase (MMLV-RT) and bacteriophage T7 RNA polymerase for which units are functionally defined as: 1 U of MMLV-RT incorporates 1 nmol of dTTP in 10 min at 37 C using 200-400 micromolar oligo dT-primed poly(A) as template, and 1 U of T7 RNA polymerase incorporates 1 nmol of ATP into RNA in 1 hr at 37 C using a DNA template containing a T7 promoter.

(82) “Hybridization Reagent” generally refers to a solution made up of reagents having concentrations in the range of about: 75-100 mM succinic acid, 2%-3.5% (w/v) LLS, 75-100 mM lithium hydroxide, 14-16 mM aldrithiol-2, 1.0-1.2 M lithium chloride, 20-1000 mM EDTA, and

2.0-4.0% (v/v) ethanol, at pH 4-5 Other formulations for a Hybridization Reagent may function equally well.

(83) “Selection Reagent” generally refers to a solution containing 600 mM boric acid, 182.5 mM sodium hydroxide, 1% (v/v) octoxynol (TRITON® X-100), at pH 8.5.

(84) “Detection Reagents” include “Detect Reagent I,” which generally refers to a solution containing 1 mM nitric acid and 32 mM hydrogen peroxide, and “Detect Reagent II,” which generally refers to a solution of 1.5 M sodium hydroxide.

Example 1: Initial Oligo Screening

(85) Objective:

(86) Non-T7 and T7 primers and probes were screened using the manual Procleix Enhanced Semi-automated System (eSAS) for transcription mediated amplification (TMA) and hybridization protection assays (HPA) in order to identify primer-probe combinations with the ability to amplify and specifically detect specifically species of *Babesia*—including *B. microti*, *B. divergens*, *B. duncani* and *B. venatorum*. The assay does not discriminate between these *Babesia* species. Reactive results indicate that a sample is positive for *Babesia*.

(87) Materials Methods:

(88) Initial primer screening was performed using TMA on the manual Procleix system using in-vitro transcripts (IVT) for *Babesia microti* (SEQ ID NO:61), *Babesia divergens* (SEQ ID NO:62), and *Babesia duncani* (SEQ ID NO:63). Reactions for this testing began at the amplification step. An assay rack consisted of 10 rows of Ten-tube units (TTUs). 75 microliters of Amplification Reagent and 10 picomoles each of one T7 promoter provider oligonucleotide from Tables 1-3 and one non-T7 primer oligonucleotide from Tables 1-3 were added to the appropriate tubes on the rack such that each combination of amplification oligomers were tested with two replicates each of *Plasmodium falciparum* IVT (SEQ ID NO:64) at 10,000 copies per reaction and *B. divergens*, *B. duncani*, and *B. microti* IVT at 15 copies per reaction. *P. falciparum* was included in initial screening as a cross reactivity specimen due to the conserved regions between *Babesia* and *Plasmodium*. *It is necessary to determine that amplification and detection systems are specific to Babesia.* To achieve the target copies per reaction, 10 µL of *P. falciparum* IVT at 1e6 c/mL diluted in a buffer was spiked into the appropriate tubes, and 10 µL of *Babesia* species IVT at 1,500 c/mL diluted in a buffer were spiked into the appropriate tubes. Various combinations of Primers were tested. This set-up allows for 10 primer combinations to be tested per rack. Once the primer combinations and IVTs were spiked, 200 µL of oil was added to each tube and then the rack was covered with sealing cards and vortexed for a minimum of 20 seconds.

(89) The rack was then incubated in a water bath at 60±1° C. for 10±1 minutes followed by incubation in a 41.5±1° C. water bath for between 9 and 20 minutes. While the rack remained in the water bath, the sealing cards were removed and 25 µL of commercially available Procleix Ultrio Plus enzyme reagent (Grifols Diagnostics Solutions, Inc.) was added to each reaction tube and then covered again with sealing cards. The rack was gently shaken to mix and then covered again with sealing cards and incubated for another 60±5 minutes in the 41.5±1° C. water bath.

(90) After incubation completed, the rack was transferred to the hybridization protection assay (HPA) area where the sealing cards were removed. 100 µL of Probe reagent consisting of 1 acridinium-ester (AE) labeled probe (Tables 1-3) added at a total desired concentration of 5e6 Relative Light Units (RLU) per reaction to a Hybridization Reagent. Probe reagent was then added to the appropriate reaction tubes. The tubes were covered with sealing cards and the rack was vortexed for a minimum of 20 seconds after which the rack was incubated in a water bath at 61±2° C. for 15±1 minutes.

(91) The rack was removed from the water bath, the sealing cards removed, and 250 µL of commercially available Procleix Ultrio Plus selection reagent (Grifols Diagnostics Solutions, Inc.) was added to each tube. The tubes were covered with sealing cards and vortexed for a minimum of 20 seconds and then returned to the 61±2° C. water bath and incubated for 10±1 minutes. After

incubation the rack was allowed to cool in a 23±4° C. water bath for a minimum of 10 minutes.

(92) For detection the TTUs are removed from the rack and loaded on to the automated Leader instrument for subsequent light off using commercially available Procleix Auto Detect 1 and 2 reagents (Grifols Diagnostics Solutions, Inc.) and the results were exported for analysis of the signal in Relative Light Units (RLU).

(93) Primers screened in Group 1a and Group 1b (Table 1) were each of SEQ ID NO:3 and SEQ ID NO:5 promoter provider oligomers paired with each of SEQ ID NOs:12 to 16, 18 and 19 to 21 using both detection probe SEQ ID NO:37 and 39. Each rack used system SEQ ID NO:1, 11 and 37 as a control.

(94) TABLE-US-00001 TABLE 1 Primers screened in Group 1a and Group 1b.
SEQ ID NO Sequence 5'-3' 1 aatttaatacgcactcactatagggagattcacctctga cagttaaatacga 3
aatttaatacgcactcactatagggagaacagttaaata cgaatgccccca 5 aatttaatacgcactcactatagggagattcacctctga
cagttaaatac 12 actacagcatggaataatga 11 cttgaatactacagcatgga 13 cttgaatactacagcatggaataa 14
acttcagcatggaataatga 15 cttgaatacttcagcatgga 16 actncagcatggaataatga (n is inosine in this
example) 18 cttgaatactncagcatgga (n is inosine in this example) 20 actttgagaaaactagagt
21 agaaaactagagtgtttcaa 39 aguaaugguuaauaggagca 37 ugaaguaggacuuugguucu

(95) Primers Screened in Group 2 (Table 2) were SEQ ID NO:3 promoter provider paired with each of SEQ ID NOs:22-31, 75 and 76 primers using SEQ ID NO:39 detection probe. Each rack used system SEQ ID NO:3, SEQ ID NO:21 and SEQ ID NO:39 as a control.

(96) TABLE-US-00002 TABLE 2 Primers screened in Group 2. SEQ ID NO:
Sequence 5'-3' 3 aatttaatacgcactcactatagggagaa cagttaaatacgaatgccccca 21 agaaaactagagtgtttcaa
24 gggtctattttgttggtt 25 tgggtctattttgttg 23 aggactttggttctattttg 75 gaagtaggactttggttctattt 76
atgaagtaggactttggttct 22 ataatgaagtaggactttg 26 ggaataatgaagtaggacttt 27 atggaataatgaagtaggac 28
atggaataatgaagtagg 29 gcatggaataatgaagtag 30 tacagcatggaataatgaag 31 tactacagcatggaataatg 39
aguaaugguuaauaggagca

(97) Primers screened in Group 3 (Table 3) were SEQ ID NO:1 promoter provider paired with each of SEQ ID NO:21, 27 and 29 primers and SEQ ID NO:38, 40 and 41 detection probes; SEQ ID NO:3 promoter provider paired with each of SEQ ID NO:21, 27 and 29 primers and SEQ ID NO:38 and 40 detection probes; and each of SEQ ID NO:3, 7 and 9 promoter provider paired with SEQ ID NO:21 primer and SEQ ID NO:39 detection probe.

(98) TABLE-US-00003 TABLE 3 Primers screened in Group 3. SEQ ID NO:
Sequence 5'-3' 3 aatttaatacgcactcactatagggagaacagt taaatacgaatgccccca 1
aatttaatacgcactcactatagggagattcac ctctgacagttaaatacga 7 aatttaatacgcactcactatagggagagcttt
cgtagtagttcgtctttaacaatc 9 aatttaatacgcactcactatagggagactttc gcagtagttcgtctttaac 21
agaaaactagagtgtttcaa 27 atggaataatgaagtaggac 29 gcatggaataatgaagtag 39 aguaaugguuaauaggagca
38 uaaugguuaauaggagcaguug 40 ggacuuugguucuaauuuuguugg 41 aaugguuaauaggagcaguuggggg
Results:

(99) Ideal candidates from screening were expected to be reactive for *Babesia* species IVTs (*B. microti*, *B. divergens* and *B. duncani*) with analyte signals consistently at 1,000,000 RLU or higher for all replicates and negative for *P. falciparum* IVT and negative samples with analyte signals below 10,000 RLU. Some candidates with RLUs near 1,000,000 RLU for *Babesia* and below 30,000 RLU for *P. falciparum* and negatives were also given consideration.

(100) For the primers and detection probes screened in Group 1a, not all primer-probe combinations consistently amplified and detected all species and some combinations resulted in high analyte signals for negative specimens (*P. falciparum* and Negative). Analyte RLU results for this group are listed in Table 4. Preferred candidates in this group were SEQ ID NO:3 promoter provider paired with SEQ ID NO:13, 20 or 21 primers and using SEQ ID NO:37 detection probe; and SEQ ID NO:3 promoter provider paired with SEQ ID NO:21 primer and SEQ ID NO:39 detection probe.

(101) TABLE-US-00004 TABLE 4 Analyte RLU Results for Group 1a. SEQ ID NO: 1 SEQ ID

NO: 3 SEQ ID NO: 11 SEQ ID NO: 12 SEQ ID NO: 11 A B C D E F 10 SEQ ID 1,362 1,558 7,937 4,149 1,858 2,602 9 NO: 39 1,751 1,944 217,421 2,083,520 624,278 1,234,110 8 1,868,814 2,073,045 1,910,998 2,237,157 1,158,006 2,142,969 7 2,296,815 2,273,471 2,348,718 2,365,590 2,259,692 2,374,484 6 5,122 3,273 4,390 5,389 4,890 3,165 5 SEQ ID 15,427 2,286 13,870 6,363 123,550 4,161 4 NO: 37 8,560 739,538 1,276,471 1,699,429 1,883,483 1,948,712 3 1,915,070 1,982,035 2,030,041 2,073,715 1,847,501 2,010,877 2 1,556,779 2,027,292 2,080,918 2,033,082 1,935,354 2,117,207 1 2,750 3,797 2,020 160,345 4,308 10,144 SEQ ID NO: 3 SEQ ID NO: 13 SEQ ID NO: 14 G H I J 10 SEQ ID 2,379 1,923 3,206 3,223 P. fal 10K c/rxn 9 NO: 39 1,512,725 1,924,925 2,028,160 1,717,151 B. div IVT 15 c/rxn 8 1,262,860 2,023,796 957,114 180,559 B. dun IVT 15 c/rxn 7 2,310,437 2,287,714 2,363,103 2,306,377 B. mic IVT 15 c/rxn 6 3,087 4,362 3,815 2,678 Neg 5 SEQ ID 5,645 2,157 21,188 10,899 P. fal 10K c/rxn 4 NO: 37 1,753,900 1,907,723 1,838,478 1,913,729 B. div IVT 15 c/rxn 3 1,929,009 1,972,774 2,050,677 1,981,200 B. dun IVT 15 c/rxn 2 2,092,557 2,077,799 1,985,182 1,993,369 B. mic IVT 15 c/rxn 1 1,727 2,368 2,807 5,288 Neg SEQ ID NO: 3 SEQ ID NO: 15 SEQ ID NO: 16 SEQ ID NO: 18 A B C D E F 10 SEQ ID 3,873 2,436 5,785 2,743 2,406 1,968 9 NO: 39 1,918,232 1,880,797 1,368,977 2,094,325 1,486,108 2,119,826 8 1,602,380 1,776,094 2,135,880 2,225,590 2,031,084 2,095,064 7 375,283 981,890 2,375,662 2,447,002 2,298,733 2,218,629 6 5,902 6,937 7,033 7,979 2,758 3,564 5 SEQ ID 7,609 150,048 314,165 25,323 1,726,513 1,674 4 NO: 37 1,942,612 1,891,974 1,684,635 1,968,192 1,669,451 1,881,606 3 1,966,629 1,993,188 2,063,195 2,071,412 1,901,864 1,960,251 2 829,468 1,477,651 2,146,350 2,147,242 2,049,500 2,063,691 1 1,553 4,723 190,368 12,971 1,949 1,627 SEQ ID NO: 3 SEQ ID NO: 20 SEQ ID NO: 21 G H I J 10 SEQ ID 1,535 2,439 2,493 2,249 P. fal 10K c/rxn 9 NO: 39 1,912,598 882,449 1,936,565 2,170,010 B. div IVT 15 c/rxn 8 1,893,023 1,800,511 2,240,797 2,087,446 B. dun IVT 15 c/rxn 7 2,463,996 2,234,843 2,323,282 2,352,415 B. mic IVT 15 c/rxn 6 2,246 1,866 2,602 1,792 Neg 5 SEQ ID 1,777 1,730 1,460 5,274 P. fal 10K c/rxn 4 NO: 37 1,842,523 1,892,595 1,944,571 1,882,381 B. div IVT 15 c/rxn 3 1,996,707 1,995,486 2,032,943 1,995,159 B. dun IVT 15 c/rxn 2 2,057,722 2,091,173 2,110,461 2,140,647 B. mic IVT 15 c/rxn 1 7,879 1,642 1,779 1,602 Neg

(102) For the primers and detection probes screened in Group 1b, not all primer-probe combinations consistently amplified and detected all species and some combinations resulted in high analyte signals for *P. falciparum*. Analyte RLU results for this group are listed in Table 5. Preferred candidates in this group were SEQ ID NO:5 promoter provider paired with SEQ ID NO:11 or 13 primers and using the SEQ ID NO:37 detection probe.

(103) TABLE-US-00005 TABLE 5 Analyte RLU Results for Group 1b. SEQ ID NO: 1 SEQ ID NO: 5 SEQ ID NO: 11 SEQ ID NO: 12 SEQ ID NO: 11 A B C D E F 10 SEQ ID 2,370 2,021 491 561 1,219 1,549 9 NO: 39 2,606,527 2,273,506 1,343 180,755 668,964 645,097 8 2,329,021 2,341,992 93,675 141,525 661,755 230,617 7 2,586,212 2,656,633 2,513,992 2,519,282 2,398,111 2,468,394 6 2,786 2,875 946 546 1,278 990 5 SEQ ID 1,086 1,296 994 848 1,488 805 4 NO: 37 1,862,765 1,798,372 392,887 965 1,378,157 1,340,279 3 1,948,808 1,873,185 609,730 1,553,691 1,684,907 1,618,765 2 2,019,896 1,956,929 1,610,113 1,432,157 1,866,395 1,912,090 1 1,670 878 965 2,036 772 1,582 SEQ ID NO: 5 SEQ ID NO: 13 SEQ ID NO: 14 G H I J 10 SEQ ID 4,246 3,048 1,726 2,454 P. fal 10K c/rxn 9 NO: 39 618,531 600,583 145,539 3,298 B. div IVT 15 c/rxn 8 439,103 421,977 7,295 48,670 B. dun IVT 15 c/rxn 7 4,661 2,419,801 43,673 16,001 B. mic IVT 15 c/rxn 6 3,573 3,852 1,243 3,789 Neg 5 SEQ ID 1,538 2,040 636,577 1,472 P. fal 10K c/rxn 4 NO: 37 1,313,625 1,403,578 41,232 1,144,411 B. div IVT 15 c/rxn 3 1,464,595 1,660,711 379,047 252,130 B. dun IVT 15 c/rxn 2 1,827,977 1,878,981 543,761 1,896,462 B. mic IVT 15 c/rxn 1 2,724 1,371 1,544 1,320 Neg SEQ ID NO: 1 SEQ ID NO: 5 SEQ ID NO: 15 SEQ ID NO: 16 SEQ ID NO: 18 A B C D E F 10 SEQ ID 6,408 5,690 1,278 1,722 1,724 2,646 9 NO: 39 144,774 478,638 207,863 4,342 167,329 1,078,348 8 197,399 17,034 260,958 607,257 172,231 934,843 7 40,088 28,810 2,472,660 2,440,145 2,404,314 2,487,783 6 1,795 2,431 1,561 911 1,192 3,614 5 SEQ ID 1,197,623 1,474 939 1,303 958 1,949 4 NO: 37 1,471,176 1,396,263 1,493,545 137,001

1,553,344 1,509,390 3 1,765,530 1,753,318 1,950,334 1,982,086 1,758,977 1,690,431 2 805,769 424,446 2,039,764 1,998,885 1,941,034 1,985,950 1 1,631 1,963 1,292 1,149 1,233 38,660 SEQ ID NO: 5 SEQ ID NO: 20 SEQ ID NO: 21 G H I J 10 SEQ ID 1,419 1,582 703 3,712 P. fal 10K c/rxn 9 NO: 39 401,756 346,658 98,181 219,007 B. div IVT 15 c/rxn 8 220,859 657,682 137,758 124,133 B. dun IVT 15 c/rxn 7 2,416,779 2,391,258 91,104 2,492,731 B. mic IVT 15 c/rxn 6 2,345 2,848 631 659 Neg 5 SEQ ID 1,492 193,967 984 1,949 P. fal 10K c/rxn 4 NO: 37 520,765 929,879 756,537 1,462,395 B. div IVT 15 c/rxn 3 1,647,079 1,357,270 334,747 1,183,608 B. dun IVT 15 c/rxn 2 2,043,012 95,200 1,937,688 1,875,971 B. mic IVT 15 c/rxn 1 956 1,542 1,262 2,454 Neg (104) For the primers and detection probes screened in Group 2, not all primer-probe combinations consistently amplified and detected all species and some combinations resulted in high analyte signals for *P. falciparum*. Analyte RLU results for this group are listed in Table 6. Preferred candidates in this group were SEQ ID NO:3 promoter provider paired with SEQ ID NO:27, SEQ ID NO:28 or SEQ ID NO:29 primers and SEQ ID NO:39 detection probe.

(105) TABLE-US-00006 TABLE 6 Analyte RLU Results for Group 2. Control SEQ ID NO: 3 SEQ ID NO: 39 A B C D E F SEQ ID NO: 21 SEQ ID NO: 24 10 5,538 2,060 3,451 912 1,957 732 9 2,777,487 1,462,676 2,411,623 42,352 261,061 36,259 8 2,090,779 2,104,742 1,619,218 344,375 87,780 46,539 7 2,422,490 2,421,930 1,995,715 392,494 1,060,155 2,468,927 6 4,111 1,487 1,423 616 698 994 SEQ ID NO: 23 SEQ ID NO: 75 5 913 886 1,458 979 613 618 4 722,789 19,023 52,416 13,234 1,273,210 130,514 3 1,099 1,330 667 5,314 2,137 1,401 2 2,228,374 1,820,781 2,162,670 1,959,300 2,524,372 1,916,040 1 1,198 608 878 556 532 1,244 SEQ ID NO: 3 SEQ ID NO: 39 G H I J SEQ ID NO: 25 SEQ ID NO: 31 10 839 344 1,230 3,330 P. fal 10K c/rxn 9 35,686 7,687 5,156 1,589,735 B. div IVT 15 c/rxn 8 16,068 22,627 110,271 1,642,035 B. dun IVT 15 c/rxn 7 473,546 1,621,876 1,933,138 2,150,212 B. mic IVT 15 c/rxn 6 667 484 508 6,630 Neg SEQ ID NO: 76 SEQ ID NO: 31 5 931 6,188 1,313 2,697 P. fal 10K c/rxn 4 901,920 816,015 1,059,461 2,138,285 B. div IVT 15 c/rxn 3 350,464 95,471 40,369 2,468,230 B. dun IVT 15 c/rxn 2 1,887,684 1,511,500 2,263,008 389,108 B. mic IVT 15 c/rxn 1 525 1,912 903 11,272 Neg SEQ ID NO: 3 SEQ ID NO: 39 A B C D E F SEQ ID NO: 22 SEQ ID NO: 26 10 924 972 8,531 5,715 1,627 1,017 9 87,406 60,599 71,115 778,850 371,058 574,028 8 3,156 3,152 5,092 150,582 154,380 75,725 7 2,201,409 2,091,981 2,098,983 2,181,859 2,111,286 2,220,157 6 1,263 1,258 1,013 3,675 1,411 1,513 SEQ ID NO: 28 SEQ ID NO: 29 5 9,482 1,7459 1,830 3,932 4,278 3,258 4 1,169,497 1,283,384 1,523,739 2,167,497 2,112,640 2,151,586 3 1,118,901 584,074 1,104,469 1,847,271 1,481,735 2,000,601 2 2,291,400 2,263,798 2,225,197 2,338,122 2,318,877 2,287,637 1 5,439 3,796 1,551 3,578 8,224 4,044 SEQ ID NO: 3 SEQ ID NO: 39 G H I J SEQ ID NO: 27 SEQ ID NO: 31 10 3,853 3,043 3,064 5,868 P. fal 10K c/rxn 9 2,101,084 1,994,330 1,804,057 2,086,334 B. div IVT 15 c/rxn 8 1,697,144 1,487,220 1,493,662 1,951,476 B. dun IVT 15 c/rxn 7 2,376,261 2,312,253 2,351,514 2,288,840 B. mic IVT 15 c/rxn 6 2,573 7,310 5,289 6,093 Neg SEQ ID NO: 30 SEQ ID NO: 31 5 7,941 10,249 397,486 2,514 P. fal 10K c/rxn 4 2,205,990 88,246 2,231,170 2,099,814 B. div IVT 15 c/rxn 3 2,143,050 2,113,739 2,037,798 1,730,715 B. dun IVT 15 c/rxn 2 2,307,754 2,272,338 2,206,558 2,230,915 B. mic IVT 15 c/rxn 1 14,896 2,781 1,362 1,916 Neg

(106) Analyte RLU results for the primers and detection probes screened in Group 3 are listed in Table 7. Primers paired with detection probes SEQ ID NO:40 and SEQ ID NO:41 yielded high analyte signals for *P. falciparum*, negative samples and *Babesia* positive samples indicating poor design of the probe. Thus, SEQ ID NO:40 and SEQ ID NO:41 and were not included in additional examples. Combinations of SEQ ID NO:3 promoter provider with detection probe SEQ ID NO:38 (regardless of the primer used in this example) yielded high analyte signal in negative samples indicating a probable primer-probe interaction, and thus these oligomers were not included in additional examples. Candidates considered for use with additional detection probes were SEQ ID NO:1 promoter provider with SEQ ID NO:27 or 29 primers and SEQ ID NO:38 detection probe. For new T7 promoter provider screened in this group, SEQ ID NO:3 and 7 paired with SEQ ID

NO:21 primer and SEQ ID NO:39 detection probe were considered. SEQ ID NO:9 promoter provider, SEQ ID NO:21 primer and SEQ ID NO:39 detection probe showed no amplification and detection of *Babesia* targets.

(107) TABLE-US-00007 TABLE 7 Analyte RLU Results For Group 3 SEQ ID NO: 38 SEQ ID NO: 41 SEQ ID NO: 1 SEQ ID NO: 1 SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID NO: 21 NO: 27 NO: 29 NO: 21 NO: 27 NO: 29 Blank A B C D E F G H I J 10 1,183 4,868 2,229 1,999,003 1,973,790 1,943,078 8 8 9 8 P.fal 10K c/rxn 9 273,967 2,140,180 896,930 3,319,370 2,458,639 1,891,471 9 9 8 8 B.div IVT 15 c/rxn 8 2,569,264 2,530,890 2,494,566 3,175,342 2,598,410 2,129,032 8 7 5 8 B.dun IVT 15 c/rxn 7 2,481,447 2,349,620 2,600,164 3,769,511 2,880,852 2,707,608 9 7 8 24 B.mic IVT 15 c/rxn 6 1,792 2,291 4,106 1,373,994 2,056,230 2,277,790 9 5 9 8 Neg 5 2,503 12,777 8,428 2,386,352 1,947,735 2,160,802 10 7 7 10 P.fal 10K c/rxn 4 2,177,455 2,061,853 1,646,064 3,297,211 2,150,824 2,802,834 9 7 7 7 B.div IVT 15 c/rxn 3 2,508,586 2,634,974 2,567,929 3,430,103 2,765,117 3,556,389 8 6 9 8 B.dun IVT 15 c/rxn 2 2,601,774 2,583,708 2,551,550 3,333,080 3,719,802 3,344,092 8 8 9 9 B.mic IVT 15 c/rxn 1 835 2,079 2,581 1,960,393 2,096,053 2,145,235 7 5 9 10 Neg SEQ ID NO: 3 SEQ ID NO: 15 SEQ ID NO: 16 SEQ ID NO: 18 SEQ ID NO: 20 SEQ ID NO: 21 A B C D E F G H I J 10 SEQ ID 3,873 2,436 5,785 2,743 2,406 1,968 1,535 2,439 2,493 2,249 P.fal 10K NO: 39 c/rxn 9 1,918,232 1,880,797 1,368,977 2,094,325 1,486,108 2,119,826 1,912,598 882,449 1,936,565 2,170,010 B.div IVT 15 c/rxn 8 1,602,380 1,776,094 2,135,880 2,225,590 2,031,084 2,095,064 1,893,023 1,800,511 2,240,797 2,087,446 B.dun IVT 15 c/rxn 7 375,283 981,890 2,375,662 2,447,002 2,298,733 2,218,629 2,463,996 2,234,843 2,323,282 2,352,415 B.mic IVT 15 c/rxn 6 5,902 6,937 7,033 7,979 2,758 3,564 2,246 1,866 2,602 1,792 Neg 5 SEQ ID 7,609 150,048 314,165 25,323 1,726,513 1,674 1,777 1,730 1,460 5,274 P.fal 10K NO: 37 c/rxn 4 1,942,612 1,891,974 1,684,635 1,968,192 1,669,451 1,881,606 1,842,528 1,892,595 1,944,571 1,882,381 B.div IVT 15 c/rxn 3 1,966,629 1,993,188 2,063,195 2,071,412 1,901,864 1,960,251 1,996,707 1,995,486 2,032,943 1,995,159 B.dun IVT 15 c/rxn 2 829,468 1,477,651 2,146,350 2,147,242 2,049,500 2,063,691 2,057,722 2,091,173 2,110,461 2,140,647 B.mic IVT 15 c/rxn 1 1,553 4,723 190,368 12,971 1,949 1,627 7,879 1,642 1,779 1,602 Neg

CONCLUSIONS

(108) The results of this initial screening demonstrated that not all primers and probes reliably amplify and detect all *Babesia* species. Some candidates were not specific to *Babesia*, showing some amplification of *P. falciparum*, while other candidates demonstrated false positives, false negatives and/or primer-probe interactions. Initial screening of amplification and detection systems identified several candidates that show sensitive and specific detection of *B. microti*, *B. divergens* and *B. duncani*. These candidates do not cross react with *P. falciparum* further demonstrating specificity of the systems. The sensitive and specific combinations from this example were considered for further sensitivity and specificity evaluation.

Example 2: Secondary Oligo Screening for the *Babesia* Assay

Objective

(109) Candidate amplification combinations identified in Example 1 for the *Babesia* Assay using the manual Procleix Enhanced Semi-automated System (eSAS) were screened on the fully automated Procleix Panther System to determine the best candidates in terms of specificity and sensitivity using *Babesia* Species in-vitro transcripts (IVT).

Materials and Methods

(110) Candidate amplification systems were tested on the automated Procleix Panther system (Grifols Diagnostics Solutions, Inc.). Combinations of amplification and detection oligomers tested are listed in Table 8. Sequences for each oligo are listed in Table 9. A total of 8 conditions were screened. Conditions screened were tested against 45 replicates of a negative panel and 6 replicates

each of diluted *Babesia* in-vitro transcripts (IVT) for *B. microti*, *B. divergens* and *B. duncani* (SEQ ID NOs:61, 62, & 63, respectively) at 30 c/mL. An assay calibrator comprising a *B. microti* IVT panel at 500 c/mL was included to determine the analyte cutoff for the run. The assay software uses the analyte cutoff to determine if samples are reactive or non-reactive. Samples with a signal to cutoff ratio of >1 are considered reactive, while those <1 are non-reactive. Assay Reagents used included the following: a Target Capture Reagent (TCR) comprising of a single Target Capture Oligo (TCO) added at a concentration of 5 pmoles per reaction; an Amplification Reagent comprising one T7 promoter primer and one non-T7 primer each added at a concentration of 10 pmoles per reaction; a Probe reagent consisting of 1 acridinium-ester (AE) labeled probe added at a total concentration of 5e6 Relative Light Units (RLU) per reaction to a solution of; Enzyme Reagent; and Selection Reagent.

(111) A second round of oligo screening was performed to follow up on candidate systems identified in Example 1. Conditions tested in this group are listed in Table 10. Sequences for each oligo are listed in Table 11. A total of 6 additional conditions were screened. Conditions screened were tested against 8 replicates of a negative sample and 7 replicates of diluted *Babesia* in-vitro transcripts (IVT) for each of *B. microti*, *B. divergens* and *B. duncani* at 100, 30, and 10 c/mL. A cross reactivity panel consisting of 8 replicates *P. falciparum* IVT at 1e6 c/mL was also tested to determine if the system is specific to *Babesia* in the presence of *Plasmodium*. An assay calibrator made from a *B. microti* IVT panel at 500 c/mL.

(112) TABLE-US-00008 TABLE 8 Conditions 1-8 Tested on the Panther System Target Capture

Condition #	Oligo	T7 Primer	Non-T7 Primer	Probe
1	SEQ ID NO: 43	SEQ ID NO: 1	SEQ ID NO: 11	SEQ ID NO: 37
2	SEQ ID NO: 43	SEQ ID NO: 1	SEQ ID NO: 27	SEQ ID NO: 39
3	SEQ ID NO: 43	SEQ ID NO: 1	SEQ ID NO: 27	SEQ ID NO: 38
4	SEQ ID NO: 43	SEQ ID NO: 1	SEQ ID NO: 13	SEQ ID NO: 37
5	SEQ ID NO: 43	SEQ ID NO: 3	SEQ ID NO: 21	SEQ ID NO: 37
6	SEQ ID NO: 43	SEQ ID NO: 7	SEQ ID NO: 21	SEQ ID NO: 37
7	SEQ ID NO: 43	SEQ ID NO: 7	SEQ ID NO: 27	SEQ ID NO: 39
8	SEQ ID NO: 43	SEQ ID NO: 5	SEQ ID NO: 21	SEQ ID NO: 37

(113) TABLE-US-00009 TABLE 9 Sequences for Oligos listed in Table 8. Reagent

Short Name	Sequence	5'-3' TCR	SEQ ID NO:
43	uaggccaauacccuaccgucc		
1	aaatataacgactcactata		
3	aaatataacgactcactata		
5	aaatataacgactcactata		
7	aaatataacgactcactata		
13	aaatataacgactcactata		
21	agaaaactagagtgtttcaa		
27	atggaataatgaagtaggac		
37	ugaaguaggacuuugguucu		
39	aguaaugguuaauaggagca		

(114) TABLE-US-00010 TABLE 10 Conditions 9-14 tested on the Panther System Target Capture

Condition	Oligo	T7 Primer	Non T7 Primer	Probe
9	SEQ ID NO: 43	SEQ ID NO: 7	SEQ ID NO: 21	SEQ ID NOs: 37 + 42
10	SEQ ID NO: 43	SEQ ID NO: 7	SEQ ID NO: 32	SEQ ID NOs: 37 + 42
11	SEQ ID NO: 43	SEQ ID NO: 7	SEQ ID NO: 33	SEQ ID NOs: 37 + 42
12	SEQ ID NO: 43	SEQ ID NO: 7	SEQ ID NO: 34	SEQ ID NOs: 37 + 42
13	SEQ ID NO: 43	SEQ ID NO: 7	SEQ ID NO: 35	SEQ ID NOs: 37 + 42
14	SEQ ID NO: 43	SEQ ID NO: 7	SEQ ID NO: 36	SEQ ID NOs: 37 + 42

(115) TABLE-US-00011 TABLE 11 Sequences for Oligos listed in Table 10. Reagent

Short Name	Sequence	5'-3' TCR	SEQ ID NO:
43	uaggccaauacccuaccgucc		
7	aaatataacgactcactata		
21	agaaaactagagtgtttcaa		
32	agaaaactagagtgtttca		
33	agaaaactagagtgtttca		
34	agaaaactagagtgtttca		
35	agaaaactagagtgtttca		
36	agaaaactagagtgtttca		
37	ugaaguaggacuuugguucu		
42	aguaggacuuugguucu		

Results:

(116) An analyte signal cutoff of 100,000 RLU was used to determine reactivity for specimens screened for Conditions 1-8 in Table 12. Samples above this cutoff were considered reactive. Conditions 1 and 4 yielded a false positive rate in negative specimens of 22% and 13% respectively indicating a primer interaction with the probe. Condition 7 was unable to detect *B. divergens* and *B. duncani* IVT at 30 c/mL at 100% reactivity. Due to the high rate of false positives seen in this example using the SEQ ID NO:1 promoter provider, additional screening was performed on alternative candidates.

(117) TABLE-US-00012 TABLE 12 Results for Conditions 1-8 Average StdDev Condition Sample % Analyte Analyte # Description Valid Reactive Reactive RLU RLU 1 *B. microti* Calibrator 3 3 100 2,895,211 95,178 Negative Buffer 45 10 22 83,538 169,164 *B. microti* 30 c/mL 6 6 100 2,785,208 192,438 *B. duncani* 30 c/mL 6 6 100 2,704,843 52,325 *B. divergens* 30 c/mL 5 5 100 2,679,799 131,010 2 *B. microti* Calibrator 3 3 100 2,164,104 85,510 Negative Buffer 45 0 0 1,419 881 *B. microti* 30 c/mL 5 5 100 2,184,976 107,021 *B. duncani* 30 c/mL 5 5 100 1,842,832 79,094 *B. divergens* 30 c/mL 5 5 100 1,710,700 484,713 3 *B. microti* Calibrator 3 3 100 5,202,778 356,597 Negative Buffer 45 0 0% 5,048 2,938 *B. microti* 30 c/mL 6 6 100 5,282,576 336,909 *B. duncani* 30 c/mL 6 6 100 4,987,308 396,702 *B. divergens* 30 c/mL 6 6 100 4,898,995 261,124 4 *B. microti* Calibrator 3 3 100 2,847,426 127,963 Negative Buffer 45 6 13 62,163 212,258 *B. microti* 30 c/mL 6 6 100 2,920,118 186,525 *B. duncani* 30 c/mL 6 6 100 2,637,399 213,549 *B. divergens* 30 c/mL 6 6 100 2,555,922 322,889 5 *B. microti* Calibrator 3 3 100 2,614,010 43,029 Negative Buffer 45 0 0 1,064 604 *B. microti* 30 c/mL 6 6 100 2,562,482 60,848 *B. duncani* 30 c/mL 6 6 100 2,317,458 47,231 *B. divergens* 30 c/mL 6 6 100 2,329,932 33,382 6 *B. microti* Calibrator 3 3 100 2,849,900 59,411 Negative Buffer 45 0 0 1,007 548 *B. microti* 30 c/mL 6 6 100 2,779,409 77,390 *B. duncani* 30 c/mL 6 6 100 2,472,381 110,379 *B. divergens* 30 c/mL 6 6 100 2,608,790 176,482 7 *B. microti* Calibrator 3 3 100 5,706,224 96,503 Negative Buffer 45 0 0 3,062 1,006 *B. microti* 30 c/mL 6 6 100 5,513,204 219,801 *B. duncani* 30 c/mL 6 2 33 124,483 155,936 *B. divergens* 30 c/mL 6 3 50 301,854 494,564 8 *B. microti* Calibrator 3 3 100 2,963,935 70,693 Negative Buffer 45 0 0 932 645 *B. microti* 30 c/mL 6 6 100 2,977,421 189,225 *B. duncani* 30 c/mL 6 6 100 2,678,642 179,586 *B. divergens* 30 c/mL 6 6 100 2,517,233 223,392

(118) In the previous round of screening, several candidates showed good sensitivity and specificity for *Babesia* species IVT. Condition 6 had good sensitivity and specificity with strong analyte signal for the *Babesia* IVT specimens tested and low analyte signal for negative specimens. Additional screening was performed on the Panther system using the SEQ ID NO:7 promoter provider paired with a number of non-T7 primers to determine the performance of these oligomer combinations. Condition 12 had the best performance of the conditions screened in this round. Condition 12 had best performance in terms of percent reactivity for all levels tested for all *Babesia* species IVTs and strongest RLU signal for all levels. Condition 12 also had the lowest analyte signal for negative specimens.

(119) TABLE-US-00013 TABLE 13 Results for Conditions 9-14. Average StdDev Condition Sample % Analyte Analyte # Description Level Valid Reactive Reactive RLU RLU 9 Negative Calibrator 0 3 0 0 3,064 2,870 *B. microti* Calibrator 500 c/mL 3 3 100 2,634,036 20,467 *B. microti* IVT 100 c/mL 7 7 100 1,782,835 124,480 30 c/mL 7 6 86 1,478,196 136,864 10 c/mL 7 5 71 1,504,124 110,673 *B. duncani* IVT 100 c/mL 7 7 100 1,615,981 207,184 30 c/mL 7 7 100 1,046,502 282,167 10 c/mL 7 6 86 817,242 310,430 *B. divergens* IVT 100 c/mL 7 7 100 1,107,329 280,864 30 c/mL 7 7 100 780,602 285,167 10 c/mL 7 3 43 312,826 199,281 Negative IC Buffer 0 8 0 0 921 1,069 *P. falciparum* IVT 1.00e6 c/mL 8 0 0 1,546 1,944 10 Negative Calibrator 0 3 0 0 2,083 1,176 *B. microti* Calibrator 500 c/mL 3 3 100 2,612,367 134,872 *B. microti* IVT 100 c/mL 7 7 100 1,841,300 261,033 30 c/mL 7 6 86 1,329,659 395,229 10 c/mL 7 4 57 708,829 275,561 *B. duncani* IVT 100 c/mL 7 7 100 1,475,991 196,296 30 c/mL 7 7 100 717,189 386,202 10 c/mL 7 4 57 471,736 348,996 *B. divergens* IVT 100 c/mL 7 7 100 910,674 330,110 30 c/mL 7 6 86 565,166

200,817 10 c/mL 7 3 43158,352 37,831 Negative IC Buffer 0 8 0 0 2,802 2,381 *P. falciparum* IVT 1.00e6 c/mL 8 0 0 3,459 3,274 11 Negative Calibrator 0 3 0 0 1,687 1,471 *B. microti* Calibrator 500 c/mL 3 3 100 2,476,171 150,100 *B. microti* IVT 100 c/mL 7 7 100 1,887,931 90,174 30 c/mL 7 7 100 1,042,715 590,288 10 c/mL 7 2 29 961,097 598,572 *B. duncani* IVT 100 c/mL 7 7 100 1,402,621 146,419 30 c/mL 7 7 100 580,415 252,668 10 c/mL 7 4 57 252,969 139,120 *B. divergens* IVT 100 c/mL 7 7 100 903,456 261,139 30 c/mL 7 7 100 484,118 271,417 10 c/mL 7 6 86 283,109 129,194 Negative IC Buffer 0 8 0 0 2,334 1,997 *P. falciparum* IVT 1.00e6 c/mL 8 0 0 2,035 1,681 12 Negative Calibrator 0 3 0 0 370 640 *B. microti* Calibrator 500 c/mL 3 3 100 2,742,289 194,078 *B. microti* IVT 100 c/mL 7 7 100 2,092,032 386,586 30 c/mL 7 7 100 1,568,851 376,918 10 c/mL 7 5 71 1,334,103 542,119 *B. duncani* IVT 100 c/mL 7 7 100 1,867,880 208,460 30 c/mL 7 7 100 1,264,832 533,439 10 c/mL 7 7 100 881,072 287,187 *B. divergens* IVT 100 c/mL 7 7 100 1,456,923 228,759 30 c/mL 7 7 100 1,013,056 162,035 10 c/mL 7 6 86 941,635 247,989 Negative IC Buffer 0 8 0 0 391 944 *P. falciparum* IVT 1.00e6 c/mL 8 0 0 479 744 13 Negative Calibrator 0 3 0 0 18 31 *B. microti* Calibrator 500 c/mL 3 3 100 2,618,344 183,612 *B. microti* IVT 100 c/mL 7 7 100 1,689,812 234,944 30 c/mL 7 7 100 1,295,755 316,505 10 c/mL 7 5 71 494,319 480,634 *B. duncani* IVT 100 c/mL 7 5 71 644,015 376,799 30 c/mL 7 7 100 1,026,959 141,543 10 c/mL 7 7 100 1,808,125 209,910 *B. divergens* IVT 100 c/mL 7 7 100 1,129,487 153,036 30 c/mL 7 7 100 670,066 221,406 10 c/mL 7 6 86 270,616 143,462 Negative IC Buffer 0 8 0 0 400 644 *P. falciparum* IVT 1.00e6 c/mL 8 0 0 1,237 1,803 14 Negative Calibrator 0 3 0 0 673 843 *B. microti* Calibrator 500 c/mL 3 3 100 2,656,960 89,493 *B. microti* IVT 100 c/mL 7 7 100 2,364,847 143,184 30 c/mL 7 6 86 2,051,745 116,278 10 c/mL 7 5 71 1,450,039 626,015 *B. duncani* IVT 100 c/mL 7 4 57 1,136,704 353,951 30 c/mL 7 7 100 1,839,238 336,415 10 c/mL 7 7 100 2,315,286 141,813 *B. divergens* IVT 100 c/mL 7 7 100 1,650,341 270,259 30 c/mL 7 7 100 1,068,065 468,460 10 c/mL 7 3 43 1,459,374 758,660 Negative IC Buffer 0 8 0 0 574 689 *P. falciparum* IVT 1.00e6 c/mL 8 0 0 941 1,394

CONCLUSIONS

(120) Several candidate systems identified on the manual eSAS system yielded good specificity and sensitivity with *Babesia* species IVTs for the *Babesia* Assay during secondary screening on the fully automated Panther system. Others combinations showed poor sensitivity (Condition 7) or specificity (Condition 1 and 4) and were not advanced to additional screening. A second round of screening on the Panther system identified a new system (Condition 12) with sensitivity in lower dilutions of *Babesia* species IVTs superior to other conditions tested and good specificity for the *Babesia* assay.

Example 3: Clinical Sample Screening

Objective

(121) Candidate amplification systems previously identified for the *Babesia* Assay on the manual Procleix Enhanced Semi-automated System (eSAS) were screened on the fully automated Procleix Panther System to determine the best candidates in terms of specificity and sensitivity using a *Babesia* clinical sample. Primers screened in this example are the same as Example 2. This experiment served as an additional test to determine the best performing candidates.

Materials and Methods

(122) Candidate amplification systems were tested on the automated Procleix Panther system. Combinations tested are listed in Table 14. Sequences for each oligo are listed in Table 15. A total of 8 conditions were screened. Conditions screened were tested using 9 replicates each of a lysed negative whole blood and diluted *Babesia*-infected clinical sample. The *Babesia* clinical sample consisted of a PCR Positive *Babesia*-infected human red blood cell (RBC) sample. The sample was received with an estimated parasite per mL value. Based upon the estimated parasite per mL value, the clinical sample was diluted in normal negative human whole blood to an estimated 30, 10, and 3 parasites per mL. The diluted *Babesia*-infected whole blood was lysed at a 1 to 6 ratio or 0.8 mL of whole blood in 4.8 mL of lysis solution Aptima Urine Transport Medium (commercially

available). The negative panel was lysed according to the same procedure. Assay Reagents used included the following: a Target Capture Reagent (TCR) comprising of a single Target Capture Oligo (TCO) added at a concentration of 5 pmoles per reaction; an Amplification Reagent comprising one T7 promoter provider and one non-T7 primer each added at a concentration of 10 pmoles per reaction; a Probe reagent consisting of 1 acridinium-ester (AE) labeled probe added at a total concentration of 5e6 Relative Light Units (RLU) per reaction to a solution of; Enzyme Reagent; and Selection Reagent.

(123) Conditions tested in this group are listed in Table 16. Sequences for each oligo are listed in Table 17. A total of 6 additional conditions were screened. Conditions were tested using 5 replicates of each of a *Babesia*-negative whole blood sample and a diluted *Babesia*-infected whole blood clinical sample. Prior to testing the negative and positive samples, the parasite concentration of the *Babesia*-infected clinical sample was estimated by using a calibration curve. The calibration curve was generated using lysates from serial dilutions of a known standard (a quantified *Babesia*-positive whole blood specimen). The clinical samples and the serial dilutions of the known standard were tested using a real-time assay. The Ttimes from each of the dilutions of the known standard were plotted against the parasites/mL to generate an equation. The equation was then used to convert Ttime to parasites/mL when the clinical sample was run using the same assay. The clinical sample was then diluted in normal negative human whole blood to an estimated 30 and 10 parasites per mL. The diluted clinical sample was lysed at a ratio of 1 mL whole blood to 3 mL lysis buffer. An assay calibrator comprising a *B. microti* IVT panel at 500 c/mL was also used. Similar to the screening described in the above examples, a single TCO was added to TCR at 5 pmoles per reaction, One T7 promoter provider and one non-T7 primer were added to Amplification Reagent at 10 pmoles per reaction each. The Probe reagent in this round of screening consisted of SEQ ID NO:37 detection probe oligomer at 2.5e6 RLU per reaction and SEQ ID NO:42 detection probe oligomer at 5e6 RLU per reaction.

(124) TABLE-US-00014 TABLE 14 Conditions 1-8 Tested on Panther Target Capture Condition #
Oligo T7 Primer Non-T7 Primer Probe 1 SEQ ID NO: 43 SEQ ID NO: 1 SEQ ID NO: 11 SEQ ID NO: 37 2 SEQ ID NO: 43 SEQ ID NO: 1 SEQ ID NO: 27 SEQ ID NO: 39 3 SEQ ID NO: 43 SEQ ID NO: 1 SEQ ID NO: 27 SEQ ID NO: 38 4 SEQ ID NO: 43 SEQ ID NO: 1 SEQ ID NO: 13 SEQ ID NO: 37 5 SEQ ID NO: 43 SEQ ID NO: 3 SEQ ID NO: 21 SEQ ID NO: 37 6 SEQ ID NO: 43 SEQ ID NO: 7 SEQ ID NO: 21 SEQ ID NO: 37 7 SEQ ID NO: 43 SEQ ID NO: 7 SEQ ID NO: 27 SEQ ID NO: 39 8 SEQ ID NO: 43 SEQ ID NO: 5 SEQ ID NO: 21 SEQ ID NO: 37

(125) TABLE-US-00015 TABLE 15 Sequences for Oligos listed in Table 14. Reagent
Short name Sequence 5'-3' TCR SEQ ID NO: 43 uaggccaauacccuaccgu
ccttataaaaaaaaaaaaaa aaaaaaaaaaaaaaaaaa Amplification SEQ ID NO: 1 aatttaatacgaactcacta
tagggagattcacctctga cagttataatagcga SEQ ID NO: 3 aatttaatacgaactcacta tagggagaacagttataata
cgaatgcccccaa SEQ ID NO: 5 aatttaatacgaactcacta tagggagattcacctctga cagttataatagcga SEQ ID NO: 7 aatttaatacgaactcacta tagggagagcttctgcagt agttcgtctttaacaaatc SEQ ID NO: 13
cttgaatactacagcatgg aataa SEQ ID NO: 21 agaaaactagagtgtttcaa SEQ ID NO: 27
atggaataatgaagtaggac Probe SEQ ID NO: 37 ugaaguaggacuuugguucu SEQ ID NO: 38
uaaugguuaauaggagcagu ug SEQ ID NO: 39 aguaaugguuaauaggagca

(126) TABLE-US-00016 TABLE 16 Conditions 9-14 tested on Panther Target Capture Condition #
Oligo T7 Primer Non-T7 Primer Probe 9 SEQ ID NO: 43 SEQ ID NO: 7 SEQ ID NO: 21 SEQ ID NOs: 37 + 42 10 SEQ ID NO: 43 SEQ ID NO: 7 SEQ ID NO: 32 SEQ ID NOs: 37 + 42 11 SEQ ID NO: 43 SEQ ID NO: 7 SEQ ID NO: 33 SEQ ID NOs: 37 + 42 12 SEQ ID NO: 43 SEQ ID NO: 7 SEQ ID NO: 34 SEQ ID NOs: 37 + 42 13 SEQ ID NO: 43 SEQ ID NO: 7 SEQ ID NO: 35 SEQ ID NOs: 37 + 42 14 SEQ ID NO: 43 SEQ ID NO: 7 SEQ ID NO: 36 SEQ ID NOs: 37 + 42

(127) TABLE-US-00017 TABLE 17 Sequences for Oligos listed in Table 16. Reagent
SEQ ID NO: Sequence 5'-3' TCR SEQ ID NO: 43 uaggccaauacccuaccgu
ccttataaaaaaaaaaaaaa aaaaaaaaaaaaaaaaaa Amplification SEQ ID NO: 7 aatttaatacgaactcacta

tagggagagctttcgagcgt agttcgtctttaacaacatc SEQ ID NO: 21 agaaaactagagtgtttcaa SEQ ID NO: 32 agaaaactagagtgtttca SEQ ID NO: 33 agaaaactagagtgtttcaa SEQ ID NO: 34 gagaaaactagagtgtttcaa SEQ ID NO: 35 gagaaaactagagtgtttcaa SEQ ID NO: 36 tgagaaaactagagtgtttc Probe SEQ ID NO: 37 ugaaguaggacuuugguucu SEQ ID NO: 42 aguaggacuuugguucu

Results:

(128) An analyte signal cutoff of 100,000 RLU was used to determine reactivity for specimens screened for Conditions 1-8 in Table 18. Samples above this cutoff were considered reactive. Conditions 1 and 4 yielded a false positive in lysed negative whole blood specimens at a rate of 11% and 33%, respectively. Conditions 2, 5 and 8 detected *Babesia* at less than 100% in lysed *Babesia*-infected clinical samples at 10 and 3 parasites per mL. Due to the high rate of false positives with SEQ ID NO:1 promoter provider and the high number of lower performing candidates, additional screening was performed to identify alternative candidates.

(129) TABLE-US-00018 TABLE 18 Average Analyte Results for Conditions 1-9 testing lysed Clinical Sample Dilutions on the Panther System. Average StdDev Condition Sample % Analyte

Analyte #	Description	Valid	Reactive	Reactive RLU	RLU 1
3	<i>B. microti</i> Calibrator	3	3	100%	
2,895,211	95,178 <i>Babesia</i> ARC	0	p/mL 9	1 11% 43,072	111,361 <i>Babesia</i> ARC
3	p/mL 9	9	100%		
2,881,463	51,926 <i>Babesia</i> ARC	10	p/mL 9	9 100%	2,947,197
42,332	<i>Babesia</i> ARC	30	p/mL 9	9	
100%	2,921,132	65,204	2 <i>B. microti</i> Calibrator	3	3 100%
2,164,104	85,510 <i>Babesia</i> ARC	0	p/mL 9	0	0%
655	63 <i>Babesia</i> ARC	3	p/mL 9	6 67%	1,263,656
996,750	<i>Babesia</i> ARC	10	p/mL 9	9	
100%	2,134,186	27,346	<i>Babesia</i> ARC	30	p/mL 9
9 100%	2,167,935	51,972	3 <i>B. microti</i> Calibrator	3	3 100%
5,202,778	356,597 <i>Babesia</i> ARC	0	p/mL 9	0 0%	1,460
140	<i>Babesia</i> ARC	3	p/mL 9	9	
100%	5,018,781	930,364	<i>Babesia</i> ARC	10	p/mL 9
9 100%	5,282,309	286,096	<i>Babesia</i> ARC	30	p/mL 9
9 100%	5,347,545	183,066	4 <i>B. microti</i> Calibrator	3	3 100%
2,847,426	127,963 <i>Babesia</i> ARC	0	p/mL 9	3 33%	248,507
581,559	<i>Babesia</i> ARC	3	p/mL 9	9 100%	2,970,739
88,830	<i>Babesia</i> ARC	10	p/mL 9	9 100%	3,007,234
76,308	<i>Babesia</i> ARC	30	p/mL 9	9 100%	3,034,392
82,795	5 <i>B. microti</i> Calibrator	3	3 100%	2,614,010	43,029
<i>Babesia</i> ARC	0	p/mL 9	0 0%	1,103	
519	<i>Babesia</i> ARC	3	p/mL 9	7 78%	1,714,505
1,012,100	<i>Babesia</i> ARC	10	p/mL 9	6 67%	
1,372,599	1,250,728	<i>Babesia</i> ARC	30	p/mL 9	9 100%
2,455,769	54,600	6 <i>B. microti</i> Calibrator	3	3 100%	2,849,900
59,411	<i>Babesia</i> ARC	0	p/mL 9	0 0%	765
72	<i>Babesia</i> ARC	3	p/mL 9	9 100%	
2,784,938	77,517	<i>Babesia</i> ARC	10	p/mL 9	9 100%
2,741,662	63,328	<i>Babesia</i> ARC	30	p/mL 9	9 100%
2,734,625	46,780	7 <i>B. microti</i> Calibrator	3	3 100%	5,706,224
96,503	<i>Babesia</i> ARC	0	p/mL 9	0 0%	3,383
1,376	<i>Babesia</i> ARC	3	p/mL 9	9 100%	5,192,042
511,510	<i>Babesia</i> ARC	10	p/mL 9	9 100%	5,240,227
135,033	<i>Babesia</i> ARC	30	p/mL 9	9 100%	5,425,945
228,652	8 <i>B. microti</i> Calibrator	3	3 100%	2,963,935	70,693
<i>Babesia</i> ARC	0	p/mL 9	0 0%	1,163	596
<i>Babesia</i> ARC	3	p/mL 9	5 56%	1,708,479	1,623,206
<i>Babesia</i> ARC	10	p/mL 9	9 100%	3,060,059	187,482
<i>Babesia</i> ARC	30	p/mL 9	9 100%	3,112,370	83,507

(130) In the previous round of screening, only a few candidates showed good sensitivity and specificity for lysed *Babesia*-infected clinical samples. Condition 6 had good specificity and sensitivity detecting the lysed clinical sample down to 3 parasites per mL at 100% reactivity and with strong analyte signal for the positive specimens tested and low analyte signal for negative specimens. Additional screening was performed on the Panther system using SEQ ID NO:7 promoter provider paired with a number of additional non-T7 primers. Conditions in this round of screening were again tested using lysed dilutions of a clinical sample. All conditions except Condition 11 detected the 30 and 10 p/mL lysates at 100% reactivity. None of conditions yielded any false positives in negative specimens.

(131) TABLE-US-00019 TABLE 19 Average Analyte Results for Conditions 9-14 testing lysed Wadsworth Center (WC) sample dilutions on Panther. Average StdDev Condition Sample % Analyte

Analyte #	Description	Level	Valid	Reactive	Reactive RLU	RLU 9
0	Negative Calibrator	0	3	3	100	
3,064	2,870 <i>B. microti</i> Calibrator	500	c/mL 3	3 100	2,634,036	20,467
<i>Babesia</i> WC	30	p/mL 5	5			

100 2,335,851 100,942 10 p/mL 5 5 100 1,970,484 170,746 Negative 0 5 0 0 1,229 1,061 10
 Negative Calibrator 0 3 0 0 2,083 1,176 *B. microti* Calibrator 500 c/mL 3 3 100 2,612,367 134,872
Babesia WC 30 p/mL 5 5 100 1,700,198 40,527 10 p/mL 5 5 100 1,639,511 112,124 Negative 0 5 0
 0 879 899 11 Negative Calibrator 0 3 0 0 1,687 1,471 *B. microti* Calibrator 500 c/mL 3 3 100
 2,476,171 150,100 *Babesia* WC 30 p/mL 5 5 100 2,083,998 176,018 10 p/mL 5 3 60 833,006
 790,769 Negative 0 5 0 0 2,237 1,256 12 Negative Calibrator 0 3 0 0 370 640 *B. microti* Calibrator
 500 c/mL 3 3 100 2,742,289 194,078 *Babesia* WC 30 p/mL 5 5 100 2,220,041 241,385 10 p/mL 5 5
 100 2,098,119 175,399 Negative 0 5 0 0 970 648 13 Negative Calibrator 0 3 0 0 18 31 *B. microti*
 Calibrator 500 c/mL 3 3 100 2,618,344 183,612 *Babesia* WC 30 p/mL 5 5 100 1,861,428 170,146
 10 p/mL 5 5 100 2,032,573 300,759 Negative 0 5 0 0 405 871 14 Negative Calibrator 0 3 0 0 673
 843 *B. microti* Calibrator 500 c/mL 3 3 100 2,656,960 89,493 *Babesia* WC 30 p/mL 5 5 100
 2,523,855 87,845 10 p/mL 5 5 100 2,376,263 191,838 Negative 0 5 0 0 2,381 4,735
 CONCLUSION

(132) Considering results for secondary screening with *Babesia* species IVTs in Example 2 and screening with clinical sample in this example, Conditions 3 and 6 were the best candidates in terms of specificity and sensitivity. Condition 12 had the best performance of the conditions screened in Conditions 9-14. Condition 12 had best performance in terms of percent reactivity for all levels tested and consistent RLU signal for all levels tested for *Babesia* species IVTs and lysed clinical sample dilutions on the Panther system. Additionally, Condition 12 had the lowest analyte signal for negative specimens.

(133) From the foregoing, it will be appreciated that, although specific embodiments have been described herein for purposes of illustration, various modifications may be made without deviating from the spirit and scope of this explicit disclosure. Accordingly, the invention is not limited by the explicit disclosure. All publications, patents, and patent applications cited herein are hereby incorporated by reference in their entireties for all purposes.

Example 4: *Babesia* IVTs and *Babesia* Positive Whole Blood Sample Screening Objective

(134) Candidate amplification systems were screened on the Procleix Panther System to determine the oligo candidates in terms of specificity and sensitivity using IVTs and a contrived *Babesia* clinical sample.

Materials and Methods

(135) Candidate amplification systems were tested on the automated Procleix Panther system. Combinations tested are listed in Table 20. Sequences for each oligo are listed in Table 21. A total of 8 conditions were screened. Conditions were screened using 16 replicates of panel members consisting of in-vitro transcript diluted in buffer for *B. microti*, *B. divergens*, *B. duncani*, and *B. venatorum* (SEQ ID NOs:61, 62, 63, & 100) at either 30 c/mL or 10 c/mL. Conditions were also screened against 12 replicates each of a contrived clinical sample. The contrived clinical sample comprised a mixture of *Babesia*-negative human whole blood and *Babesia microti*-infected hamster whole blood (acquired from the American Red Cross (ARC)). In short, *Babesia microti* infected hamster blood having a predetermined parasitemia value was diluted with *Babesia*-negative human whole blood to yield an estimated 4 parasites/mL (p/mL) of an infected blood mixture. The infected blood mixture was then lysed at a 1 to 3 ratio (here using 0.9 mL of whole blood in 2.7 mL of an aqueous solution of 100 mM TRIS, 30 mM magnesium chloride, and 6% (v/v) LLS, at pH 7.5). The lysed infected blood mixture is referred to herein as a hamster blood lysate. In some test conditions hamster blood lysate was further diluted in negative lysate to a level equating to 0.01 parasites/mL. A negative lysate was prepared according to the same procedure. An assay positive calibrator consisted of a *B. microti* in-vitro transcript (IVT) (SEQ ID NO:61) panel at 500 c/mL. Negative calibrator and negative buffered specimens contained only buffer. Assay Reagents used consisted of: Target Capture Reagent (TCR) comprising of a single Target Capture Oligo (TCO) added at a concentration of approximately 5 pmoles per reaction; Amplification Reagent

comprising of T7 and NT7 primers added at a concentration of approximately 5 pmoles each per reaction; Probe reagent consisting of acridinium-ester (AE) labeled probes added at a total concentration of approximately 1e6 Relative Light Units (RLU) per reaction. Commercially available Procleix Ultrio Plus Enzyme and Selection reagents were used.

(136) TABLE-US-00020 TABLE 20 Conditions 1-8 Tested on Panther Condition # Oligo Combination 1 SEQ ID NO: 89, SEQ ID NO: 7, SEQ ID NO: 82, SEQ ID NO: 84, SEQ ID NO: 34, SEQ ID NO: 86, SEQ ID NO: 91, SEQ ID NO: 42, SEQ ID NO: 98 2 SEQ ID NO: 87, SEQ ID NO: 7, SEQ ID NO: 82, SEQ ID NO: 84, SEQ ID NO: 34, SEQ ID NO: 86, SEQ ID NO: 91, SEQ ID NO: 42, SEQ ID NO: 98 3 SEQ ID NO: 43, SEQ ID NO: 87, SEQ ID NO: 89, SEQ ID NO: 82, SEQ ID NO: 84, SEQ ID NO: 34, SEQ ID NO: 86, SEQ ID NO: 91, SEQ ID NO: 42, SEQ ID NO: 98 4 SEQ ID NO: 43, SEQ ID NO: 87, SEQ ID NO: 89, SEQ ID NO: 7, SEQ ID NO: 82, SEQ ID NO: 84, SEQ ID NO: 91, SEQ ID NO: 42, SEQ ID NO: 98 5 SEQ ID NO: 43, SEQ ID NO: 87, SEQ ID NO: 89, SEQ ID NO: 7, SEQ ID NO: 82, SEQ ID NO: 86, SEQ ID NO: 91, SEQ ID NO: 42, SEQ ID NO: 98 6 SEQ ID NO: 43, SEQ ID NO: 87, SEQ ID NO: 89, SEQ ID NO: 7, SEQ ID NO: 82, SEQ ID NO: 84, SEQ ID NO: 34, SEQ ID NO: 86, SEQ ID NO: 91 7 SEQ ID NO: 43, SEQ ID NO: 87, SEQ ID NO: 89, SEQ ID NO: 7, SEQ ID NO: 82, SEQ ID NO: 84, SEQ ID NO: 34, SEQ ID NO: 86, SEQ ID NO: 92 8 SEQ ID NO: 43, SEQ ID NO: 87, SEQ ID NO: 89, SEQ ID NO: 7, SEQ ID NO: 82, SEQ ID NO: 84, SEQ ID NO: 34, SEQ ID NO: 86, SEQ ID NO: 93

(137) TABLE-US-00021 TABLE 21 Sequences for Oligos listed in Table 20. Seq # Reagent Class Sequence (5'-3') SEQ ID NO: 43 TCR Capture Oligo uaggccaauacccuaccgucctttaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa aa SEQ ID NO: 87 Capture Oligo aaagacuugauuucucucaaggtttaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa aaaaaa SEQ ID NO: 89 Capture Oligo caagaaagagcuaucuaucugucaaucctttaaaaaaaaaaaaaaaaaaaaaa aaaaaaaa SEQ ID NO: 7 Amplification T7 Primer aatttaatacgactcactatagggagagctttcgagtagttcgtctttaac aaatc SEQ ID NO: 82 T7 Primer aatttaatacgactcactatagggagaggcaaatgctttcgagtagttigt cttaaca SEQ ID NO: 84 Non-T7 primer gcggttaattccagctccaatag SEQ ID NO: 34 Non-T7 primer gagaaaactagagtgtttcaa SEQ ID NO: 86 Non-T7 primer cttgaatacticagca SEQ ID NO: 91 Probe Probe aguaggacuuugguuct SEQ ID NO: 92 Probe aguaggacuuugguuc SEQ ID NO: 93 Probe aguaggacuaauugguuc SEQ ID NO: 42 Probe aguaggacuuugguucu SEQ ID NO: 98 Probe aguaggacuaauugguucu

Results:

(138) An analyte signal cutoff calculated was calculated using negative and positive calibrators and the Panther Software. Test results yielding a signal to cutoff ratio of greater than or equal to 1.0 were considered reactive. All conditions tested detected *B. microti*, *B. divergens*, *B. duncani* and *B. venatorum* as well as the hamster blood lysate panel members to some degree (Table 22). All conditions detected all species at 100% reactivity at the 30c/mL level except Condition 5 in which 15 out of 16 replicates were detected. For conditions 1-5 all hamster blood lysate panels at 4 p/mL were 100% reactive except Condition 2 in which 8 out of 12 replicates were detected. For Conditions 6-8, which tested hamster blood lysates at 0.01 p/mL, results show 14/16, 15/16, and 14/16 positive. There were no false positive in negative reactions.

(139) TABLE-US-00022 TABLE 22 Average Analyte Results for Conditions 1-8 testing. SD SD Condition Sample % Analyte Analyte Analyte Analyte # Description c/mL Valid Reactive Reactive RLUs RLUs S/CO S/CO 1 Negative 0 3 0 0.0 0 0 0.00 0.00 Calibrator Positive 500 3 3 100.0 1,310,257 45,626 33.33 1.16 Calibrator *B. microti* 30 c/mL 16 16 100.0 691,469 302,377 17.59 7.69 10 c/mL 16 15 93.8 648,137 282,342 16.49 7.18 *B. duncani* 30 c/mL 16 16 100.0 1,100,887 309,481 28.01 7.87 10 c/mL 16 16 100.0 973,540 246,634 24.77 6.27 *B. divergens* 30 c/mL 16 16 100.0 1,099,778 294,789 27.98 7.50 10 c/mL 16 16 100.0 959,134 335,825 24.40 8.54 *B. venatorum* 30 c/mL 16 16 100.0 1,052,533 280,903 26.78 7.15 10 c/mL 16 16 100.0 902,987 309,501 22.97 7.87 Negative 0 c/mL 16 0 0.0 0 0 0.00 0.00 Buffer Hamster Blood 4 12 12 100.0 1,307,118 28,853 33.25 0.73 Lysate Hamster Blood 0 12 0 0.0 4 15 0.00 0.00 Lysate 2 Negative 0 3

0 0.0 14 24 0.00 0.00 Calibrator Positive 500 3 3 100.0 1,360,358 88,289 33.32 2.16 Calibrator
B. microti 30 c/mL 16 16 100.0 990,774 405,235 24.27 9.93 10 c/mL 16 13 81.3 674,210 358,804
16.52 8.79 *B. duncani* 30 c/mL 16 16 100.0 1,261,270 234,066 30.90 5.73 10 c/mL 16 16 100.0
1,010,323 412,156 24.75 10.10 *B. divergens* 30 c/mL 16 16 100.0 1,136,882 281,770 27.85 6.90 10
c/mL 16 16 100.0 1,138,086 288,726 27.88 7.07 *B. venatorum* 30 c/mL 16 16 100.0 1,095,242
357,885 26.83 8.77 10 c/mL 16 16 100.0 979,611 405,233 24.00 9.93 Negative 0 c/mL 16 0 0.0 0 0
0.00 0.00 Buffer Hamster Blood 4 12 8 66.7 1,148,331 589,637 28.13 14.44 Lysate Hamster Blood
0 12 0 0.0 220 763 0.01 0.02 Lysate 3 Negative 0 3 0 0.0 0 0 0.00 0.00 Calibrator Positive 500 3
3 100.0 1,247,077 16,587 33.33 0.44 Calibrator *B. microti* 30 c/mL 16 16 100.0 996,284 147,903
26.63 3.95 10 c/mL 16 14 87.5 733,412 225,774 19.60 6.03 *B. duncani* 30 c/mL 16 16 100.0
1,154,290 46,421 30.85 1.24 10 c/mL 16 16 100.0 1,051,223 93,259 28.10 2.49 *B. divergens* 30
c/mL 16 16 100.0 1,160,454 42,968 31.02 1.15 10 c/mL 16 16 100.0 1,067,057 104,387 28.52 2.79
B. venatorum 30 c/mL 16 16 100.0 1,120,866 52,999 29.96 1.42 10 c/mL 16 16 100.0 988,682
127,271 26.43 3.40 Negative 0 c/mL 16 0 0.0 0 0 0.00 0.00 Buffer Hamster Blood 4 12 12 100.0
1,153,646 24,699 30.84 0.66 Lysate Hamster Blood 0 12 0 0.0 0 0 0.00 0.00 Lysate 4 Negative 0 3
0 0.0 0 0 0.00 0.00 Calibrator Positive 500 3 3 100.0 689,988 39,272 33.33 1.90 Calibrator *B.*
microti 30 c/mL 16 16 100.0 236,235 167,806 11.41 8.11 10 c/mL 16 11 68.8 142,663 122,889 6.89
5.94 *B. duncani* 30 c/mL 16 16 100.0 331,217 55,106 16.00 2.66 10 c/mL 16 16 100.0 174,519
88,419 8.43 4.27 *B. divergens* 30 c/mL 16 16 100.0 589,500 20,005 28.48 0.97 10 c/mL 16 16
100.0 482,888 106,825 23.33 5.16 *B. venatorum* 30 c/mL 16 16 100.0 560,636 66,993 27.09 3.24
10 c/mL 16 16 100.0 406,685 129,080 19.65 6.24 Negative 0 c/mL 16 0 0.0 0 0 0.00 0.00 Buffer
Hamster Blood 4 12 12 100.0 870,137 62,684 42.04 3.03 Lysate Hamster Blood 0 12 0 0.0 0 0 0.00
0.00 Lysate 5 Negative 0 3 0 0.0 0 0 0.00 0.00 Calibrator Positive 500 3 3 100.0 1,041,315
125,263 33.33 4.01 Calibrator *B. microti* 30 c/mL 16 15 93.8 306,912 218,362 9.82 6.99 10 c/mL
16 11 68.8 243,155 159,289 7.78 5.10 *B. duncani* 30 c/mL 16 16 100.0 748,753 281,035 23.97 9.00
10 c/mL 16 16 100.0 751,922 286,380 24.07 9.17 *B. divergens* 30 c/mL 16 16 100.0 807,915
258,741 25.86 8.28 10 c/mL 16 16 100.0 643,482 256,352 20.60 8.21 *B. venatorum* 30 c/mL 16 16
100.0 792,659 264,484 25.37 8.47 10 c/mL 16 16 100.0 747,633 243,096 23.93 7.78 Negative 0
c/mL 16 0 0.0 0 0 0.00 0.00 Buffer Hamster Blood 4 12 12 100.0 1,410,648 33,678 45.16 1.08
Lysate Hamster Blood 0 12 0 0.0 84 210 0.00 0.01 Lysate 6 Negative 0 c/mL 3 0 0.0 0 0 0.00 0.00
Calibrator Positive 500 c/mL 3 3 100.0 524,314 5,119 33.33 0.33 Calibrator *B. microti* 30 c/mL 16
16 100.0 507,560 25,719 32.26 1.64 *B. divergens* 30 c/mL 16 16 100.0 471,886 64,616 30.00 4.11
B. duncani 30 c/mL 16 16 100.0 502,786 46,565 31.96 2.96 *B. venatorum* 30 c/mL 16 16 100.0
495,758 19,503 31.51 1.24 Hamster Blood 0.01 p/mL 16 14 87.5 337,759 102,178 24.48 7.41
Lysate Negative 0 c/mL 16 0 0.0 145 387 0.00 0.00 Buffer 7 Negative 0 c/mL 3 0 0.0 254 439 0.01
0.02 Calibrator Positive 500 c/mL 3 3 100.0 668,273 11,200 32.91 0.55 Calibrator *B. microti* 30
c/mL 16 16 100.0 654,530 36,463 32.24 1.80 *B. divergens* 30 c/mL 16 16 100.0 652,634 44,688
32.14 2.20 *B. duncani* 30 c/mL 16 16 100.0 628,767 49,861 30.97 2.46 *B. venatorum* 30 c/mL 16
16 100.0 620,345 63,099 30.55 3.11 Hamster Blood 0.01 p/mL 16 15 93.8 397,026 182,905 23.35
10.76 Lysate Negative 0 c/mL 16 0 0.0 91 151 0.00 0.00 Buffer 8 Negative 0 c/mL 3 0 0.0 280 486
0.02 0.03 Calibrator Positive 500 c/mL 3 3 100.0 521,485 14,518 32.74 0.91 Calibrator *B. microti*
30 c/mL 16 16 100.0 460,267 52,434 28.90 3.29 *B. divergens* 30 c/mL 16 16 100.0 423,530 61,700
26.59 3.87 *B. duncani* 30 c/mL 16 16 100.0 428,292 43,574 26.89 2.74 *B. venatorum* 30 c/mL 16
16 100.0 415,273 64,983 26.07 4.08 Hamster Blood 0.01 p/mL 16 14 87.5 198,996 127,811 13.96
8.97 Lysate Negative 0 c/mL 16 0 0.0 817 885 0.00 0.00 Buffer

CONCLUSION

(140) Oligo combinations tested herein exemplify specific and sensitive capture, amplification, and detection of *Babesia* species including *B. microti*, *B. divergens*, *B. duncani*, and *B. ventorum* and demonstrated detection of *Babesia microti* parasite in whole blood.

(141) TABLE-US-00023 SEQ ID NO. Sequence (5' to 3') Comments 1

aatttaatacgcactcactatagggagattcacctctgacaggttaaatacga T7 promoter primer 2
ttcacctctgacaggttaaatacga SEQ ID NO: 1 without promoter sequence 3
aatttaatacgcactcactatagggagaaacaggttaaatacgaatgccccca T7 promoter primer 4
acaggttaaatacgaatgccccca SEQ ID NO: 3 without promoter sequence 5
aatttaatacgcactcactatagggagattcacctctgacaggttaaatac T7 promoter primer 6 ttcacctctgacaggttaaatac
SEQ ID NO: 5 without promoter sequence 7
aatttaatacgcactcactatagggagagctttcgagtagttcgctcatttaacaaatc T7 promoter primer 8
gctttcgagtagttcgctcatttaacaaatc SEQ ID NO: 7 without promoter sequence 9
aatttaatacgcactcactatagggagagctttcgagtagttcgctcatttaac T7 promoter primer 10
ctttcgagtagttcgctcatttaac SEQ ID NO: 9 without promoter sequence 11 cttgaatactacagcatgga
Non-T7 primer 12 actacagcatggaataatga Non-T7 primer 13 cttgaatactacagcatggaataa Non-T7
primer 14 acttcagcatggaataatga Non-T7 primer 15 cttgaatacttcagcatgga Non-T7 primer 16
actncagcatggaataatga Non-T7 primer, wherein “n” means a or g or c or t/u, or
unknown, or other (WIPO Standard ST.25 (1998), Appendix 2, Table 1) 17
actwcagcatggaataatga Non-T7 primer 18 cttgaatactncagcatgga Non-T7 primer, wherein “n”
means a or g or c or t/u, or unknown, or other (WIPO Standard ST.25 (1998),
Appendix 2, Table 1) 19 cttgaatactwcagcatgga Non-T7 primer 20 actttgagaaaactagagtg Non-
T7 primer 21 agaaaactagagtgtttcaa Non-T7 primer 22 ataatgaagtaggactttgg Non-T7 primer 23
aggactttggttctattttg Non-T7 primer 24 ggttctattttgttggtt Non-T7 primer 25 tggttctattttgttggtt Non-
T7 primer 26 ggaataatgaagtaggacttt Non-T7 primer 27 atggaataatgaagtaggac Non-T7 primer
28 atggaataatgaagtagg Non-T7 primer 29 gcatggaataatgaagtag Non-T7 primer 30
tacagcatggaataatgaag Non-T7 primer 31 tactacagcatggaataatg Non-T7 primer 32
agaaaactagagtgtttca Non-T7 primer 33 agaaaattagagtgtttcaa Non-T7 primer 34
gagaaaactagagtgtttcaa Non-T7 primer 35 gagaaaattagagtgtttcaa Non-T7 primer 36
tgagaaaactagagtgtttc Non-T7 primer 37 ugaaguaggacuuugguucu Probe 38
uaaugguuaauaggagcaguug Probe 39 aguaaugguuaauaggagca Probe 40 ggacuuugguucuaauuuuguugg
Probe 41 aaugguuaauaggagcaguuggggg Probe 42 aguaggacuuugguucu Probe 43
uaggccaauacccuaccgucctttaaaaaaaaaaaaaaaaaaaaaaaaaaaa Target capture 44
uaggccaauacccuaccgucc SEQ ID NO: 43, without capture tail sequence 45
atggaataatgaagtag Same sequence in SEQ ID NOs: 27, 28, 29 and 31 46
agaaaactagagtg Same sequence in SEQ ID NOs: 20, 21, 32, 34 and 36 47
agaaaattagagtg Same sequence in SEQ ID NOs: 33 and 35. 48 agaaaaytagagtg
Substitutes y for c in SEQ ID NO: 46 and y for t in SEQ ID NO: 47,
wherein “y” means c or t/u (WIPO Standard ST.25 (1998), Appendix 2, Table 1) 49
agaaaactagagtgtttc Same sequence in SEQ ID NOs: 19, 20, 21, 32, 34 and 36. 50
agaaaattagagtgtttc Same sequence in SEQ ID NOs: 33 and 35. 51 agaaaaytagagtgtttc
Substitutes y for c in SEQ ID NO: 49) and y for t in SEQ ID NO: 50,
wherein “y” means c or t/u (WIPO Standard ST.25 (1998), Appendix 2, Table 1) 52
tactacagcatggaataa Same sequence in SEQ ID NO: 13 and 31. 53 actncagcatgga
Residue “n” means a or g or c or t/u, or unknown, or other (WIPO Standard
ST.25 (1998), Appendix 2, Table 1), and “other preferably means “i” or inosine 54
actwcagcatgga 55 actacagcatgga Same sequence as SEQ ID NO: 54, except that “n”
is “a”. 56 gtatttaactgt Same sequence in SEQ ID NOs: 1, 3 and 5. 57
Gtatttaactgtcagaggtgaa Same sequence in SEQ ID NOs: 1 and 5. 58
aatttaatacgcactcactatagggaga T7 promoter 59 aguaggacuuugguucu Same sequence in SEQ
ID NO: 42. 60 uaaugguuaauaggagca Same sequence in SEQ ID NOs: 38 and 39. 61
gggcgaauuggguaccgggccccccucgaggucgacgcuuaguauaagcuuuu *B. microti* IVT
uacagcgaaacugcgaauggcucauuuuuacaguuuauaguuuuuuugauguucgu Sequence
uuuacauaggauaaccgugguaauucuaagggcuauuacaugcucgaggcgcuuuu
cgcguggcguuuuuuagacuuuuaaccaacccuucggguuauucggugauucauuu

aaauuagcgauucgcauuggcggcggauguaucuucaaguucugaccua
ucagcuuugggacgguaaggguauuuggccuaccggggcgacgacgggugacggggaa
uuggggguucgauuccggagagggagccugagaaaacggcuaccacaucaaggaag
gcagcagggcgcgcaauuacccaauccugacacagggagguagugacaagaaauaa
caauacaggggcuuaaagucuuuguaauuuggaauugauugggaaucuaaaaccuucca
gaguaucaauuggaggggcaagucuggugccagcagccgcgguaauuccagcucca
auagcguauauuaaaguuugugcaguuuaagaagcucguaguugaauuucugccuu
gucauuaaucucgcuuuccgagcguuuuuuuauugacuuggcaucuuucuggauuu
ggugccuucggguacuauuuuuccaggauuuacuauugagaaaaacuagaguguuuca
aacaggcauucgccuugaauacuacagcauggaauaauagaaguaggacuungguu
cuauuuuuguuugguauuugagccagaguaauugguuauuaggagcaguugggggca
uucguauuuuacugucagagggugaaauucuuagauuuguuuaaagacgaacuacug
cgaaagcauuugccaaggauuuuucuuuaaucaagaacgaaaguuagggggaucg
aagacgaucagauaccgucguaguuccuaaccauaaacuaugccgacuagagauugg
aggucgucaguuuaaacgacuccuucagcaccuugagagaaaaucaaagucuuugg
guucuggggggaguauggucgcaagucugaaacuuaaagggaauugacgggaagggc
accaccaggcguggagccugcggcuuaauuugacucaaacacgggaaaccucaccag
guccagacauagagaggauugacagauugauagcucuucuuugaugaauu 62
gggcgaauuggguaccggggccccccucgaggucgacgguaucgaaagcuugau *B. divergens* IVT
aucgaauuccugcagcccgggggauccaaccugguugauccugccaguagucuaa Sequence
ugcuugucuuaaagauuaagccaugcaugucuaaguacaaacuuuuuacggugaa
acugcgaauggcucuuacaacaguuauaguuuucuuugguauucguuuuccaugg
auaaccgugcuauuuguaagggcuaauacaaguucgaggccuuuuggcggcguuua
uuaguucuaaaaccauuccuuuuggguuuucggugauucauaaaacuugcgaau
cgcauuuuuugcgauggaccauucaaguuuucugacccaucagcuugacgguaagg
guauuuggccuaccgaggcagcaacgggguaacgggggaauuaggguucgauuccgga
gagggagccugagaaacggcuaccacauccaaggaaggcagcagggcgcgcaauua
cccaauccugacacagggagguagugacaagaaauaacaauacagggcaauugucu
uguaauuuggaaugauggugaccuaaaaccucaccagaguaacaauuggagggcaa
gucuggugccagcagccgcgguaauuccagcuccaauagcguauauuaaacuugu
ugcaguuaaaaagcucguaguugaauuuuugcgugguguuauuugacuauug
ucgagauugcacuucgcuuuuugggauuuaucccuuuuacuuugagaaaauuaga
guguuucagcagacuuuugucuuugaauacuucagcauggaauaauagaguagga
cuuuggguucuuuuuguuugguuugugaaccuuaguaauugguuauuaggaacggg
uggggggcauucguauuuuacugucagaggugaaaauucuuagauuuuguuaaagacg
aacuacugcgaaagcauuugccaaggacguuuuucuuuaaucaagaacgaaaguuag
gggaucgaagacgaucagauaccgucguaguuccuaaccauaaacuaugccgacuag
ggauuaggagggucgucuuuuuuccgacuccuucagcaccuugagagaaucuaagu
cuuuggguucugggggggaguauggucgcaaggcugaaacuuaaagggaauugacg
gaagggcaccaccaggcguggagccugcggcuuaauuugacucaaacacggggaaac
ucaccagguccagacaauuguuaggauugacagauugauagcucuucuuugauucu uuggguggugcgcc 63
gggcgaauuggguaccggggccccccucgaggucgacguugauccugccaguagu *B. duncani* IVT
cauauugcuugucuuaaagauuaagccaugcaugucuaaguauaaacuuaauaug Sequence
gugaaacugcgaauggcuaauuacaacaguuauaguuuuauuugaaagucguuuuu
acauggauaaccgugcuauuuguaagggcuaauacaugcucgaggccuuggcuccu
gucuuggcugcguuuauuagacucgaaaccuucccgcuugcgguacucggugauu
cauaauaaaaauugcgaaucgcauggcuuuugccggcgauugguucuucaaguuuc
ugaccuaucagcuuuggacgguaaggguauuuggccuaccggggcagcgacggguua
cgggggaauuaggguucgauuccggagagggagccugagaaacggcuaccacaucu
aagggaaggcagcagggcgcgcaauuacccaauacggacaccgugagguagugacaa

gaaauaacaauacaggggcuuaaaagcuuuuguaauuggaauugauggggaauccaaacc
cuuccagaguaucuuuuggaggggcaagucuggugccagcagccgcgguaauucca
gcuccaauagcguauauuaaacuuuguugcaguuaaaaagcucguaguugaacuuc
ugccgcuuuggccuuucguuccccuuuggggguuuucguucgccuggguggcuuaccuc
uggcgguugguucuccauuugccaguuuuacuuugagaaaauuagaguguuucaa
gcagggcuuuugccuugaauacuucagcauggaauaauaaaguaggacuuugguuc
uauuuugguugguuucaggaccaaaguaauugguuaauaggaacaguuggggggcau
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agacgaucagauaccgucguaguccuaacuauaaaacuaugccgacuagagauugga
ggucgucauuuuuuuacgacuccuucagcaccuugagagaaaucaaagucuuuggg
uucugggggggaguauggucgcaaggcugaaaacuuuaaggaaauagcgggaaggga
ccaccaggcguggagccugcgggcuuaauuuugacucaacacgggggaaccucaccagg
uccagacauaguuaaggauugacagauugauagcucgaauu 64

ggggcgaauggguaccguagaaacugcgaacggcucuuuuuacaguuauagucu *P. falciparum* IVT
acuugacauuuuuuuuuuaagggauaacuacggaaaagcuguagcuaauacuugcu Sequence

[illegible]

ugaaguaggacuuugguucuaauuuuguugguuauugagccagaguaauggguuaa uaggagcaguugggggg 66
ttcacctctgacagttaaatacgaatgccccaa Contains SEQ ID NOs: 56 and 57, which the
target hybridising sequence can contain 67 tactncagcatggaataatgaagtaggactttgg Contains SEQ
ID NOs: 45 & 69, wherein “n” means a or g or c or t/u, or unknown, or
other (WIPO Standard ST.25 (1998), Appendix 2, Table 1) 68 cttgaatactncagcatggaataatga
Contains SEQ ID NOs: 53, 54 or 55, which the target hybridising sequence can
contain, wherein “n” means a or g or c or t/u, or unknown, or other (WIPO
Standard ST.25 (1998), Appendix 2, Table 1) 69 atggaataatg Contained in SEQ ID
NO: 67 70 actttgagaaaaytagagtgtttcaa Contains SEQ ID NOs: 46, 47, 48, 49, 50,
and 51, wherein “y” means c or t/u (WIPO Standard ST.25 (1998), Appendix 2,
Table 1) 71 aacctggttgatcctgccagtagtcatatgcttgctctaaagattaagccatgcattgcttagtat GenBank

Accession aagcttttatacagcgaaactgcgaatggctcattaaaacagttatagttgatgttcgtttt No: AY693840.1

acatggataaccgtggtaattctagggctaatacatgctcgaggcgcttttcgctggcggtt *Babesia microti*
attagactttaaccaacccttcgggtaatcgggtgattcataataaattagcgaatcgcatggctt isolate Gray 18S
tgccggcgatgtatcattcaagtttctgacctatcagctttggacggtaggggtattggcctacc ribosomal RNA
ggggcgacgacgggtgacggggaattgggggtcgattccggagagggagcctgagaaa gene, partial
cggctaccacatctaaggaaggcagcaggcgcgcaaattacccaatcctgacacagggag sequence

gtagtgacaagaaataacaatacagggcctaaagtcttgaattggaatgatgggaatctaaa
ccctcccagagtatcaattggaggggcaagtctgggtgccagcagccgcggttaattccagctc
caatagcgtatattaaagtgttgacagtaagaagctcgtagttgaatttctgccttgtcattaatc
tcgcttcgagcgcttttttattgacttggcatcttctggatttgggtgccttcgggtactattttcca
ggatttactttgagaaaactagagtgttcaaacaggcattcgccttgaatactacagcatgga
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cgaaagcatttgccaaggatgtttcattaatcaagaacgaaagttaggggatcgaagacga
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agaagtcgtaacaaggtttccgtaggtgaacctgcggaaggatcattc 72

aacctgggtgatcctgccagtagtcatatgcttgtcttaaagattaagccatgcatgtctaagta GenBank Accession
caaactttttacggtgaaactgcgaatggctcattacaacagttatagtttctttgggtattcgtttt No: AY789076.1

ccatggataaccgtgctaattgtagggctaatacaagttcgaggccttttggcggcggtttatta *Babesia divergens*
gttctaaaaccatcccttttgggttctcggtgattcataataaacttgcgaatcgcaatttttgcgat 18S ribosomal RNA
ggaccattcaagtttctgacctatcagcttgacggtaggggtattggcctaccgaggcagcaa gene, partial
cgggtaacggggaattaggggtcgtattccggagagggagcctgagaaacggctaccacat sequence

ccaaggaaggcagcaggcgcgcaaattacccaatcctgacacagggaggtagtgacaag
aaataacaatacagggcaattgtcttgaattggaatgatggtagcctaaacctcaccagag
taacaattggagggcaagtctgggtccagcagccgcggttaattccagctccaatagcgtata
ttaaacttgttcagttaaaaagctcgtagtgaatttttgcgtgggtgtaattattgactaatgtc
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ccttatcacttaaaggaaggagaagtcgtaacaagggttccgtaggtgaacctgcggaagg atcattc 73
ccttggtgatcctgccagtagcatatgcttgtcttaaggattaagccatgcatgtctaagtata GenBank Accession
aacttttatatggtgaaactgcgaatggctcattacaacagttatagtttatttgaaagtcgtttt No: AY027815.1
acatggataaccgtgctaattgtagggctaatacatgctcgaggccttggtctgtcttggt *Babesia* sp. WA1
gcgtttattagactcgaaccttcccgttgcggactcggtagtcataataaatttgcgaatc isolate CA5 18S
gcatggcttttgcggcgatgggtcattcaagttctgacctatcagcttggacggtaggggtat ribosomal RNA
tggcctaccggggcagcgacgggtaacgggggaattaggggtcattccggagaggggagc gene, partial
ctgagaaacggctaccacatctaaggaaggcagcaggcgcgcaaattaccaatacggac sequence
accgtgaggtagtgaagaataacaatacagggttaagcttgaattggaatgatggg
aatccaaacccctccagagtatcaattggagggaagtcgttggtgccagcagccgcggtaa
ttccagctccaatagcgtatattaaacttgagcagttaaaaagctcgtagtgaacttctgccg
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attaaaacagttatagctacttgacattttattataaggataactacggaaaagctgtagctaa No: JQ627151.1
tacttgctttattatcctttgattttatctttggataagtatttggtaggccttataagaaaaaagtta *Plasmodium*
ttaacttaaggaattataacaaagaagtaacacgtaataaattttattttattagtgtgatcaatc *falciparum* isolate
gagtttctgacctatcagctttgatgttagggatttggcctaacatggctatgacgggtaacg SF3 18S ribosomal
gggaattagagttcattccggagagggagcctgagaaatagctaccacatctaaggaag RNA gene, partial
gcagcaggcgctaaattaccaattctaaaaagagaggtagtgaagaataacaatg sequence
caaggccaatttttggttttgaattggaatgggtgggaatttaaaccttcccagagtaacaatt
ggaggggcaagtctggtgccagcagccgcggtaattccagctccaatagcgtatattaaaatt
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tgattgaaaagcttcttagaggaacattgtgtgtctaacacaaggaagtttaaggcaacaaca
ggctctgtgatgtccttagatgaactaggctgcacgcgtgctacactgatataataacgagtttt taaaaat 75 gaagtaggactttgggttctattt
Non-T7 primer 76 atgaagtaggactttgggttct Non-T7 primer 77
aacctgggtgatcctgccagtagtcatatgcttgccttaagattaagccatgcatgtctaagta GenBank Accession
caaactttttacggtgaaactgcgaatggctcattacaacagttatagtttctttgggtattcgtttt No: AY046575.1
ccatggataaccgtgctaattgtagggcctaatacaagttcgaggccttttggcggcggtttatta *Babesia* sp. EU1
gttctataaccacccttttgggtttcggtgattcataataaaactgcgaatcgcaattattgcat clone BAB20 18S
ggaccattcaagtttctgacctatcagcttgacggtaggggtattggcctaccgaggcagcaa ribosomal RNA
cgggtaacggggaattaggggtcgattccggagagggagcctgagaaacgggtaccacat gene, complete
ccaaggaaggcagcaggcgcgcaattacccaatcctgacacagggaggtagtgacaag sequence
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gggtgaattattcggaccgtggcttttccgattcgtcgggtttgcctaggggaagtctcgtgaacct
tatcacttaaaggaaggagaagtcgtaacaaggtttccgtaggtgaacctgcagaaggatca agc 78
gyygccycgguaggccaauaccuaccguccaaagcugaur 79 gyygccycggtaggccaataacctaccgtccaaagctgatr 80
aatttaatacgaactcactatagggagaggcaaagtcttgcagtagttngtctttaaca T7 promoter primer,
wherein “n” means a or g or c or t/u, or unknown, or other (WIPO Standard
ST.25 (1998), Appendix 2, Table 1) 81 ggcaaagtcttgcagtagttngtctttaaca SEQ ID NO:
80 without promoter sequence, wherein “n” means a or g or c or t/u, or
unknown, or other (WIPO Standard ST.25 (1998), Appendix 2, Table 1) 82
aatttaatacgaactcactatagggagaggcaaagtcttgcagtagttigtctttaaca T7 promoter primer. “i” means
inosine. 83 ggcaaagtcttgcagtagttigtctttaaca SEQ ID NO: 82 without promoter
sequence. “i” means inosine. 84 gcgtaattccagctccaatag Non-T7 primer 85 cttgaatactncagca
Non-T7 primer, wherein “n” means a or g or c or t/u, or unknown, or other
(WIPO Standard ST.25 (1998), Appendix 2, Table 1) 86 cttgaatacticagca Non-T7
primer. “i” means inosine. 87 aaagacuuugauuucucucaagggtttaaaaaaaaaaaaaaaaaaaaaaaaa
Target capture 88 aaagacuuugauuucucucaagg SEQ ID NO: 87 without capture tail
sequence 89 caagaaagagcuaucuaucugucaaucctttaaaaaaaaaaaaaaaaaaaaaaaaa Target capture aa
90 caagaaagagcuaucuaucugucaaucc SEQ ID NO: 89 without capture tail sequence 91

aguaggacuuugguuct Probe 92 aguaggacuuugguuc Probe 93 aguaggacuuugguuc Probe 94
aguaggacuuugguuc Probe, wherein “x” means (i) u; or (ii) ua 95
actttgagaaaaytagagtgtttcaaacaggccatttgccttgaatactncagcatggaataat “n” means a or g or
gaagtaggactttggttctattttgttggtt c or t/u, or unknown, or other (WIPO Standard ST.25
(1998), Appendix 2, Table 1) 96
ggcaaagtctttcgcagtagtngtctttaacaaatctaagaatttcacctctgacagttaaata “n” means a or g or
cgaatgcccccaa c or t/u, or unknown, or other (WIPO Standard ST.25 (1998),
Appendix 2, Table 1) 97 ggcaaagtctttcgcagtagtngtctttaacaaatc “n” means a or g or c
or t/u, or unknown, or other (WIPO Standard ST.25 (1998), Appendix 2, Table 1) 98
aguaggacuuugguucu Probe 99 aguaggacuuugguucu Probe, wherein “x” means (i) u; or
(ii) ua 100 GGGCGAAUUGGGUACCGGGCCCCCCCCUCGAGGUCGAC *B. venatorum* IVT
GGUAUCGAUAAGCUUGAUUAUCGAAUUCUGCAGCCCG
GGGGAUCCAACCUGGUUGAUCCUGCCAGUAGUCAUAU
GCUUGUCUUAAGAUAUAAGCCAUGCAUGUCUAAGUAC
AACUUUUUACGGUGAAACUGCGAAUGGCUCAUUACA
ACAGUUUAAGUUUCUUUGGUAAUUCGUUUUCCAUGGA
UAACCGUGCUAAUUGUAGGGCUAAUACAAGUUCGAG
GCCUUUUGGCGGCGUUUAUUAGUUCUAUAACCACCCU
UUUGGUUUUCGGUGAUUCAUAAUAAACUCGCGAAUC
GCAAUUUAUUGCGAUGGACCAUUCAGUUUCUGACCC
AUCAGCUUGACGGUAGGGUAUUGGCCUACCGAGGCAG
CAACGGGUAACGGGGAAUUAAGGGUUCGAUUCGGAG
AGGGAGCCUGAGAAACGGCUACCACAUCCAAGGAAGG
CAGCAGGCGCGCAAUUAACCAAUCCUGACACAGGGA
GGUAGUGACAAGAAAUAACAAUACAGGGCAAUUGUC
UUGUAAUUGGAAUGAUGGUGACCUAAACCCUCACCAG
AGUAACAAUUGGAGGGCAAGUCUGGUGCCAGCAGCCG
CGGUAAUUCAGCUCCAUAUAGCGUAUAUUAACUUGU
UGCAGUUAAAAAGCUCGUAGUUGAAUUCUGCGUUA
UCGAGUUUAUUGACUCUUGUCUUUAAUCGAUUUCGCUU
UUGGGAAUUUAUCCCUUUUUACUUUGAGAAAAUUAGA
GUGUUUCAAGCAGACUUUUGUCUUGAAUACUUCAGCA
UGGAUAUAUAGAGUAGGACUUUGGUUCUAUUUUGUU
GGUUUUUGAACCUUAGUAAUGGUUAAUAGGAACGGU
UGGGGGCAUUCGUAAUUUAACUGUCAGAGGUGAAAUU
CUUAGAUUUUGUUAAGACGAACUACUGCGAAAGCAU
UUGCCAAGGACGUUCCAUAUAUCAAGAACGAAAGUU
AGGGGAUUCGAAGACGAUCAGAUACCGUCGUAGUCCUA
ACCAUAAACUAUGCCGACUAGGGAUUGGAGGUCGUCA
UUUUUCCGACUCCUUCAGCACCUUGAGAGAAAUCAA
GUCUUUGGGUUCUGGGGGGAGUAUGGUCGCAAGGCU
GAAACUUAAGGAUUGACGGAAGGGCACCACCAGGC
GUGGAGCCUGCGGCUAAUUAUUGACUCAACACGGGGAA
ACUCACCAGGUCCAGACAAUGUUAGGAUUGACAGAUU
GAUAGCUCUUUCUUGAUUCUUUGGGUGGUGCGGCC 101 ctttcgcagtagtngtctttaac
Wherein “n” means a or g or c or t/u, or unknown, or other (WIPO Standard
ST.25 (1998), Appendix 2, Table 1).

Claims

1. A combination of at least two oligomers for determining the presence or absence of *Babesia* in a sample, said oligomer combination comprising first and second amplification oligomers for amplifying a target region of *Babesia* target nucleic acid, wherein (a) the first amplification oligomer comprises a first target-hybridizing sequence that consists of SEQ ID NO:8, wherein the first amplification oligomer is a promoter primer or promoter provider further comprising a promoter sequence located 5' to the first target-hybridizing sequence; and (b) the second amplification oligomer comprises a second target-hybridizing sequence that consists of SEQ ID NO:34.
2. The combination of claim 1, wherein the promoter sequence is a T7 promoter sequence.
3. The combination of claim 2, wherein the first amplification oligomer nucleotide sequence consists of SEQ ID NO:7.
4. The combination of claim 1, further comprising at least one capture probe oligomer, wherein the at least one capture probe oligomer comprises a target-hybridizing sequence covalently attached to a sequence or moiety that binds to an immobilized probe, wherein said target-hybridizing sequence consists of SEQ ID NO:44, SEQ ID NO:88, or SEQ ID NO:90.
5. The combination of claim 4, wherein the capture probe oligomer sequence comprises or consists of SEQ ID NO:43, SEQ ID NO:87, or SEQ ID NO:89.
6. The combination of claim 1, further comprising a detection probe oligomer configured to specifically hybridize to a *Babesia* amplification product amplifiable by the first and second amplification oligomers.
7. The combination of claim 6, wherein the detection probe oligomer comprises a detection probe target-hybridizing sequence consisting of the sequence selected from the group consisting of: SEQ ID NO:37, SEQ ID NO:42, SEQ ID NO:91, SEQ ID NO: 92, SEQ ID NO:93, and combinations thereof.
8. The combination of claim 7, wherein the detection probe comprises a fluorescent label and a quencher.
9. The combination of claim 1, wherein the at least two amplification oligomers further comprise: (a) a third amplification oligomer comprising a third target-hybridizing sequence that consists of SEQ ID NO:83; (b) a fourth amplification oligomer comprising a fourth target-hybridizing sequence that consists of SEQ ID NO:84; and, (c) a fifth amplification oligomer comprising a fifth target-hybridizing sequence that consists of SEQ ID NO:86.
10. The combination of claim 9, further comprising at least one capture probe oligomer, wherein the at least one capture probe oligomer comprises a target-hybridizing sequence covalently attached to a sequence or moiety that binds to an immobilized probe, wherein said target-hybridizing sequence consists of SEQ ID NO:44, SEQ ID NO:88, or SEQ ID NO:90.
11. The combination of claim 1, further comprising a detection probe oligomer, wherein the detection probe oligomer comprises a detection probe target-hybridizing sequence consisting of a sequence selected from the group consisting of SEQ ID NO: 91, SEQ ID NO:92, and SEQ ID NO:93.
12. A combination of at least two oligomers for determining the presence or absence of *Babesia* in a sample, said oligomer combination comprising first and second amplification oligomers for amplifying a target region of *Babesia* target nucleic acid, wherein (a) the first amplification oligomer comprises a first target-hybridizing sequence that consists of SEQ ID NO:8, wherein the first amplification oligomer is a promoter primer or promoter provider further comprising a promoter sequence located 5' to the first target-hybridizing sequence; (b) the second amplification oligomer comprises a second target-hybridizing sequence that consists of SEQ ID NO:34; and (c) a detection probe oligomer configured to specifically hybridize to a *Babesia* amplification product amplifiable by the first and second amplification oligomers, wherein the detection probe oligomer comprises a 2' methoxy modification on at least one nucleotide residue member of the nucleotide

sequence.

13. A combination of at least two oligomers for determining the presence or absence of *Babesia* in a sample, said oligomer combination comprising first and second amplification oligomers for amplifying a target region of *Babesia* target nucleic acid, wherein (a) the first amplification oligomer comprises a first target-hybridizing sequence that consists of SEQ ID NO:8, wherein the first amplification oligomer is a promoter primer or promoter provider further comprising a promoter sequence located 5' to the first target-hybridizing sequence; (b) the second amplification oligomer comprises a second target-hybridizing sequence that consists of SEQ ID NO:34; and (c) a detection probe oligomer configured to specifically hybridize to a *Babesia* amplification product amplifiable by the first and second amplification oligomers, wherein the detection probe comprises a chemiluminescent label or a fluorescent label.
